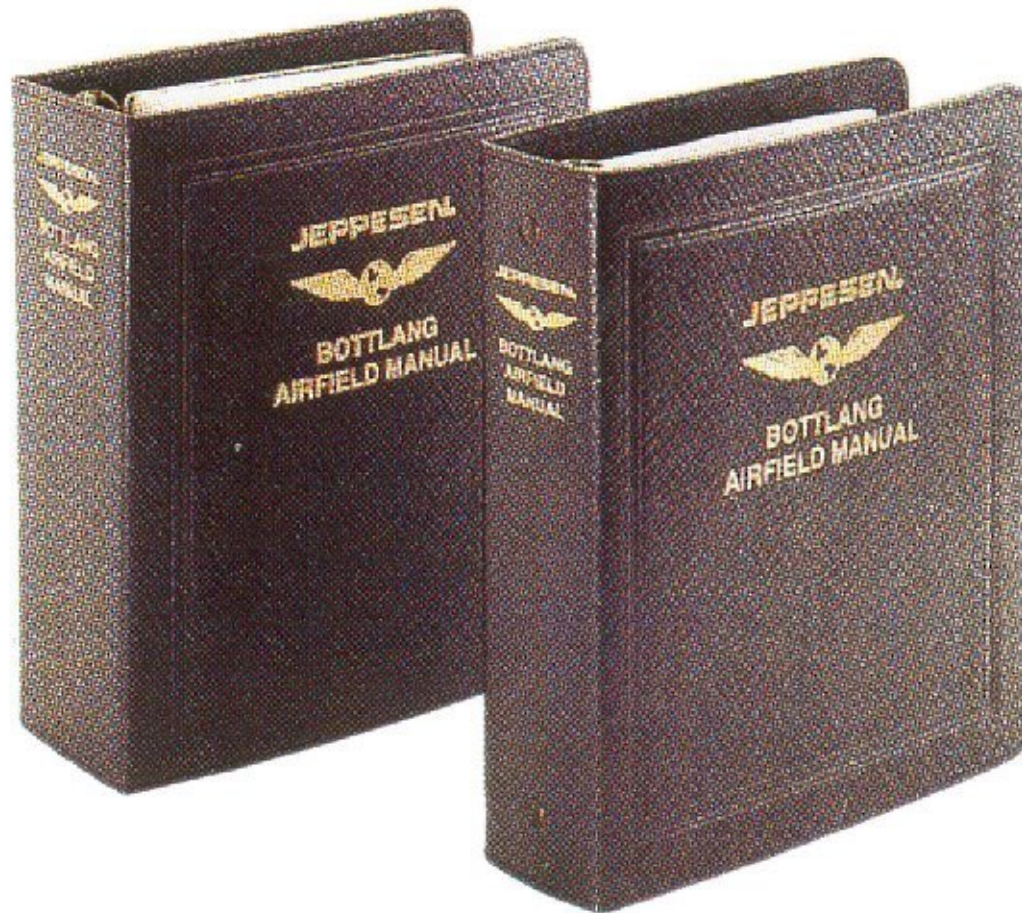


JEPPESEN



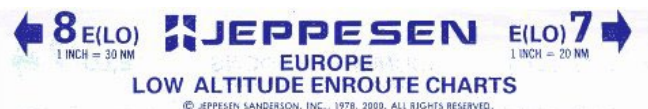
PRESENTATION

2 manuals:

1. First manual: you can find EN ROUTE charts, and different information about Meteorology, the airport, the radio aids, etc

2. Second manual: You can find TERMINAL CHARTS (departure, arrival, different procedures...)

DÉSIGNATION ET PRÉSENTATION DES CARTES



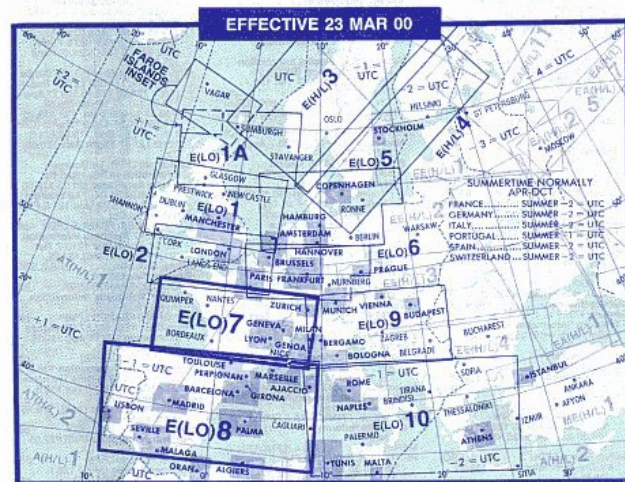
AIRWAYS/ROUTES/CONTROLLED AIRSPACE shown on these charts are generally effective up to the upper limits of the low airspace of each country tabulated below. Refer to HIGH ALTITUDE CHARTS for operations above the upper limits of low altitude airspace. Flying outside of ATS routes is prohibited within Algeria.

LIMITS AND CLASSIFICATIONS OF DESIGNATED AIRSPACE					
	CLASS	LOWER - RNAV - UPPER		CLASS	LOWER - RNAV - UPPER
ALGERIA	(G)	up to FL 245	PORTUGAL	(G)	up to FL 245
FRANCE	(G,D)	up to FL 195	SPAIN	(A/E/G)	up to FL 245
GERMANY	(C,D,E,G)	GND - FL 95 - FL 245 (FL 195 Alpine Area)	SWITZERLAND	(C,D,E,G)	up to FL 195
ITALY	(D,E,F,G)	GND - FL 95 - FL 195			

REVISION DATA

CHART E(LO) 7 10 MAR 00 ATS system revised within France and Switzerland.

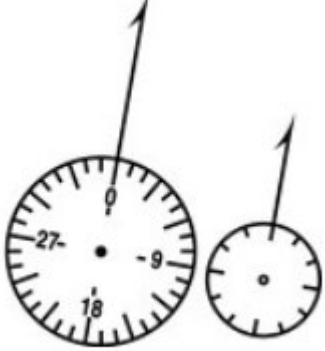
CHART E(LO) 8 10 MAR 00 ATS system revised within Portugal and Spain.



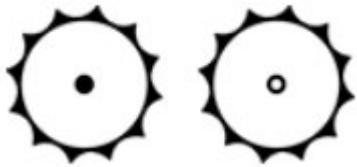
COMMUNICATIONS

TABULATION LEGEND - -
BOLD NAME - Voice call. **Light Names/Abbreviations** - Identifying names/abbreviations not used in radio call. **G** - Guard only. ***** - Part-time operation. **X** - On request. **(R)** - Radar Capability. **C** - Clearance Delivery. **Cpt** - Clearance (Pre-Info Proc.) **pSC (LFL)** - Charted location is shown by Area chart initials in parenthesis and/or by quarter panel number-letter combination. Common. **EMERGENCY 121.5** is not listed. Refer to Glossary and Abbreviations in Introduction pages for further explanations.
SSB - All HF communications listed below have single side band capability unless indicated otherwise.

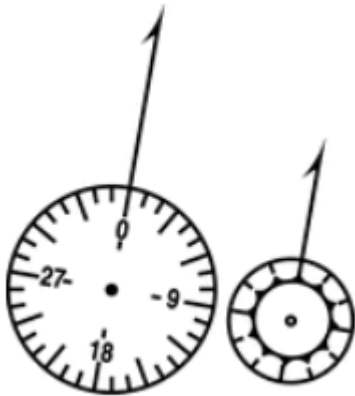
A CORUNA, SPAIN LECO	05A	Avignon, France LFNT	(LFML,LFMN)	BRIARE, FRANCE LFEI	02B
Santiago App 130.2 A Coruna *Twr		Puyat, *Information 123.35 (Air to Air)		Chailon, Briare *Information (AFIS)	
15.3 *Gnd 121.7		AVOIRON, FRANCE LFDA	02B	125.4	
AGEN, FRANCE LFDA	02D/0B	Avord AB, Avord *App 119.7 *Twr		BRIVE, FRANCE LFBV	02D
La Garonne, ATIS 129.6 Agen *App		122.1		La Roche, Brive *Information (AFIS)	



VOR (VHF Omnidirectional Range)



**TACAN (Tactical Air Navigation) or DME
(Distance Measuring Equipment)**



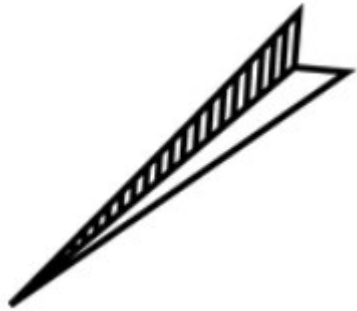
VORTAC/VORDME



NDB (Nondirectional Radio Beacon)



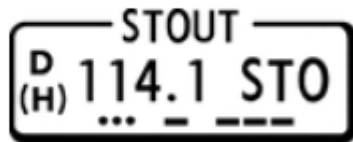
Magnetic north ticks on navigational facilities fit compass roses on IFR Enroute Chart



LOC,

Navaid identification is given in shadow box when navaid is airway or route component, with frequency, identifier, and Morse Code. DME capability is indicated by a small “D” preceding the VOR frequency at frequency paired nav aids. VOR and VORTAC navaid operational ranges are identified (when known) within the navaid

NAVAID IDENTIFICATION



TRINITY AFRS
1490



KADENA
D 112.0 KAD
N26 22.4 E127 48.0
335 KD
N26 20.0 E127 44.8

BENBECULA
D 114.4 BEN
(DME not Co-located)

Armed Forces Radio Station

**Fréquences: VOR, ADF + coordonnées
Indicatif et indicatif morse**

When VOR and TAC/DME antennas are not co-located, a notation “DME not Co-located” is shown below the navaid box.

Off-airway navaids are unboxed on Low and High/Low charts. TACAN/DME channel is shown when VOR navaid has a frequency paired DME capability. When an L/MF navaid performs an enroute function, the Morse Code of its identification letters are shown

MOODY
113.3 VAD
TAC-80

KENNEY
254 ENY

LIPTON
TAC-88 LPT
(114.1)

122.2-122.45-5680
RIVER
D 114.6 RIV
... ..

Off-airway nav aids are unboxed on Low and High/Low charts. TACAN/DME channel is shown when VOR nav aid has a frequency paired DME capability. When an L/MF nav aid performs an enroute function, the Morse Code of its identification letters are shown.

When TACAN or DME are not frequency paired with the VOR, the TACAN is identified separately. The “Ghost” VOR frequency, shown in parentheses, enables civilian tuning of DME facility.

River Radio transmits on 114.6 and transmits and receives on 122.2, 122.45 MHz and HF frequency 5680.

HIWAS
MIAMI WX *122.0
MIAMI
D 115.9 MIA
N25 57.8 W080 27.6

**HIWAS — Service d'avis
météorologiques dangereux
en vol. Diffusions SIGMETS,
AIRMETS en continu sur la
fréquence VOR.**



**Restricted airspace. The accompanying
label indicates it as prohibited, restricted,
danger, etc.**



**Training, Alert, Caution, and
Military Operations Areas.**

• ED (R)-7



Dot indicates permanent activation on some chart series.

RESTRICTED AIRSPACE DESIGNATION

A:	Alert
T:	Training
C:	Caution
W:	Warning
D:	Danger
TRA:	Temporary Reserved Airspace
P:	Prohibited
TSA :	Temporary Segregated Area
R:	Restricted
MOA:	Military Operations Area

AIRPORTS

Civil		Military		
IFR	VFR	IFR	VFR	
				Airports
				Seaplane Base
				Heliports

(LAA) : LAA Local Airport Advisory

(AFIS) : AFIS (Aerodrome Flight Information Service)

(ALA) : Authorized Landing Area

AIRWAY AND ROUTE COMPONENTS

AIRWAY AND ROUTES CENTER LINES



Airway/Route



Diversionary Route



Overlying High Altitude Airway/Route



Oceanic Transition Route



RNAV Airway/Route



Compulsory Reporting Point



Non-Compulsory Reporting Point



Low Altitude Compulsory Reporting Point



Low Altitude non- Compulsory Reporting Point



Holding Pattern. DME figures, when provided, give the DME distance of the fix as the first figure followed by the outbound limit as the second figure.



Length of holding pattern in minutes when other than standard

LIMON
V-8 7500 NW
(MRA 7000)

Fix name with Minimum Crossing Altitude (MCA) showing airway, altitude, and direction, and Minimum Reception Altitude (MRA).

△ — 095° →

LF bearings forming a fix are to the navaid

△ ← 296° —

VHF radials forming a fix are from the navaid.

△ ← 296° $\frac{\text{BOR}}{116.8}$

VHF frequency and identifier included when off chart or remoted.

△ $\frac{\text{ABC} \dots}{294}$ — 095° →

LF frequency, identifier and Morse Code included when off chart or remoted.

△ — ←

Arrow along airway points from the navaid designating the reporting point.

△ **D55/MAZ**

Fix formed by 55 DME from MAZ navaid.

V 168

Airway and route designators. Negative (white letters in black) designators are used for distinction.

ATS

ATS-Designated route without published identifier

AWY 4

AWY-Airway

BR 7

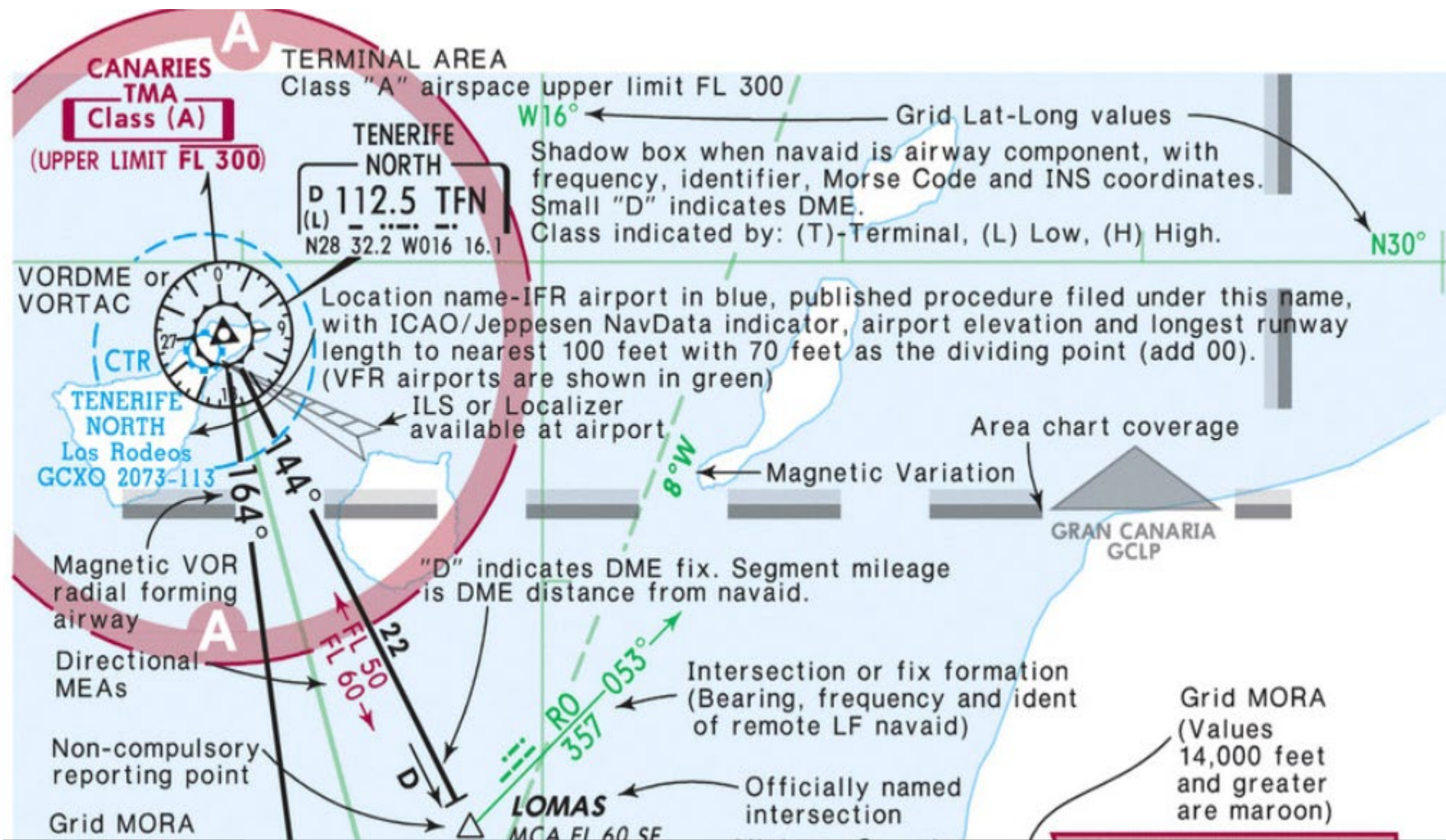
BR-Bahama Route, Canada Bravo Route

D

Direct Route

DOM

DOM-Domestic Route. Use by foreign operators requires special authorization.



Grid MORA
(Values less than
14,000 feet are
green)

145

High Altitude Route
Included for
orientation only

Arrow from facility
designating reporting
point

Compulsory
reporting point
with name and
INS coordinates

Intersection or fix
formation (Bearing,
frequency and ident
of remote VHF navaid;
Morse code shown when
navaid is outside of
chart neatline)

ROYAL
N25 00.0
W015 47.2

CWT
115.1

105

Route usability by non
B-RNAV equipped aircraft
(within Europe only)

MCA FL 60 SE

MEA
change

Segment mileage

Changeover Point

"D" with distance
indicates total DME
distance from navaid.

CABOJ

N26 08.3
W014 38.6
MRA 7000

141

Uncontrolled
airspace
(Class G)

Holding pattern

Total mileage
between navaids

Airway Designator

Flight planning relevant restrictions for
ATS routes on Eastern Hemisphere
chart series are published on respective
EROUTE pages

FL 90 45

G851

Intersection
Minimum Crossing
Altitude (MCA)
for SE-bound flights

77

Special use airspace

are maroon)

GM(D)-24

FL 250
GND

NAME
112.5 ABC
N26 20.0 W014 05.0

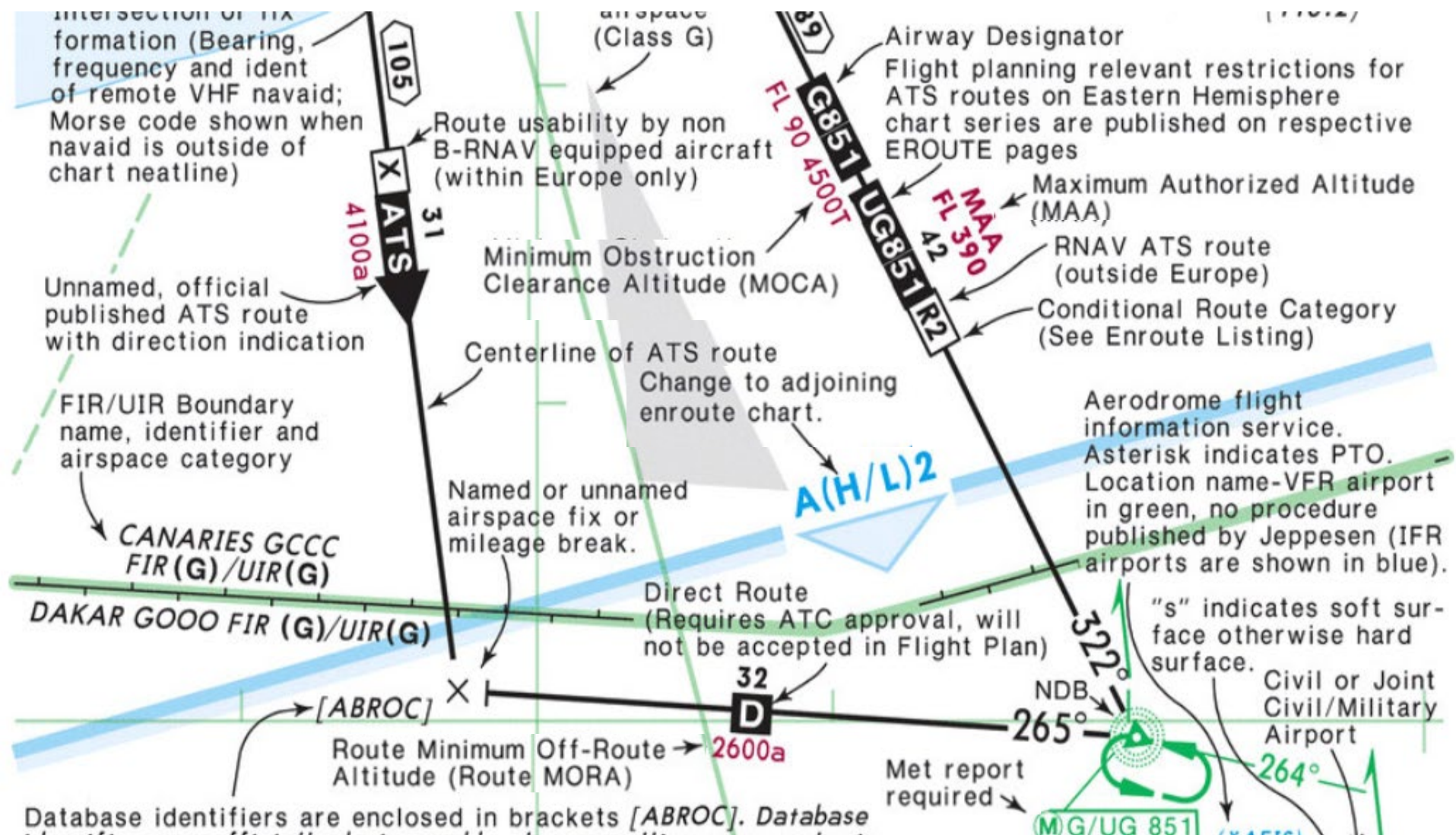
Minimum Reception
Altitude (MRA)

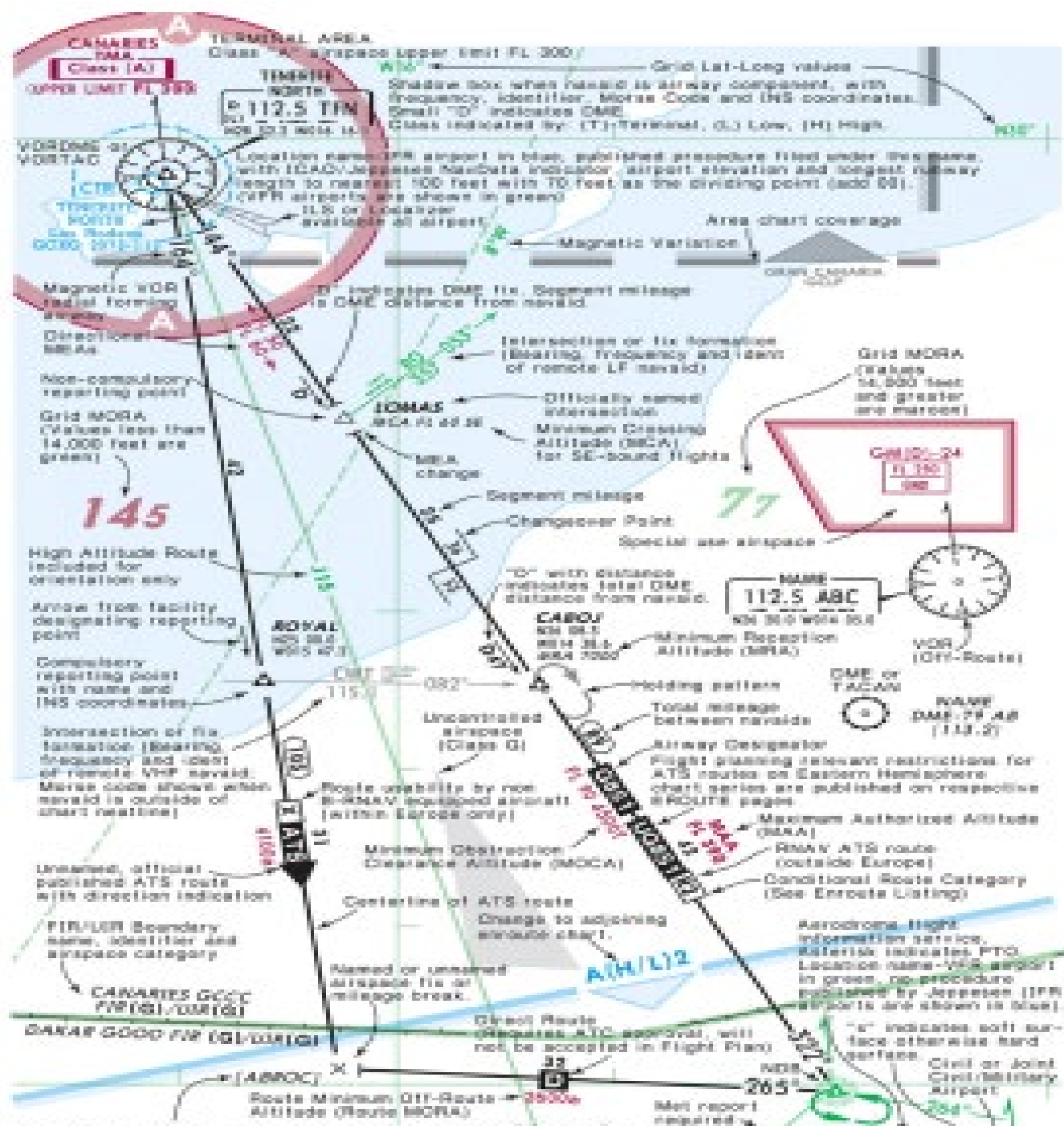
VOR
(Off-Route)

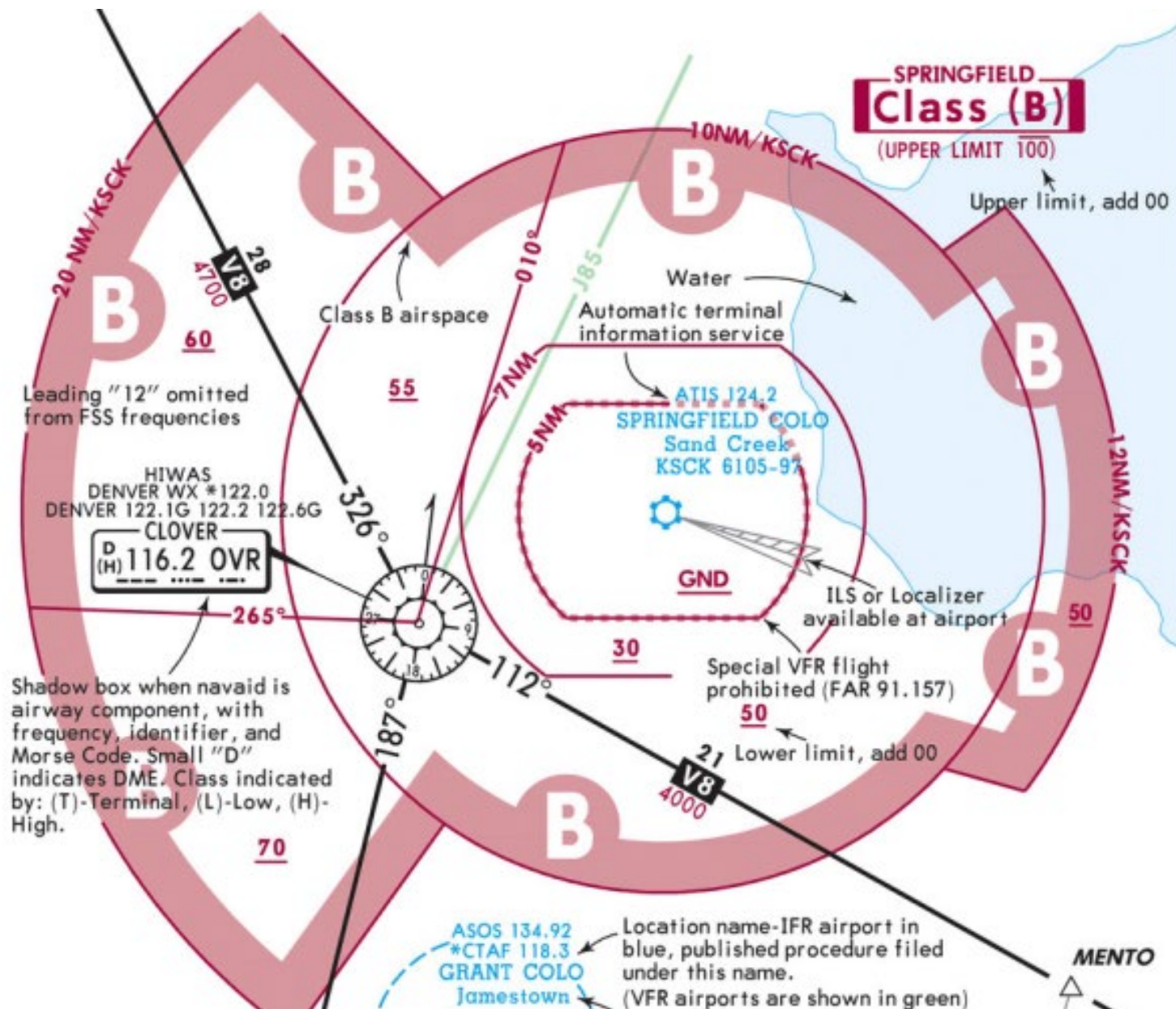
DME or
TACAN



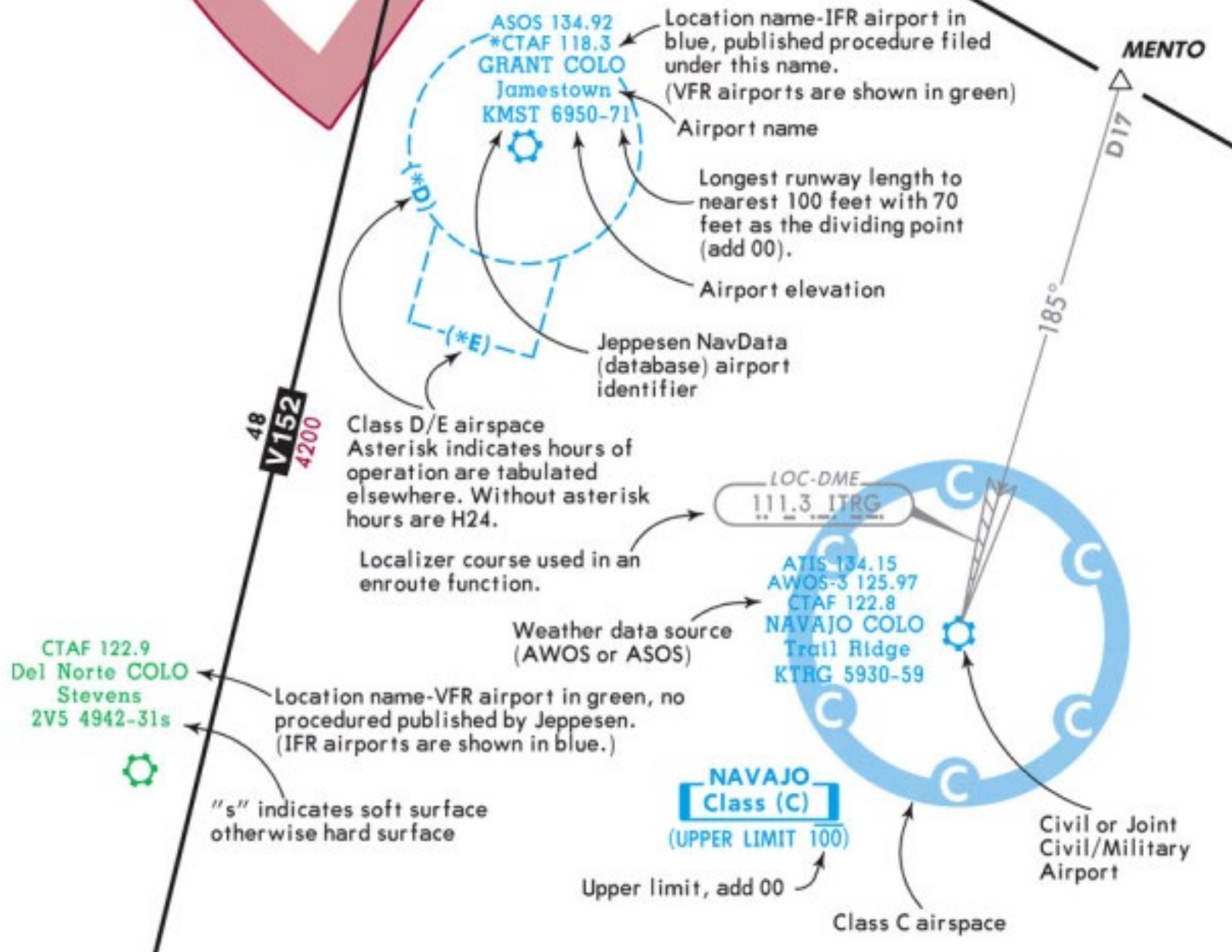
NAME
DME-79 AB
(113.2)







70



TERMINAL AREA

Class "A" airspace upper limit FL 300

FL 300
CANARIES TMA(A)

TENERIFE

NORTH

D 112.5 TFM

(L)
N28 32.2 W016 16.1

VOR, VORDME
or VORTAC

TENERIFE
NORTH
Los Rodeos
GCMO

IFR Airport of Entry with at least one
paved runway exceeding 6000' with
ICAO/Jeppesen NavData indicator

Segment mileage

Magnetic VOR
radial forming
airway

Directional
MEAs

Non-compulsory
reporting point

Grid MORA
(Values less than
14,000 feet are
green)

145

"D" indicates DME fix. Segment mileage
is DME distance from navaid.

164°
144°
122
150
60

D

LOMAS
MCA FL 370 SE

MEA
change

Segment mileage

"D" with distance
indicates total DME

Magnetic Variation

Shadow box when navaid is airway component, with
frequency, identifier, Morse Code and INS coordinates.
Small "D" indicates DME.

Class indicated by: (T)-Terminal, (L) Low, (H) High.

Grid Lat-Long values

GRAN CANARIA
GCLP

Area chart coverage

Intersection or fix formation
(Bearing, frequency and ident
of remote LF navaid)

Officially named
intersection
Minimum Crossing
Altitude (MCA)
for SE-bound flights

Grid MORA
(Values
14,000 feet
and greater
are maroon)

GM(D)-24

77

Special use airspace

145

77

Segment mileage

Special use airspace

"D" with distance indicates total DME distance from navaid.

Arrow from facility designating reporting point

ROYAL
N25 00.0
W015 47.2

Compulsory reporting point with name and INS coordinates

CWT 115.1

CABOJ
N26 08.3
W014 38.6

Time zone boundary with conversion values to UTC

+1 = UTC UTC

Holding pattern

Uncontrolled airspace (Class G)

Changeover Point

Intersection or fix formation (Bearing, frequency and ident of remote VHF navaid; Morse code shown when navaid is outside of chart neatline)

Route usability by non B-RNAV equipped aircraft (within Europe only)

Flight planning relevant restrictions for ATS routes on Eastern Hemisphere chart series are published on respective EROUTE pages

Maximum Authorized Altitude (MAA)

RNAV ATS route (outside Europe)

Conditional Route Category (See Enroute Listing)

Total mileage between navaids

Unnamed, official published ATS route with direction indication

RVSM airspace boundary with designator

RVSM AIRSPACE
CANARIES GCCC UIR (G)

Centerline of ATS route

Named or unnamed airspace fix or mileage break.

A(HI)2

ATAR

145

77

Segment mileage

Special use airspace

"D" with distance indicates total DME distance from navaid.

Arrow from facility designating reporting point

ROYAL
N25 00.0
W015 47.2

CABOJ
N26 08.3
W014 38.6

Time zone boundary with conversion values to UTC

Compulsory reporting point with name and INS coordinates

CWT
115.1

082°

DAY

Holding pattern

+1 = UTC UTC

Intersection or fix formation (Bearing, frequency and ident of remote VHF navaid; Morse code shown when navaid is outside of chart neatline)

Uncontrolled airspace (Class G)

Route usability by non B-RNAV equipped aircraft (within Europe only)

Changeover Point

Flight planning relevant restrictions for ATS routes on Eastern Hemisphere chart series are published on respective EROUTE pages

Maximum Authorized Altitude (MAA)

RNAV ATS route (outside Europe)

Conditional Route Category (See Enroute Listing)

Total mileage between navaids

Unnamed, official published ATS route with direction indication

RVSM airspace boundary with designator

RVSM AIRSPACE
CANARIES GCCC UIR (G)

Centerline of ATS route

Named or unnamed airspace fix or mileage break.

FL 370
FL 390
MAA 42

FL 370

FL 390

MAA 42

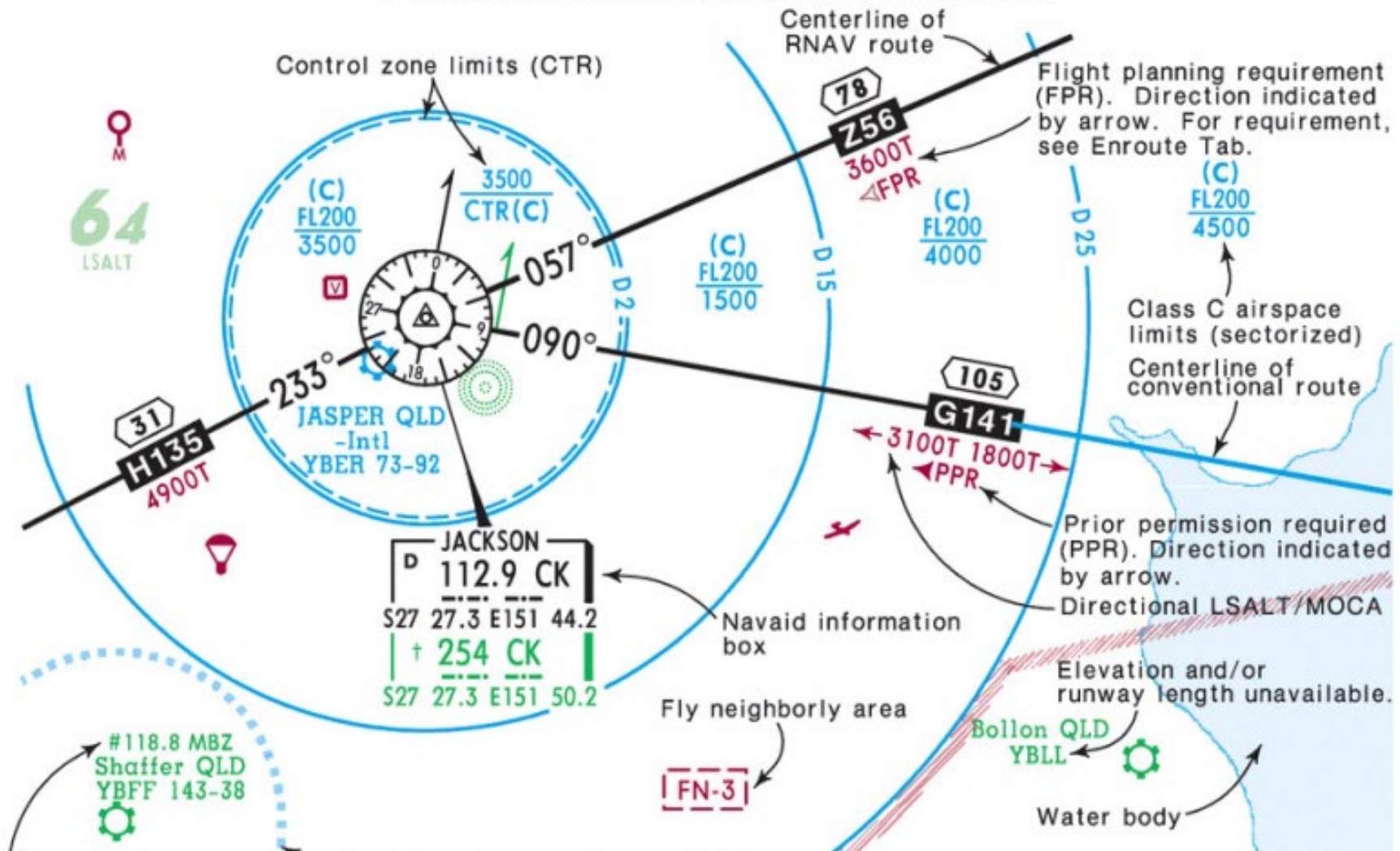
UG851 R2

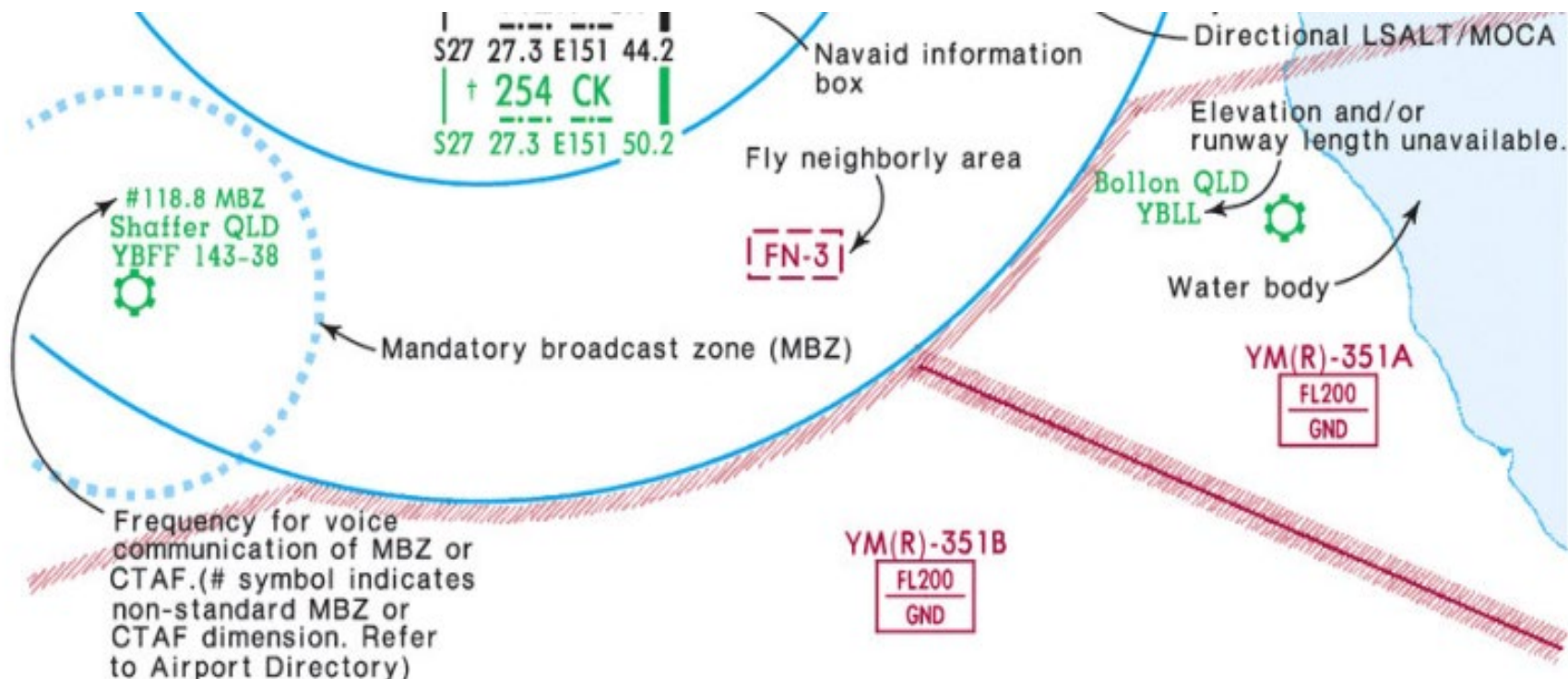
89

A(HI)2

ATAR

The symbology explained on these pages pertain specifically to Australia Enroute and Area charts.





SPECIAL ACTIVITY AREAS

- ✈ Ultra-light activity above 500' AGL.
- 🚁 Hang glider activity above 5000' AGL.
- ✈ Model aircraft activity above 300' AGL.
- 🎈 Meteorology balloon ascents.
- 🎈 Manned balloon ascents.
- 🪂 Parachute jumping area.

AIR TRAFFIC SERVICE UNITS & BOUNDARIES



-  Hang glider activity above 5000' AGL.
-  Model aircraft activity above 300' AGL.
-  Meteorology balloon ascents.
-  Manned balloon ascents.
-  Parachute jumping area.
-  Glider Operations.
-  Gliders Launching.

 Airport within VHF range of responsible ATS unit.

Non-standard CTAF and MBZ, see airport directory for dimensions.

† † Navaid limitation, see Radio Aids page AU-37 (applicable only for Australia domestic services).

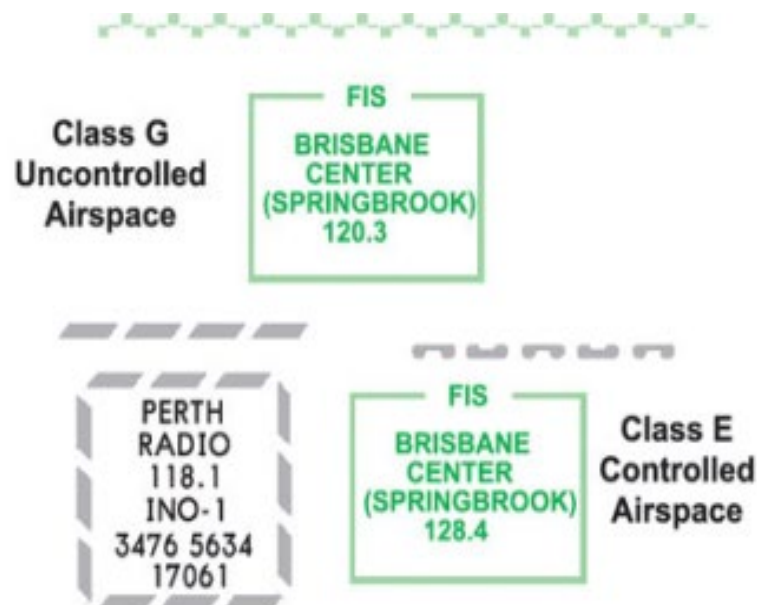
REPORTING POINTS (AUSTRALIA)

COMPULSORY for all aircraft.

  All altitude
  Low altitude

ON-REQUEST 300 KT TAS or more.
 COMPULSORY Under 300 KT TAS.

  All altitude



ROUTE DESIGNATORS

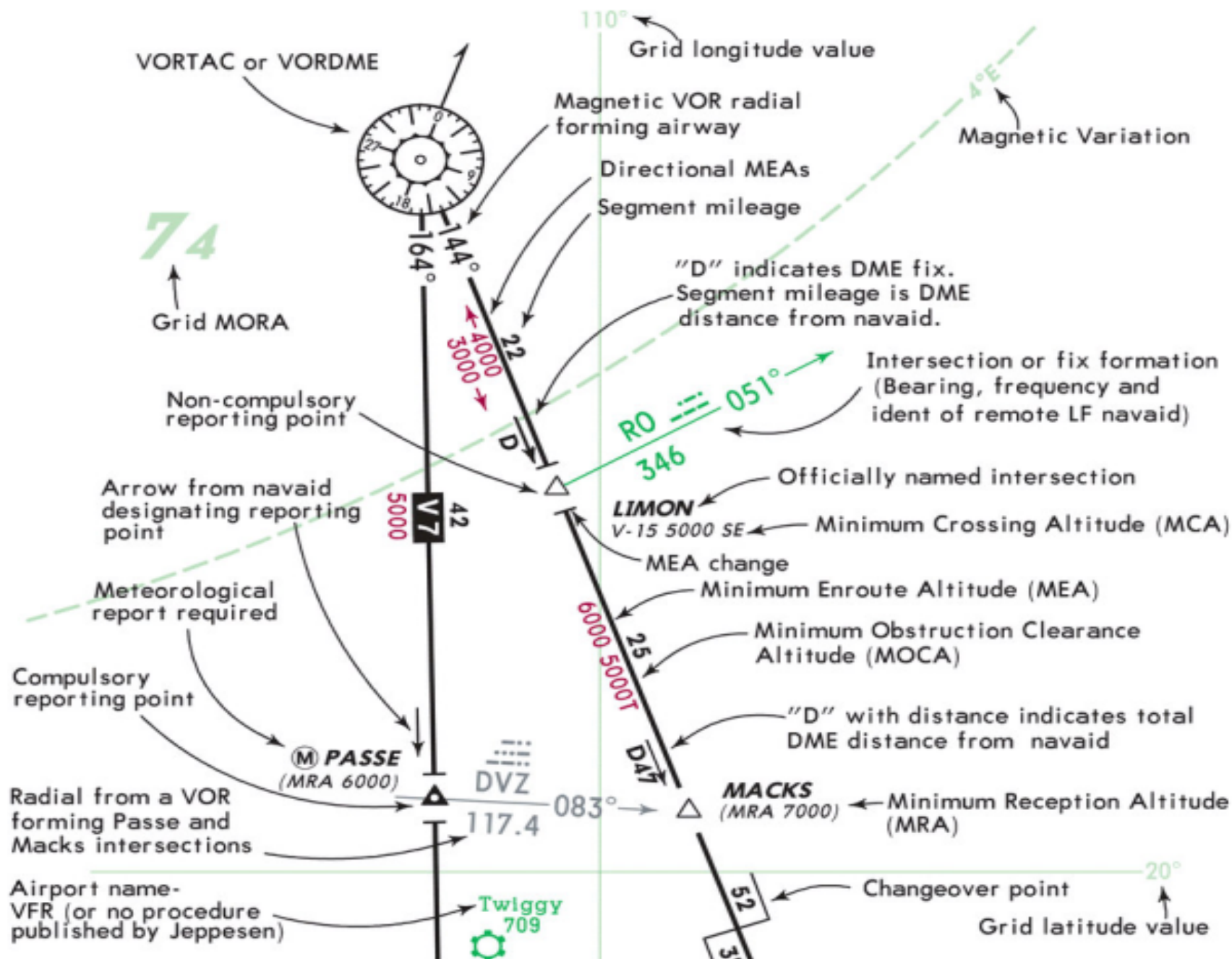
Conventional Routes:

A,B,G,R: Regional
 H (one-way), J (two-way): Domestic
 V (one-way), W (two-way):
 Predominantly low-level domestic

RNAV Routes:

L,M,N: Regional (Tasman)
 Q: 180°-359° domestic
 Y: 360°-179° domestic
 T: Two-way domestic
 Z: Two-way low-level domestic

 JEPPESEN.



Airport name-
VFR (or no procedure
published by Jeppesen)

Twiggy
709

Changeover point
Grid latitude value

Maximum Authorized Altitude
(MAA)

Airway designator

Total mileage
between navaids

Local Airport Advisory
service available

Location name- IFR
airport (Procedures
published by Jeppesen
filed under this name)

Airport name
Airport elevation
Civil or Joint
Civil/Military
airport

6000

33

(LAA)
WALDO IND
Vogt
1345

Change to adjoining
enroute chart at
this point.

2

[ABROC]

Database identifiers are enclosed in brackets [ABROC]
Database identifiers are officially designated by the
controlling state authority or they may be derived by
Jeppesen. In either case, these identifiers have no ATC function

and should not be used in filing flight plans nor should they be used when
communicating with ATC. They are shown only to enable the pilot to maintain
orientation when using charts in concert with database navigation systems.

52
37

MAA 6000
MAA 13000

42

V15

89

315°

255°

NDB

30

V7

105

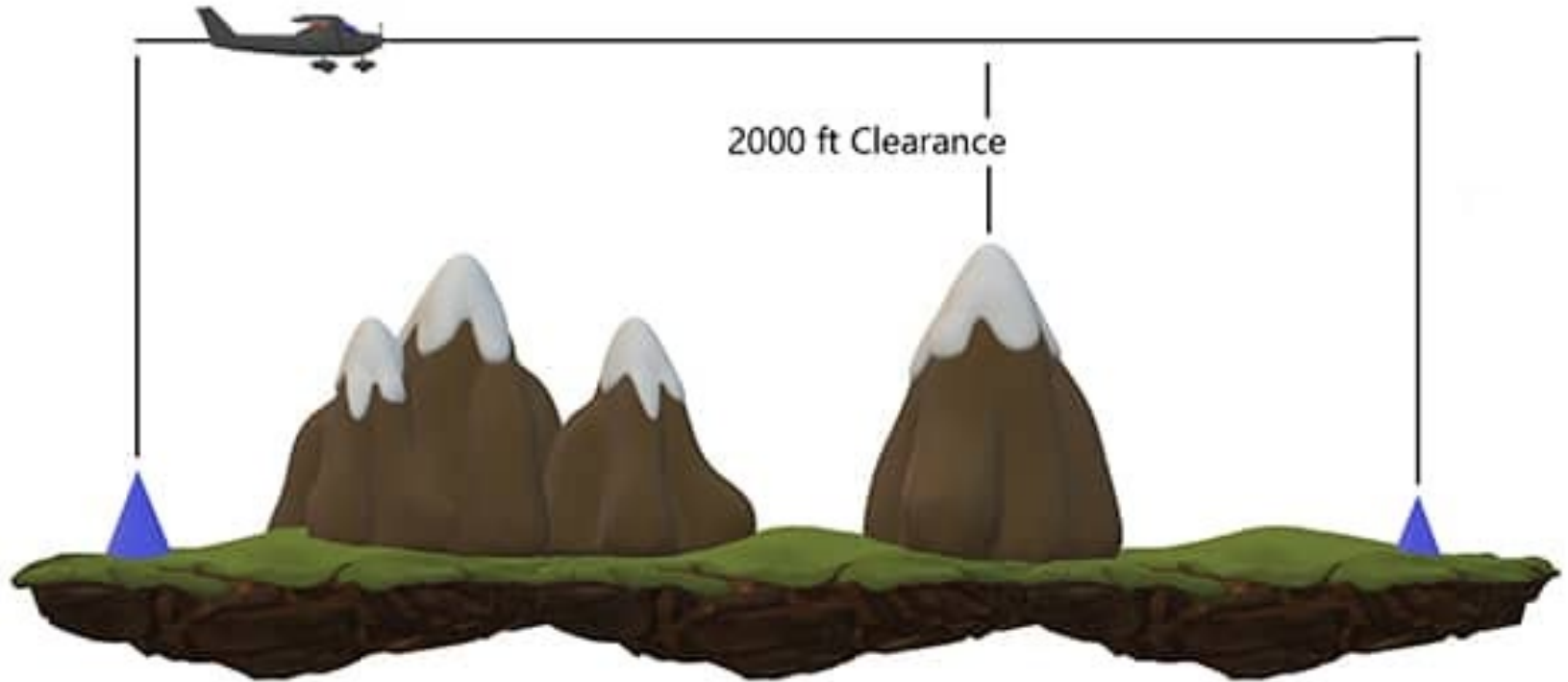
4000a

Route Minimum Off-Route
Altitude (Route MORA)

Centerline of airway

JEPPESSEN.

MSA, MEA, MOCA, MORA, MVA. What are they?

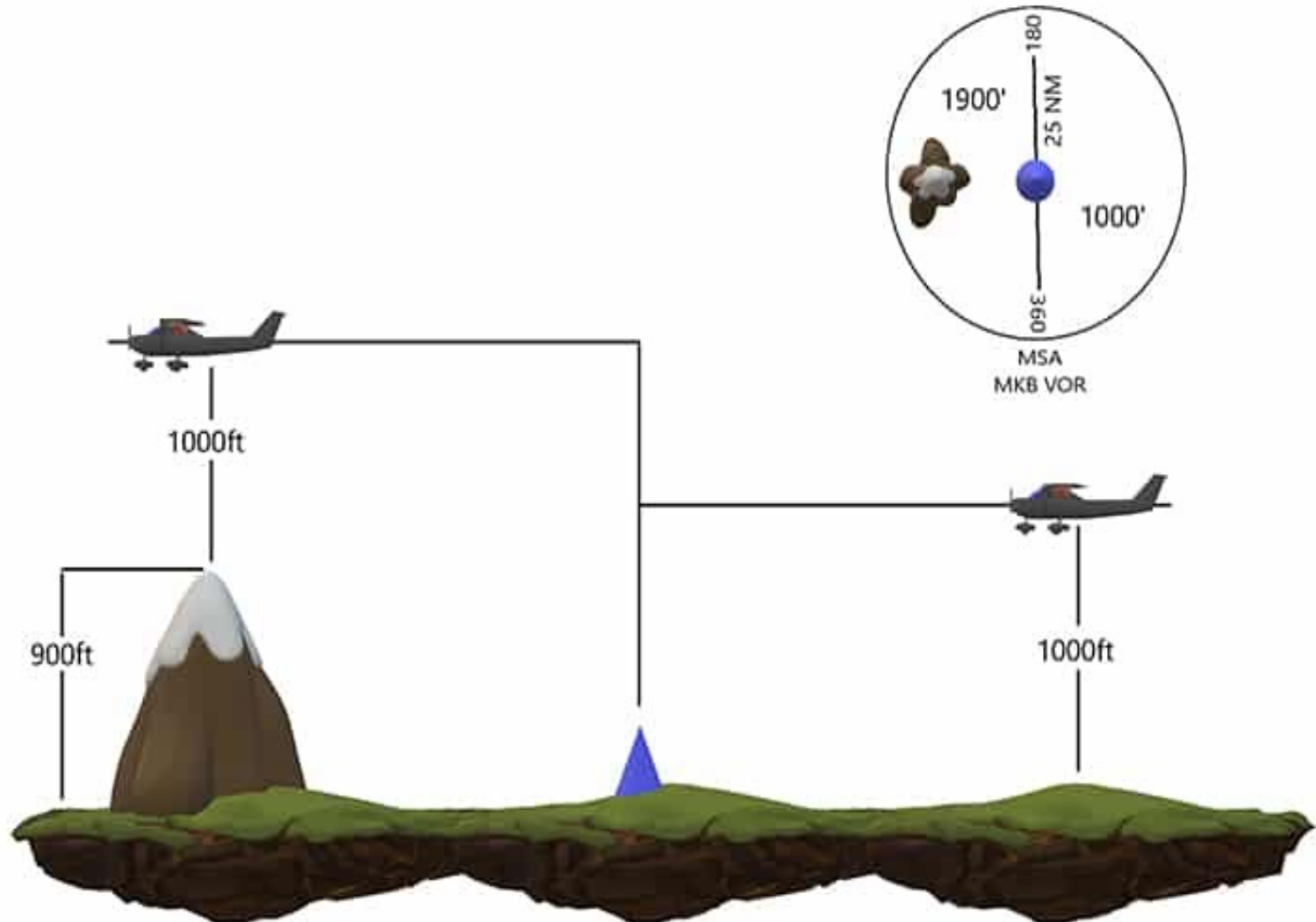


MSA (Minimum Sector Altitude)

MSA is the minimum safe altitude within a 25 nm (Nautical Mile) radius of a navigational aid. MSA will give you 1000 ft separation from the highest terrain within that sector. We normally see the MSA on our Approach plates for runways.

Can we descend below MSA? Yes, provided we are under radar control and we still remain above the radar altitudes. (Remember it is still your responsibility to make sure you don't fly into terrain!)

MSA (Minimum Sector Altitude)

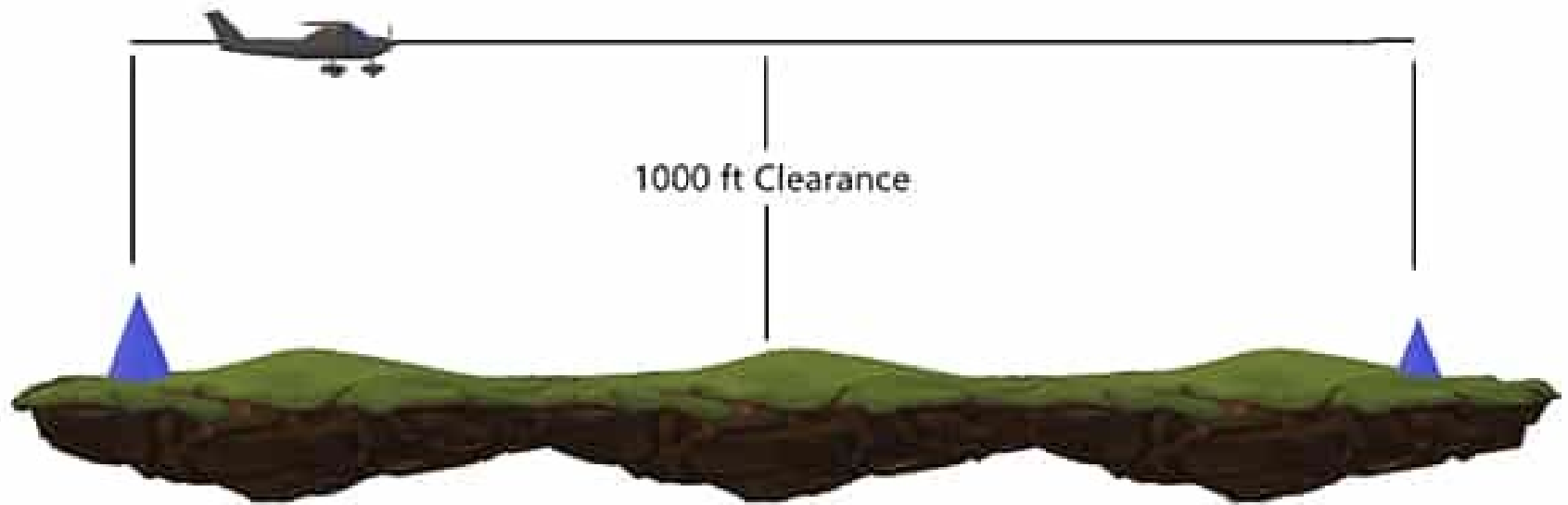


MEA

La MEA ou **Minimum en route Altitude** est l'**altitude** la plus basse entre deux balises, qui permet à la fois le franchement des obstacles et la réception des aides radio. Celle ci peut-être définie pour l'ensemble de la voie aérienne (airway) ou une portion de cette dernière, elle assure:

- MFO 1 000 ft
- 5 NM de part et d'autre de la voie aérienne.
- Assure la réception des Moyens radio.

MEA



MORA

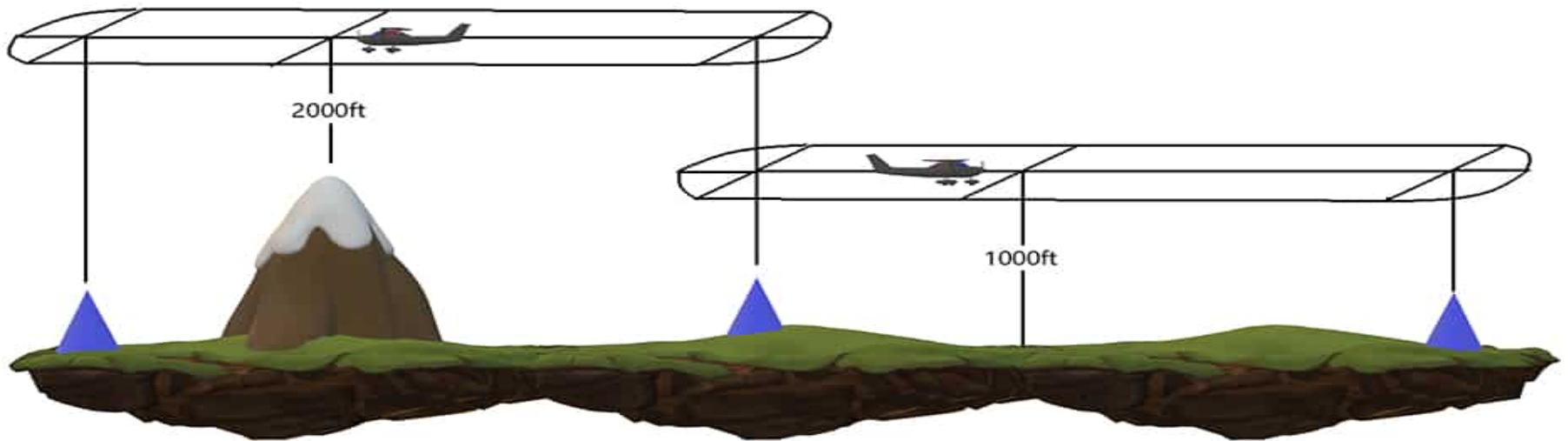
MINIMUM OF ROUTE ALTITUDE

- **MFO 1 000 ft (obstacles inférieurs à 5000 ft)**
- **MFO 2 000 ft (obstacles > 5 000 ft)**
- **10 NM de part et d'autre de la voie aérienne**

MORA (suite)

MORA will give you separation from terrain up to 10 nm off the route center line and 10 nm radius around the ends of the route.

MORA will give you 1000 ft separation from terrain in non mountainous areas and 2000 ft separation in mountainous areas



MORA

- **MORA** ou Minimum Off Route Altitude est une altitude minimale de vol calculée par Jeppesen à partir des cartes usuelles **ONC** ou **WAC**.

Il existe deux types de **MORA** qui sont :

- a - la **MORA** de route Exemple : **9800a** ;
- b - la **MORA** de grille Exemple : **98**.

Les valeurs **MORA** de route sont calculées sur la base d'une surface s'étendant sur **10 NM** de chaque côté de l'axe de la route et incluant un arc de cercle de **10 NM** au-delà du moyen radio / point de compte rendu ou du point de mesure de distance définissant le segment de route.

MORA (suite)

(Minimum Off-Route Altitude)

Elle garantit une marge de franchissement d'obstacles de 1000 ft en dessous de 5000 ft MSL et de 2000 ft au-dessus de 5001 ft MSL dans une zone de 10 NM de part et d'autre de la voie aérienne.

En dehors des voies aériennes se référer à la Grid MORA altitude calculée par Jeppesen. Les valeurs sont indiquées par chaque maille de la grille formée par les méridiens et les parallèles.

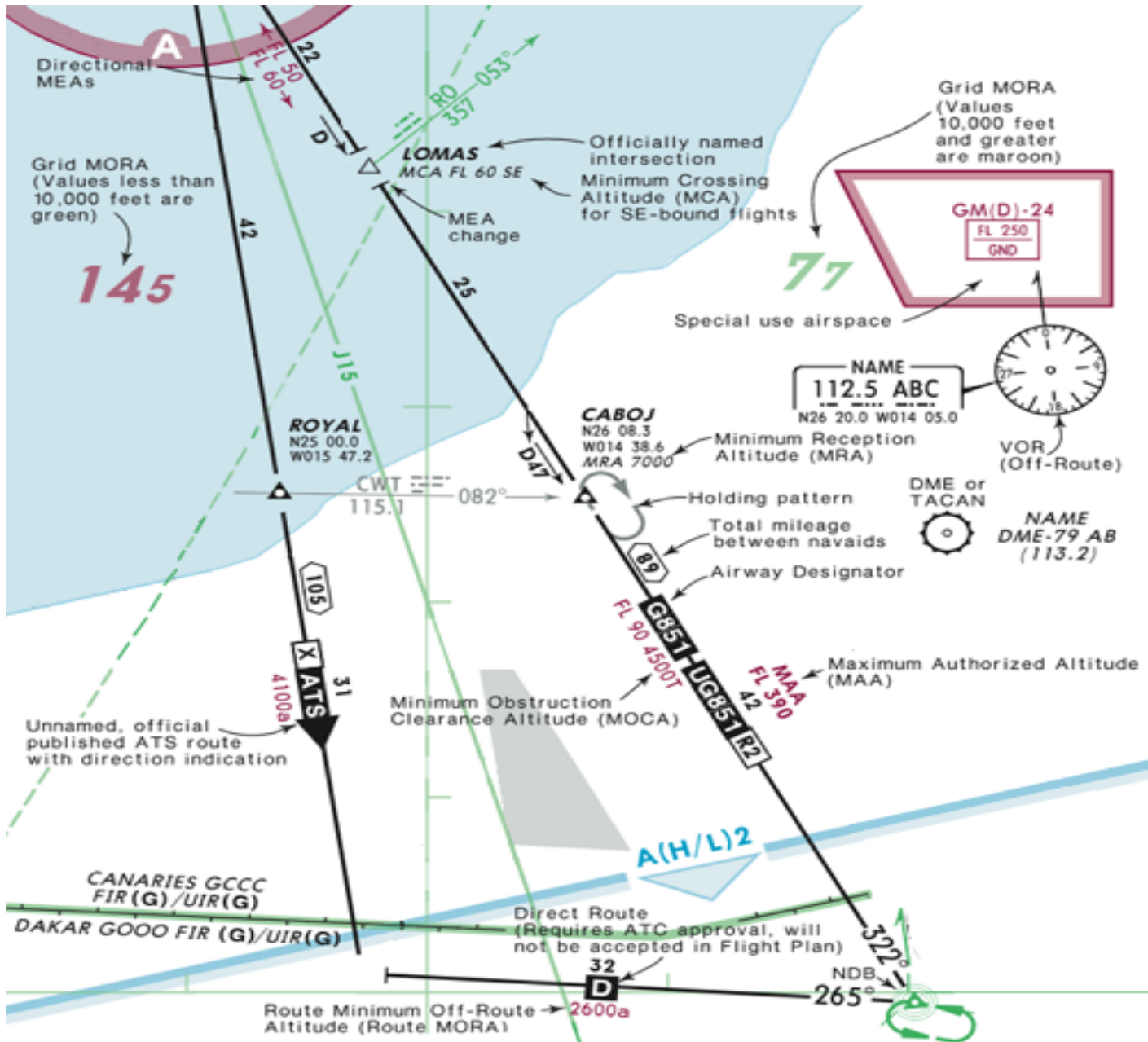
MORA (suite)

(Minimum Off-Route Altitude)

Les valeurs sont indiquées en milliers et centaines de pieds (en omettant les deux derniers chiffres afin d'éviter une surcharge de la carte).

Ces valeurs sont de couleur verte si la valeur est inférieure à 10 000 ft et marron si la valeur est supérieure à 10 000 ft.

MORA (suite)



Carte extraite de la documentation Jeppesen

MORA (suite)

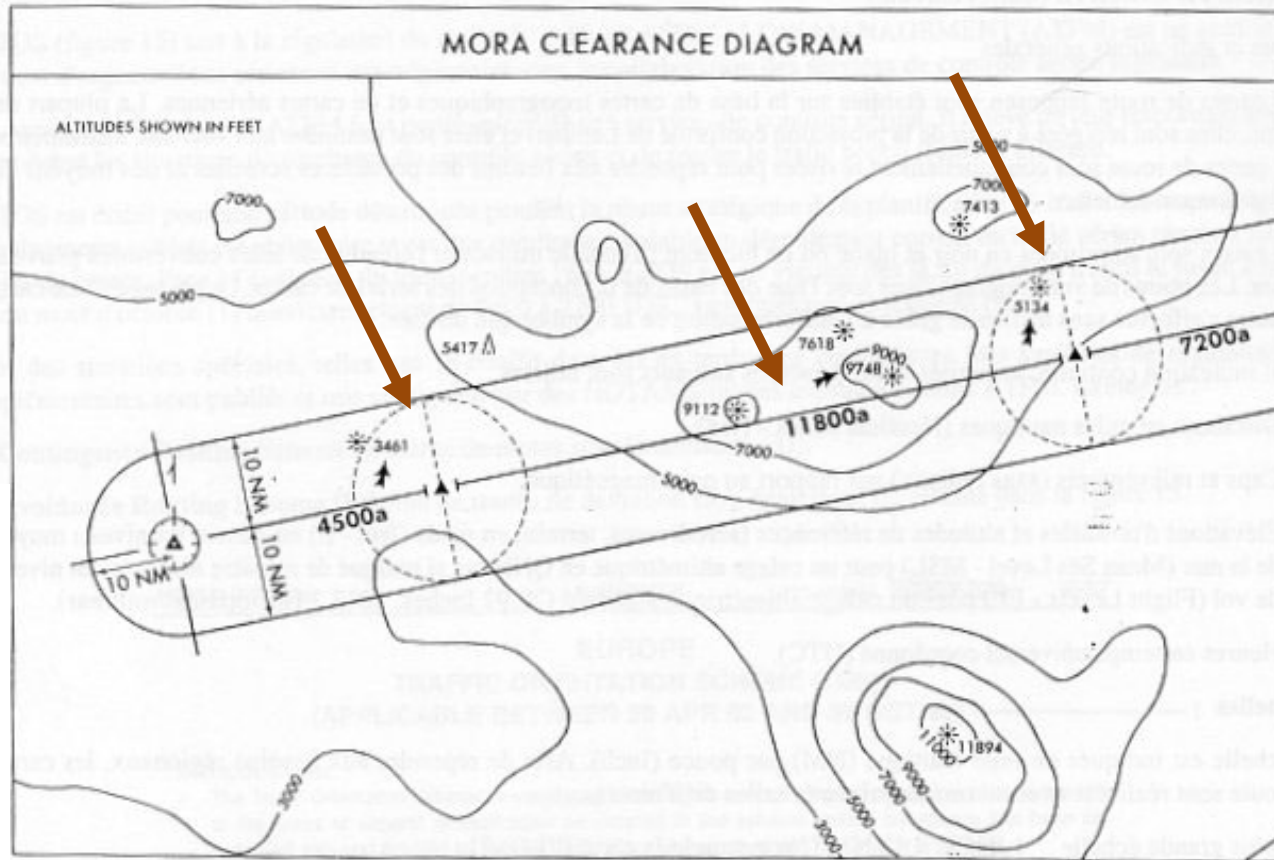


Fig. 16 : Ce dessin indique la procédure utilisée pour déterminer la MORA.

exemple :	Obstacle critique	3461 ft/MSL
	Arrondi au cent supérieur	3500 ft/MSL
	Marge de franchissement des obstacles	1000 ft
	Route MORA	4500 ft/MSL

GRID MORA

(Grid Minimum Off-Route

Altitude)

GRID MORA gives you terrain separation within latitude and longitude lines. GRID MORA will give you 1000 ft separation from terrain when terrain is less than 5001 ft and 2000 ft separation when terrain is more than 5000 ft.

GRID MORA: provides clearance within a LAT/LONG section on a chart. These are given in hundreds of feet, for example 79 would mean a GRID MORA of 7900ft

Grid MORA
(Values less than
14,000 feet are
green)

145

42

Grid MORA
(Values
14,000 feet
and greater
are maroon)

77

MOCA

Minimum Obstruction Clearance altitude

MOCA will give you a minimum altitude above terrain and guarantee VOR reception within 22 nm from the beacon. MOCA will give you 1000 ft separation from terrain when terrain is less than 5001 ft and 2000 ft separation when terrain is more than 5000 ft.

MOCA

Minimum Obstruction Clearance altitude

- **MOCA** ou Minimum Obstruction Clearance Altitude assure le franchissement des obstacles sur une route **ATS** , tronçon de route **ATS** , ou autre route directe , et pour Les Etats Unis la réception adéquate des signaux de navigation **VOR** dans un rayon de **22 NM** de la station **VOR** utilisé pour définir l'itinéraire.
- La **MOCA** est la somme de :
 - a - l'altitude maximale des obstacles ou du relief, la plus élevée des deux,

MOCA

Minimum Obstruction Clearance altitude

- b - plus **1000 ft** pour une altitude jusqu'à **6000 ft** inclus,
 - c - ou **2000 ft** pour une altitude excédant **6000ft** arrondie aux **100 ft** suivants.
- La plus faible **MOCA** devant être indiquée s'élève à **2000 ft**.

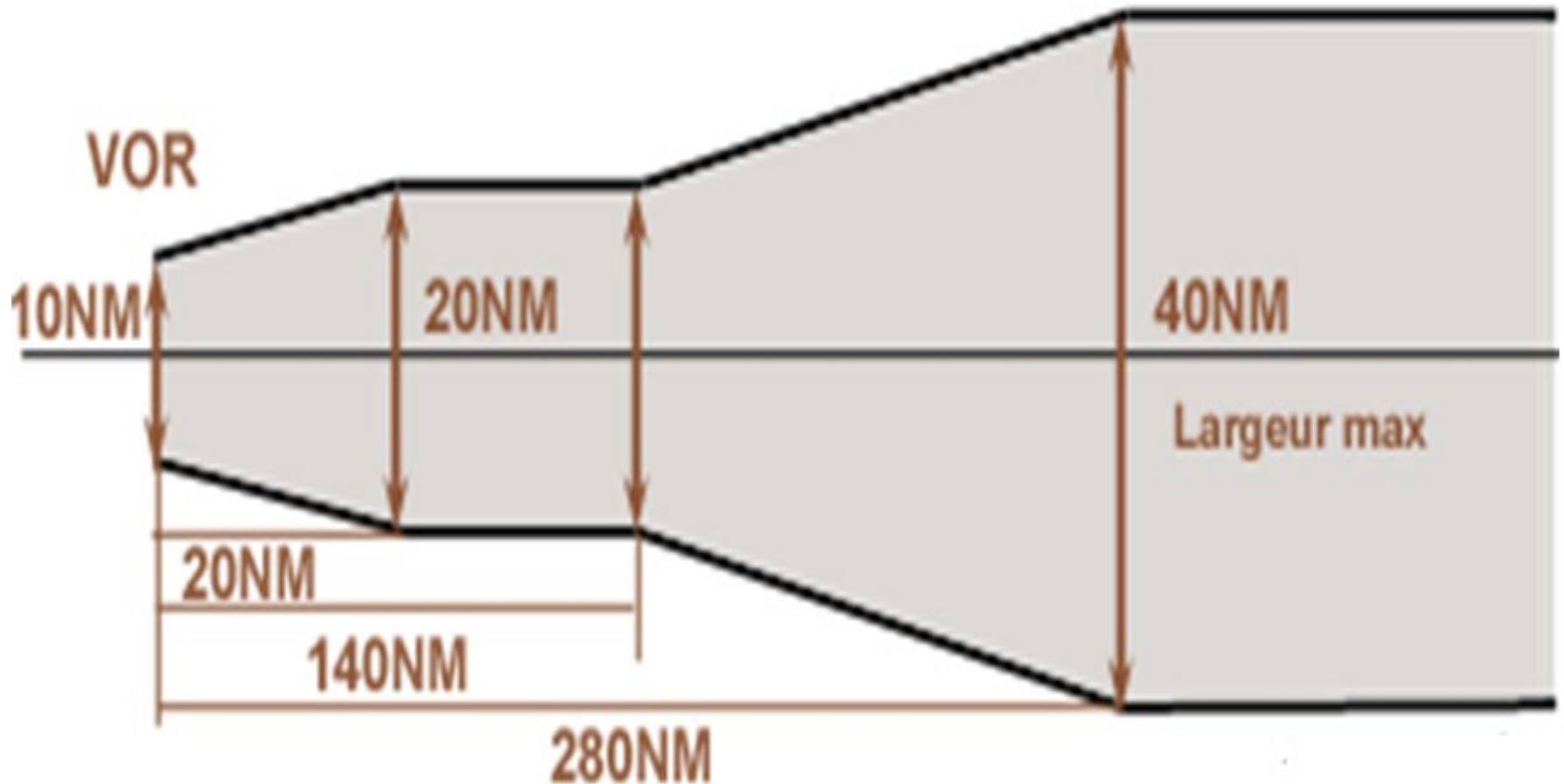
MOCA

Minimum Obstruction Clearance altitude

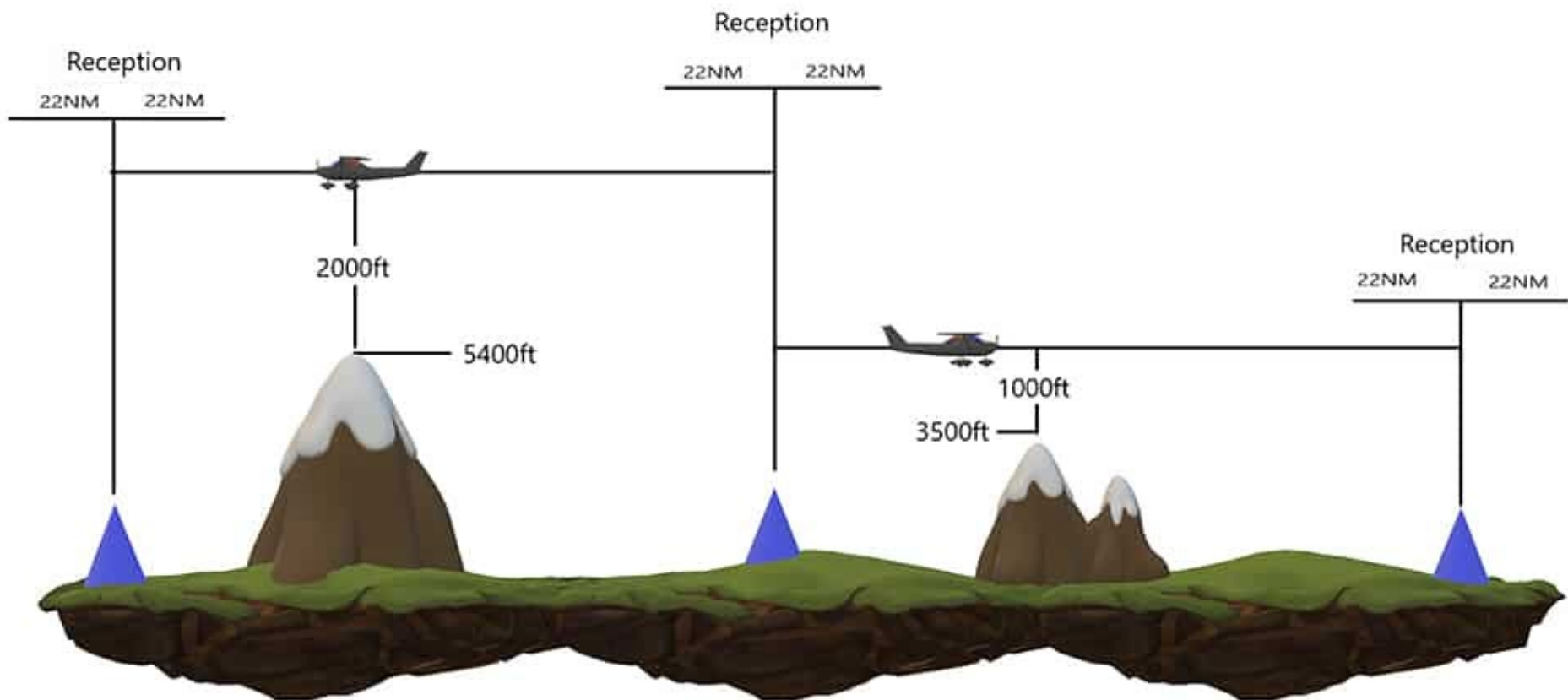
La largeur du couloir partant d'une station **VOR** est définie par une bordure qui commence à **5 NM** de part et d'autre du **VOR**, puis diverge de **4°** par rapport à l'axe pour atteindre une largeur de **20 NM** à **70 NM** de distance, puis devient parallèle jusqu'à une distance de **140 NM**, puis diverge à nouveau de **4°** pour atteindre la largeur maximale de **40 NM**, à **280 NM** du **VOR**. À partir de ce point, la largeur reste constante

MOCA

Minimum Obstruction Clearance altitude

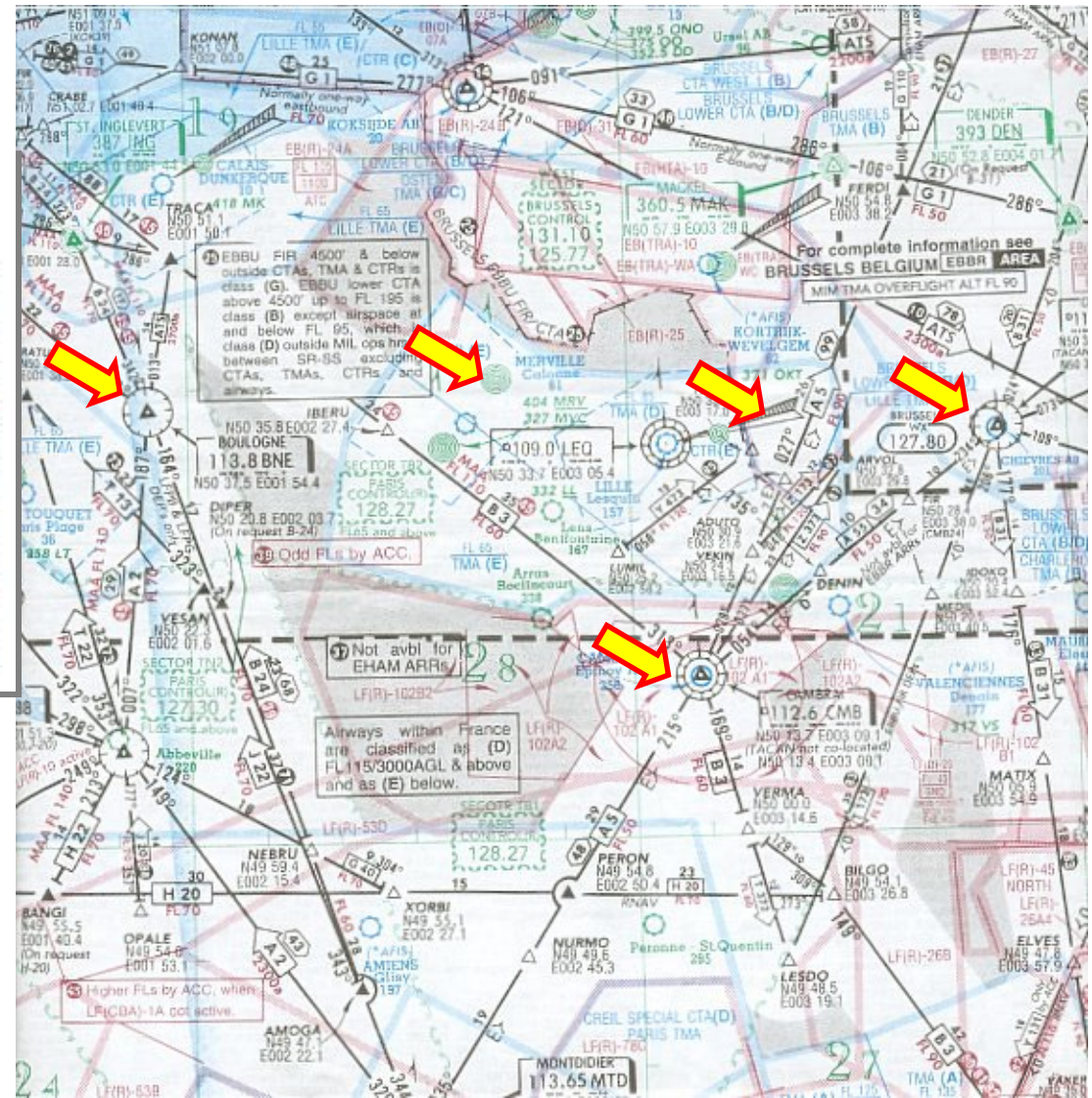
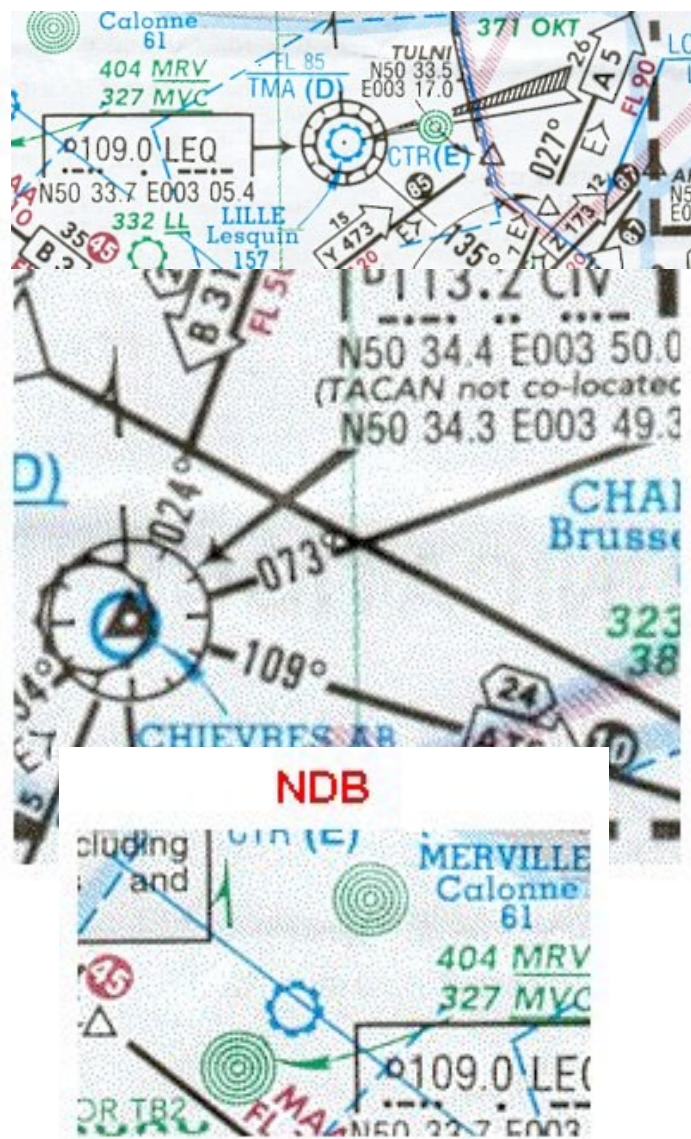


MOCA

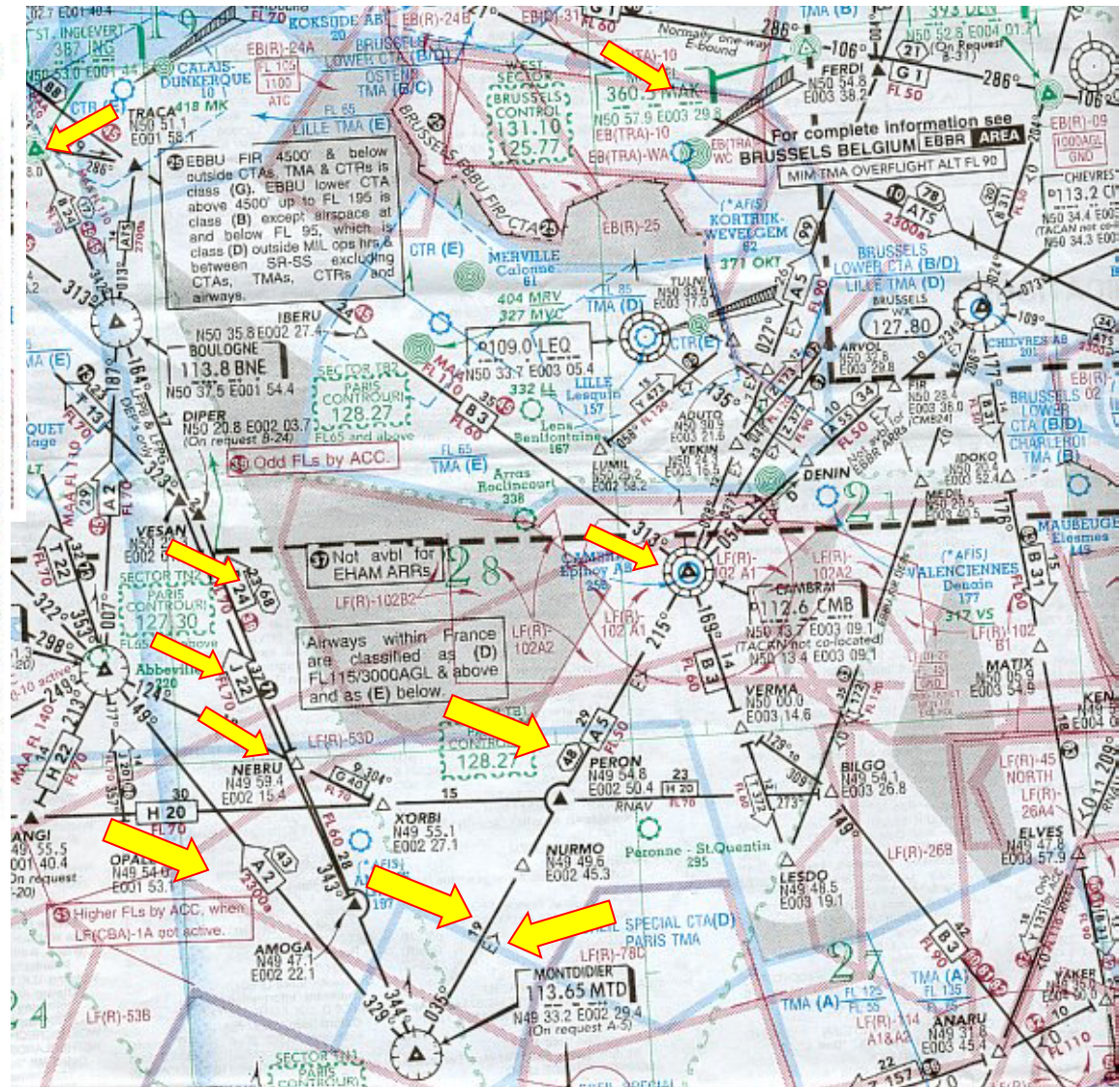


- [illegible]

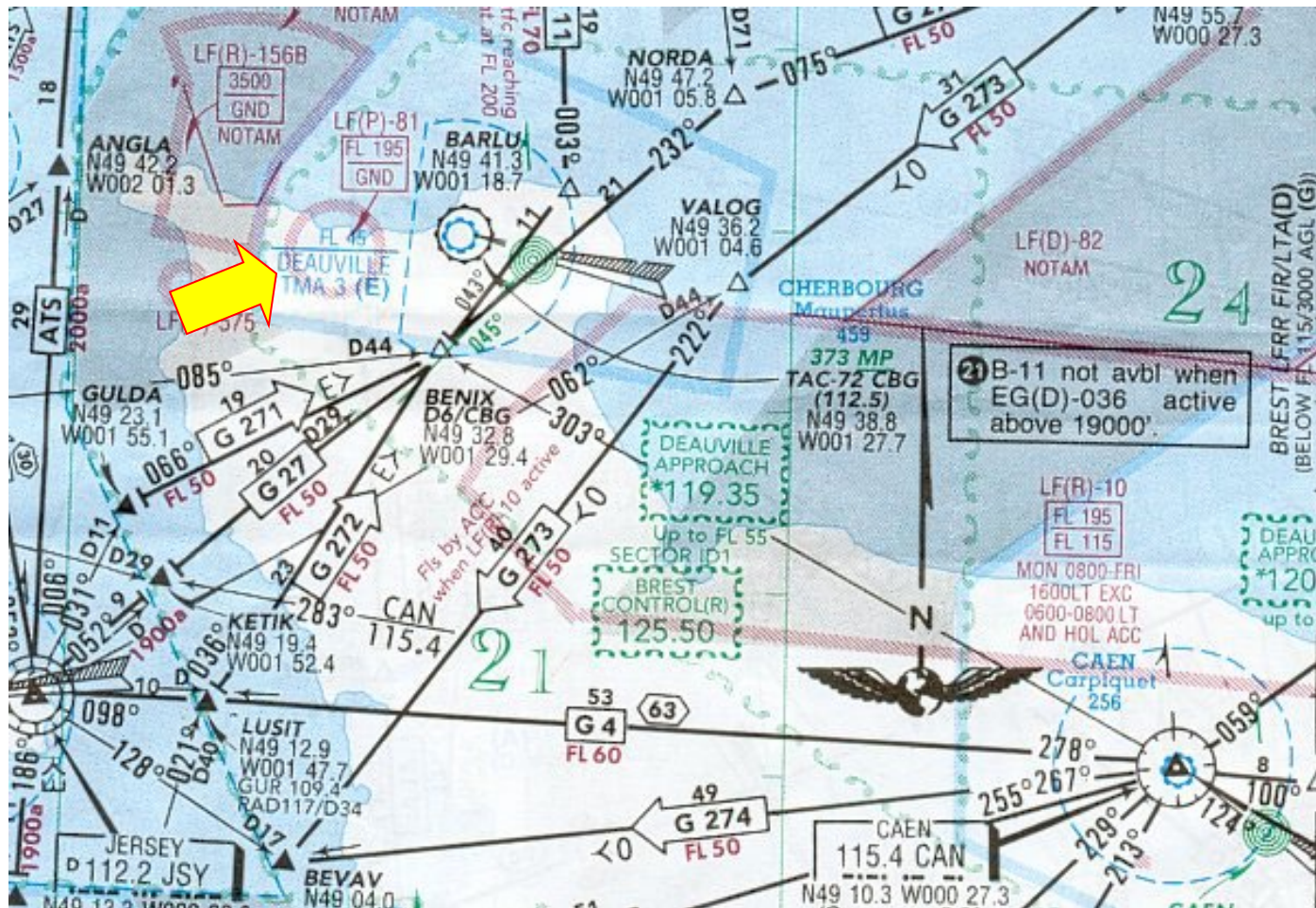
Moyens radioélectriques



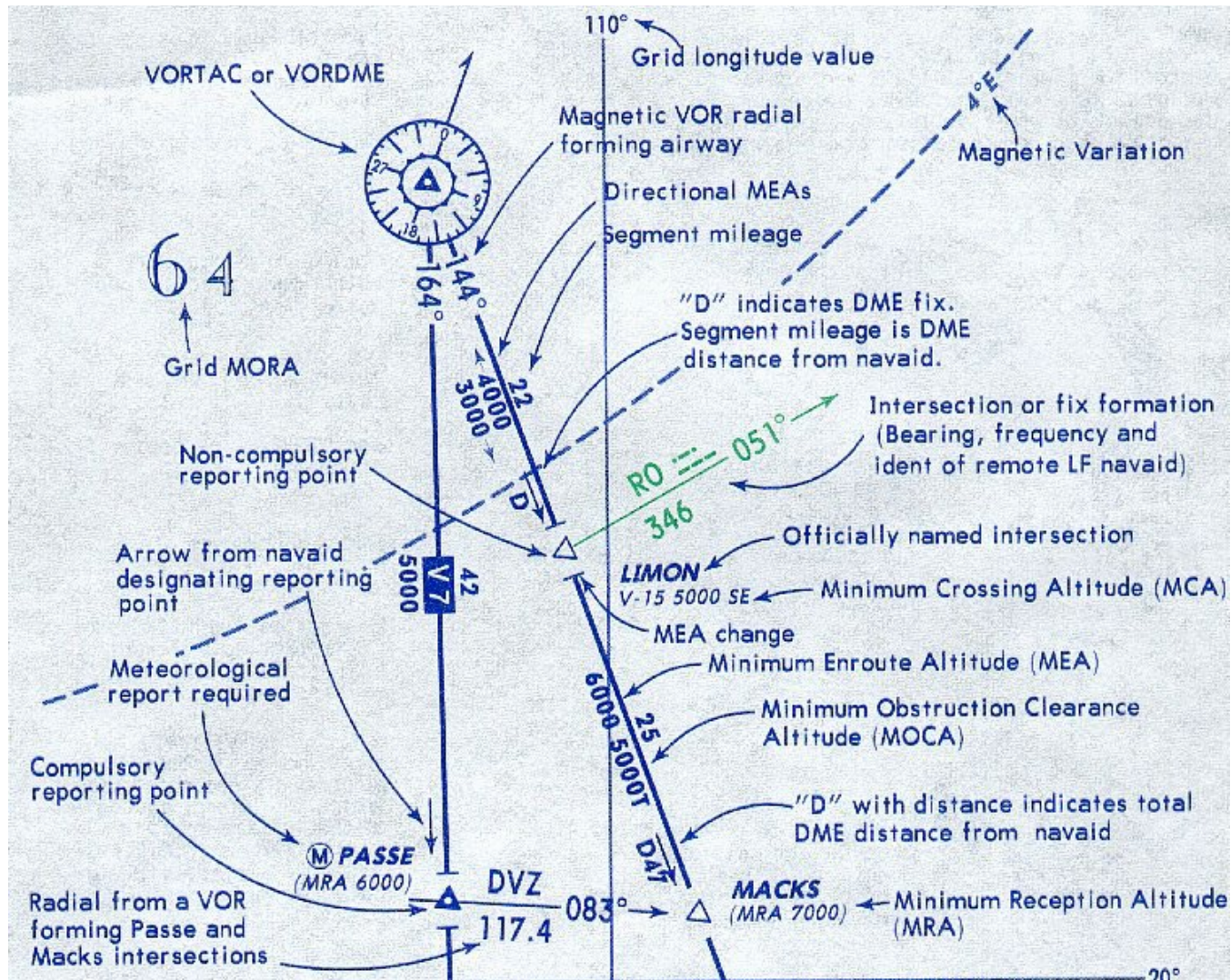
ve



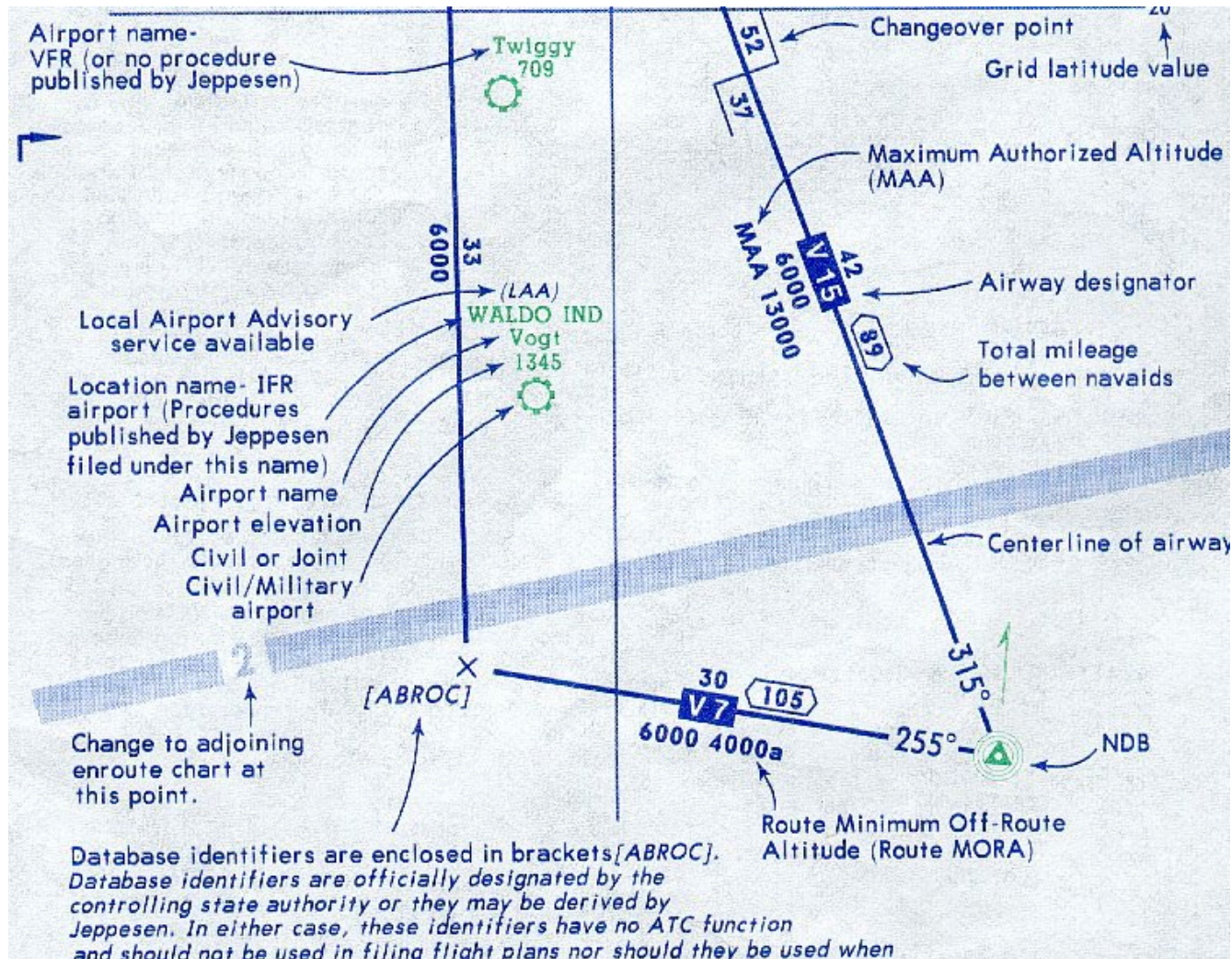
Zone interdite



ENROUTE CHART LEGEND



ENROUTE CHART LEGEND



Aérodromes

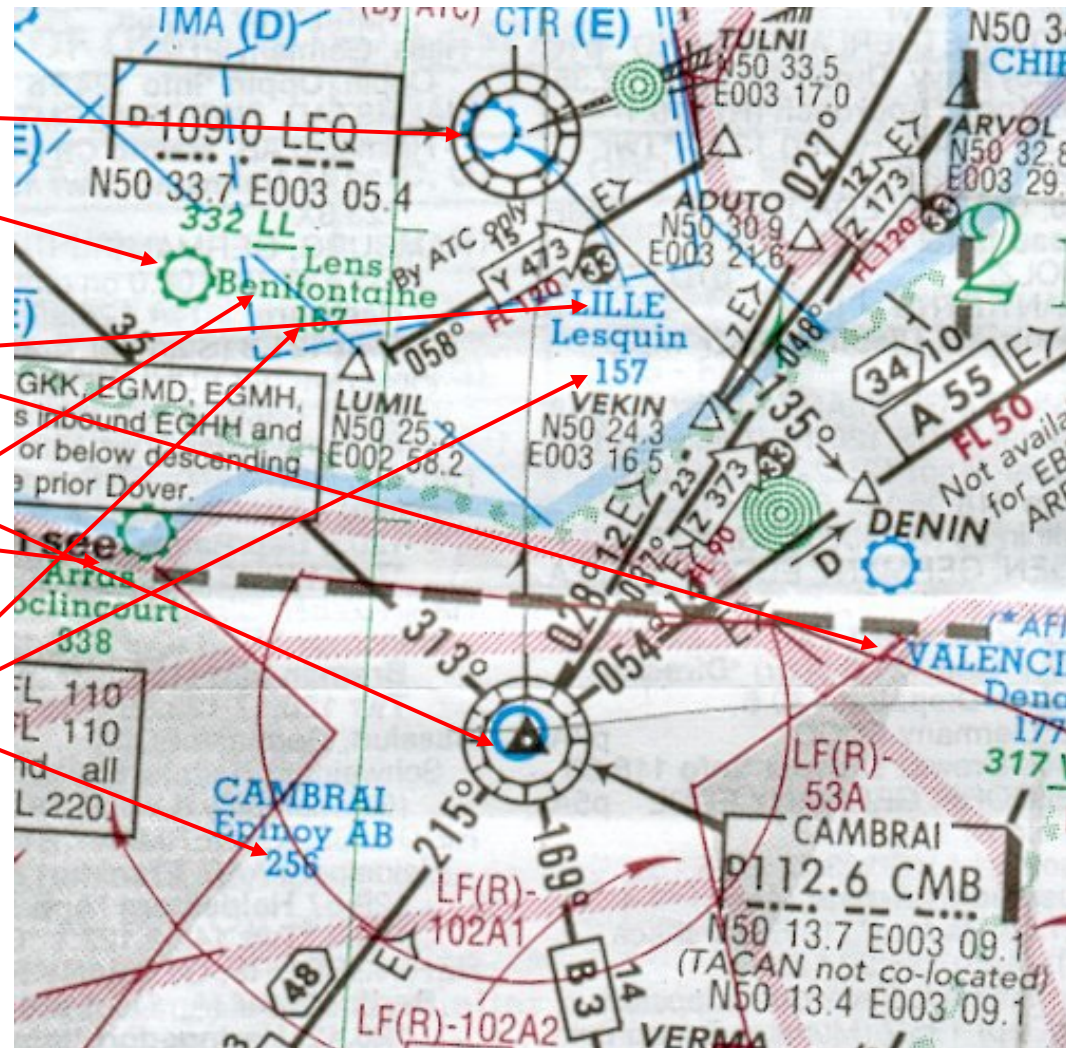
•AERODROME CIVIL

•Nom de ville de l'aérodrome en majuscules pour lequel les cartes Jeppesen sont publiées

•Nom de l'aérodrome en minuscules : pas de cartes d'approches Jeppesen disponibles

•Altitude du point de référence de l'aéroport

•Aérodrome militaire



Espaces aériens réglementés

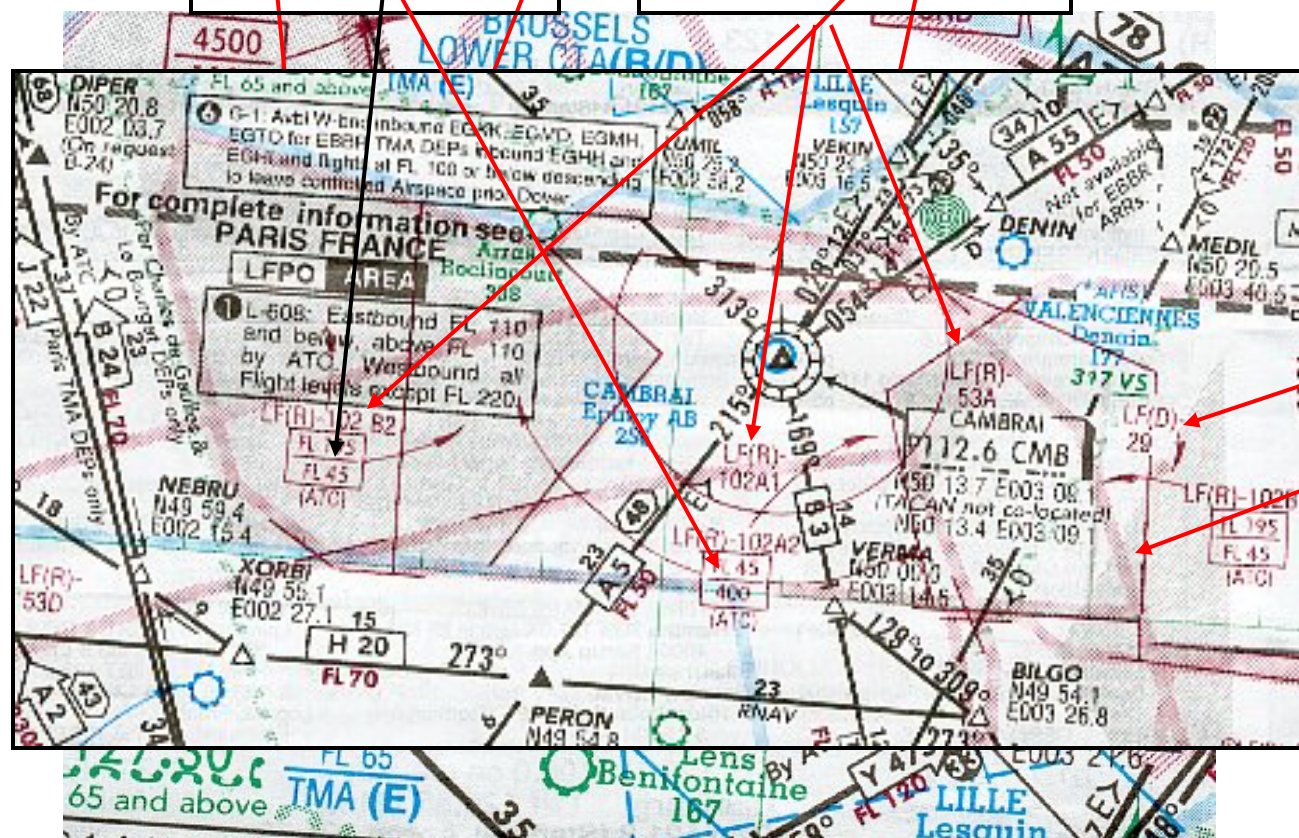
• Limite CTR

• Limite TMA

• Zone belge, limites

• Limites verticales

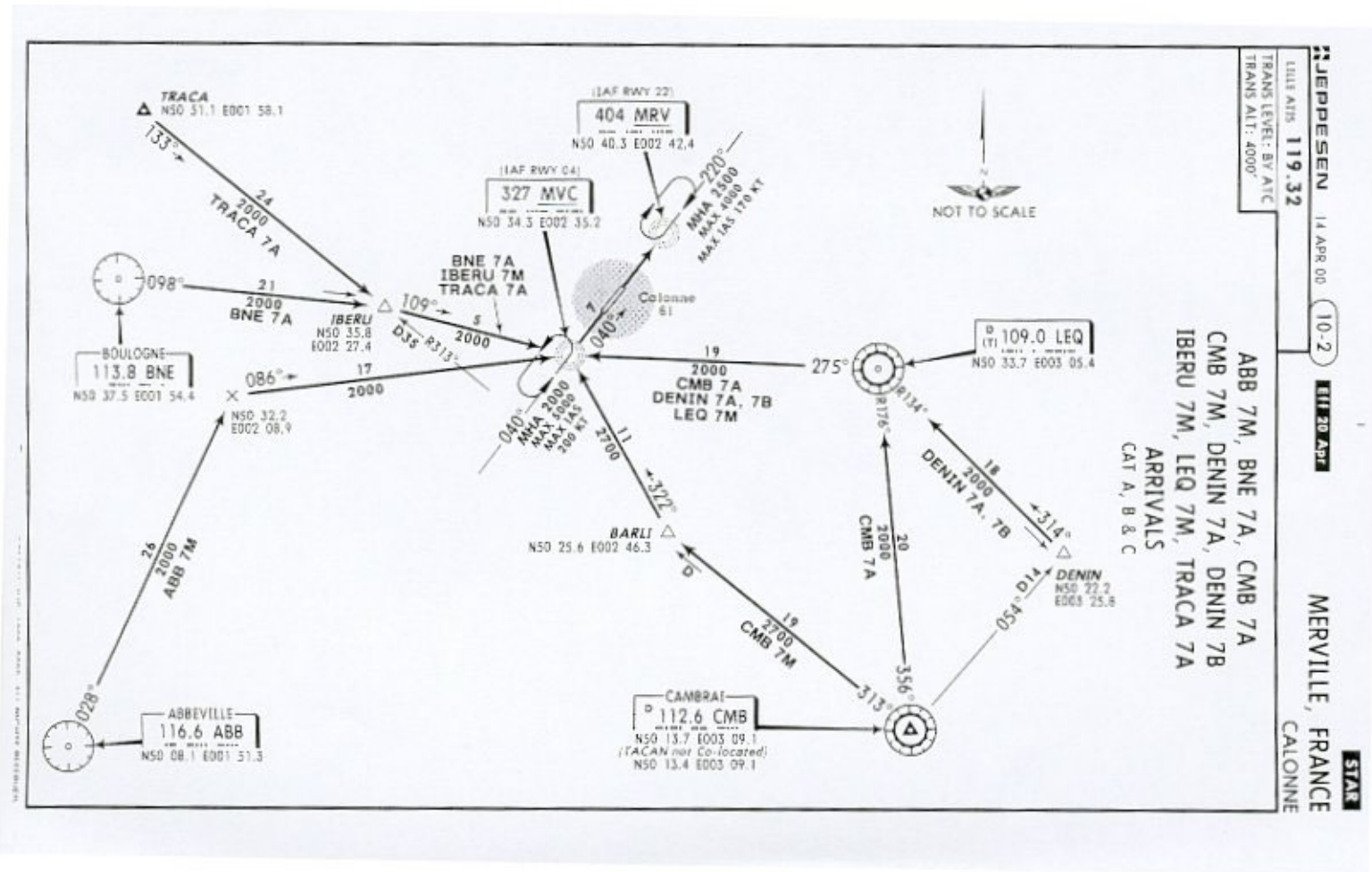
• Indicatif de la zone



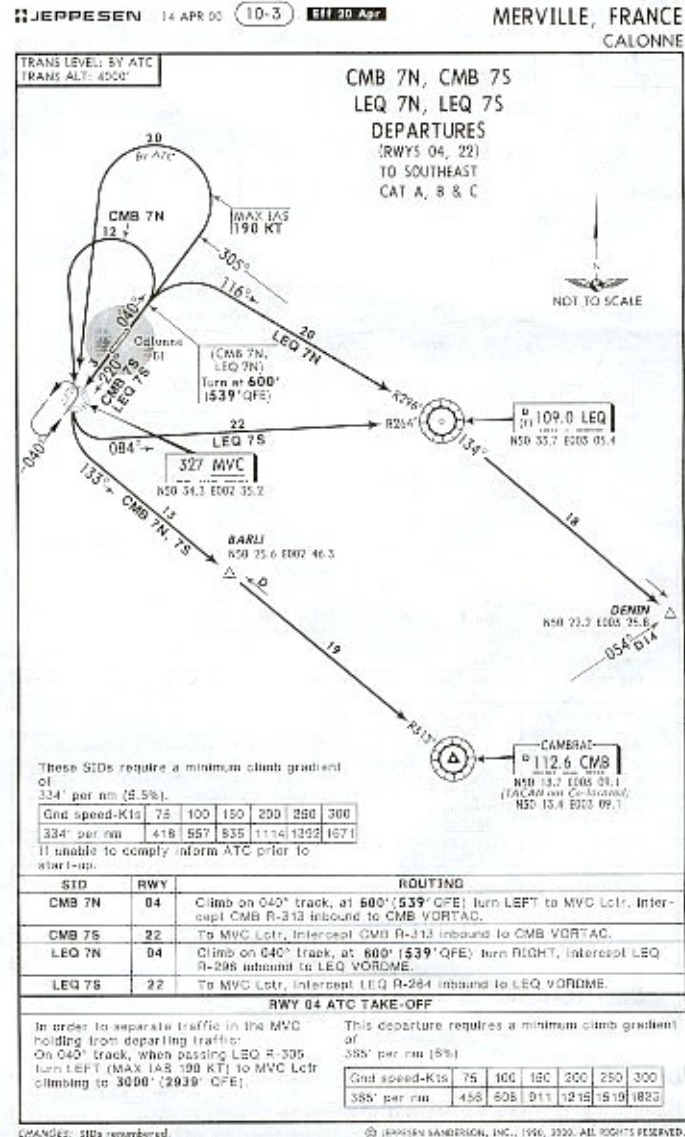
Area



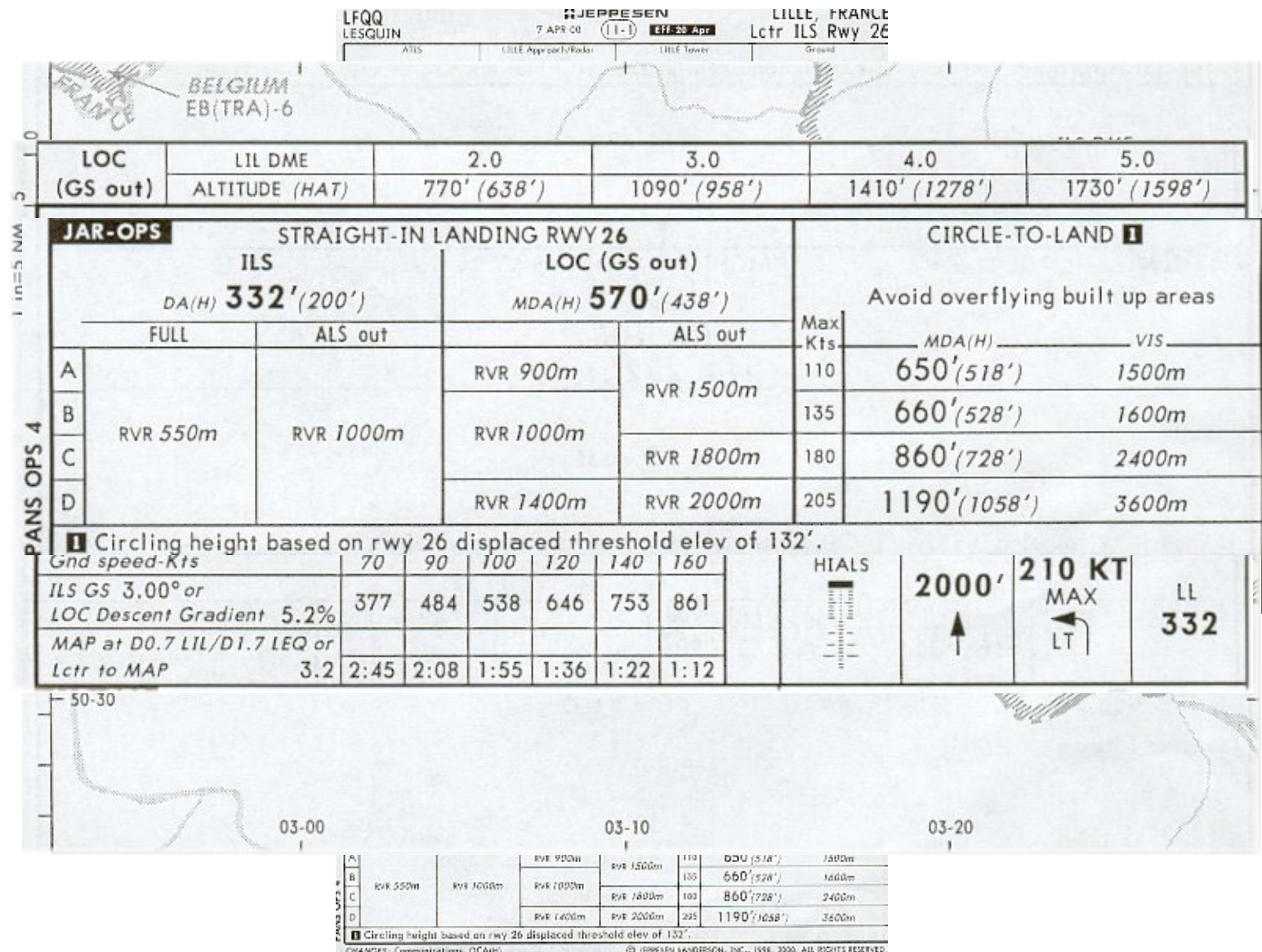
STAR



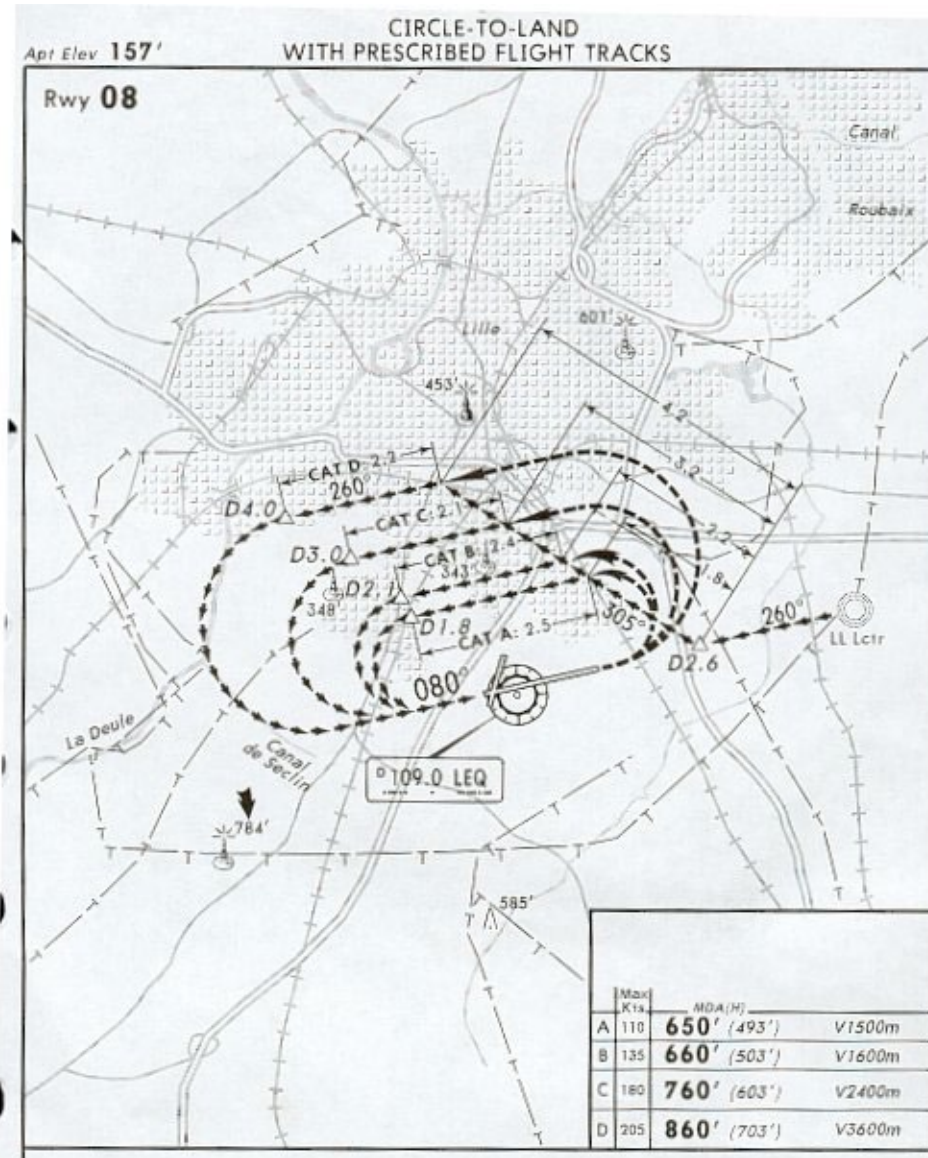
SID



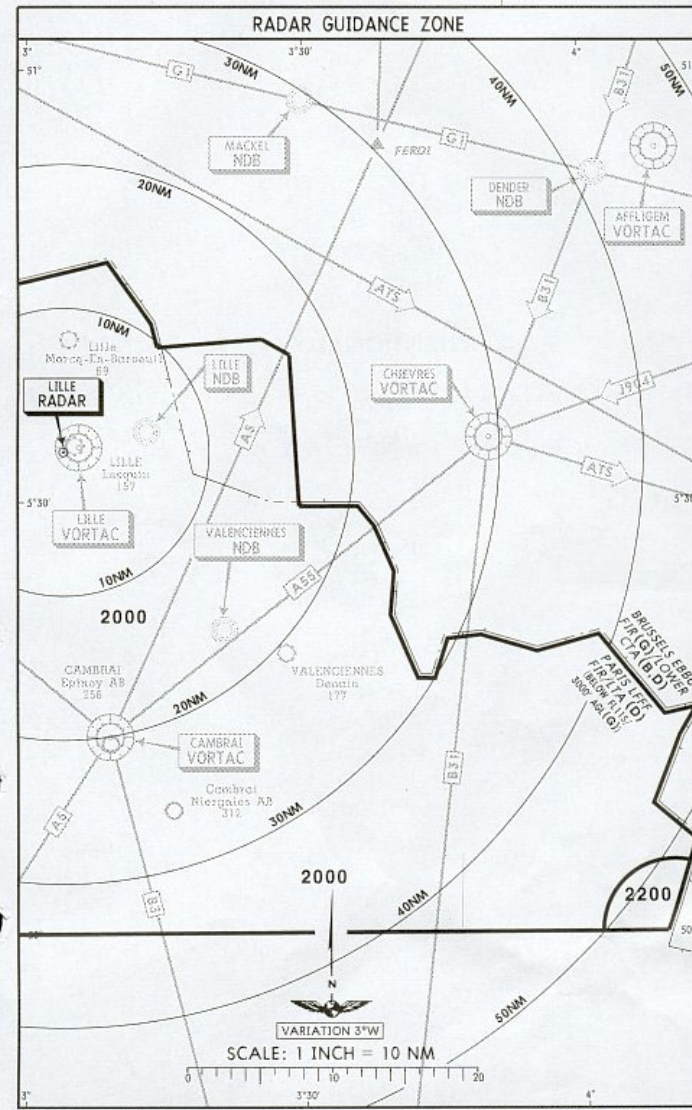
Procédure d'approche aux instruments



Approche indirecte



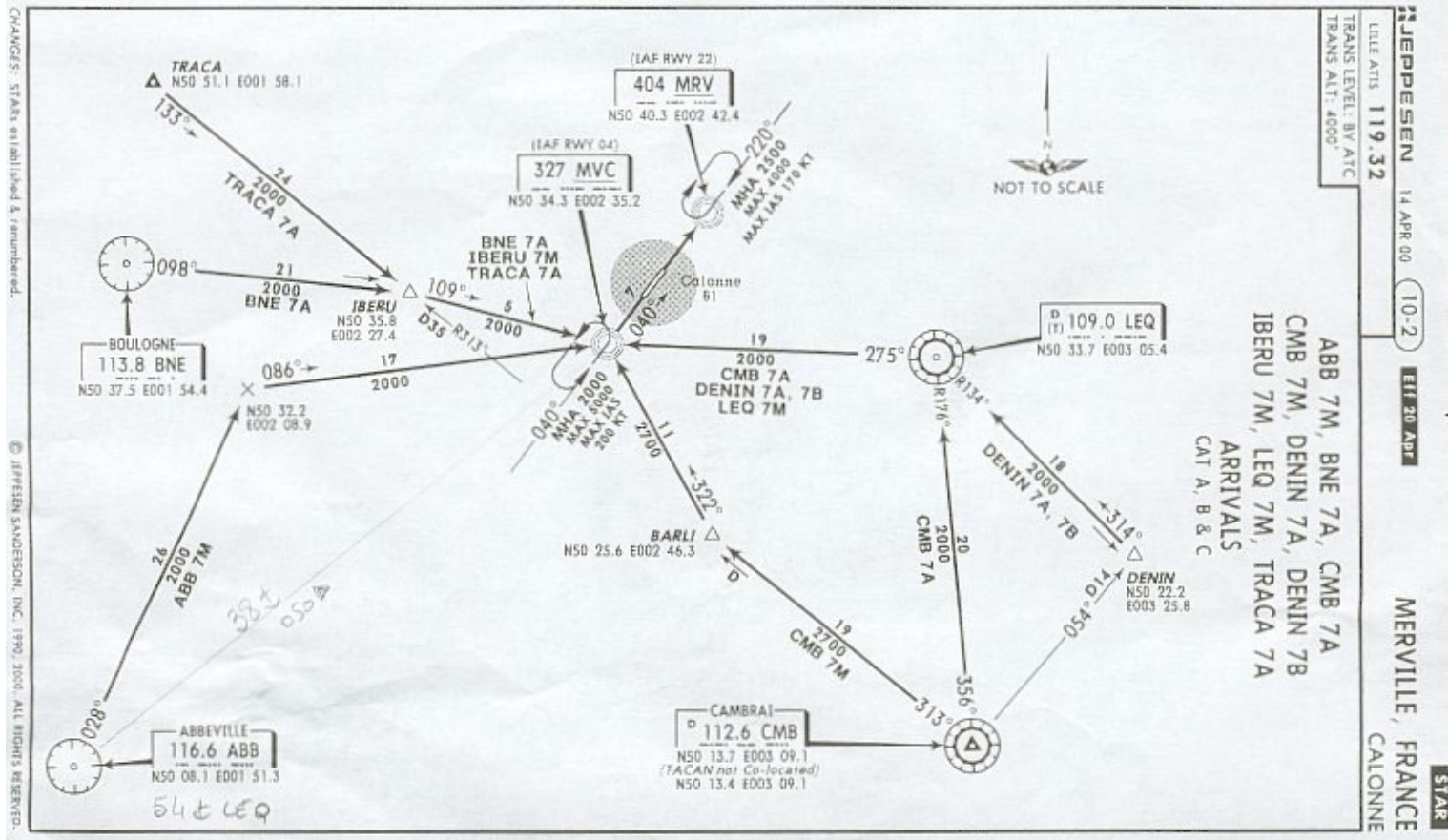
Radar landing minimum



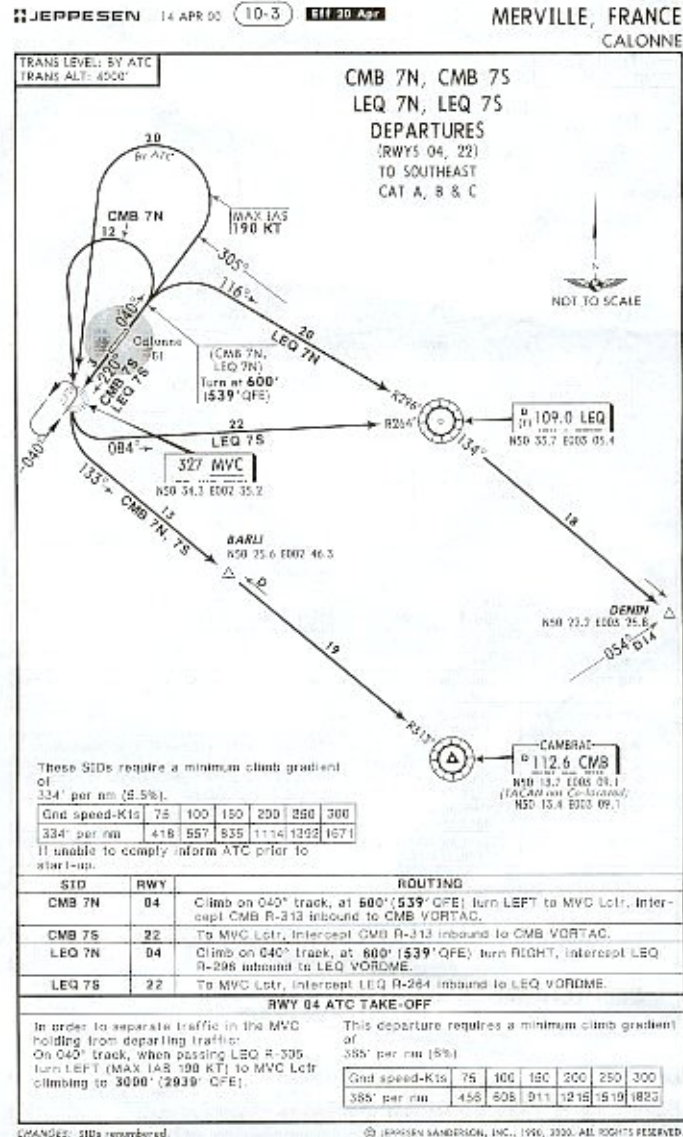
Chapitre 3

- Fiches d'aérodrome Merville
 - STAR
 - SID CMB LEQ
 - SID BNE TRACA
 - Aérodrome
 - Lctr RWY 04
 - Lctr RWY 22

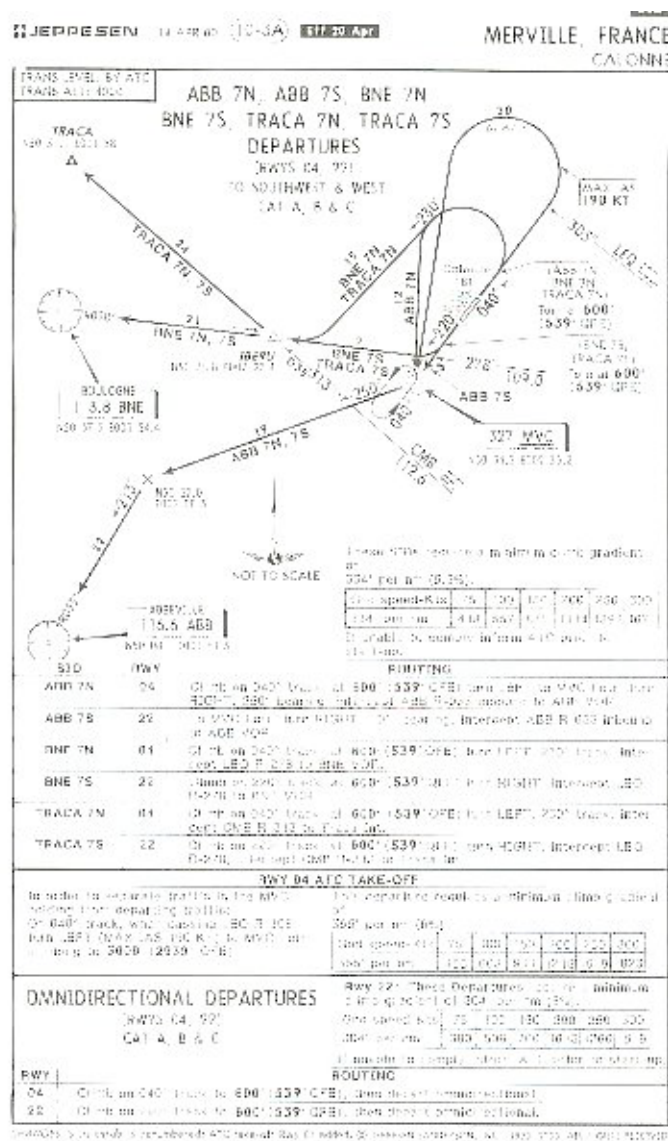
STAR



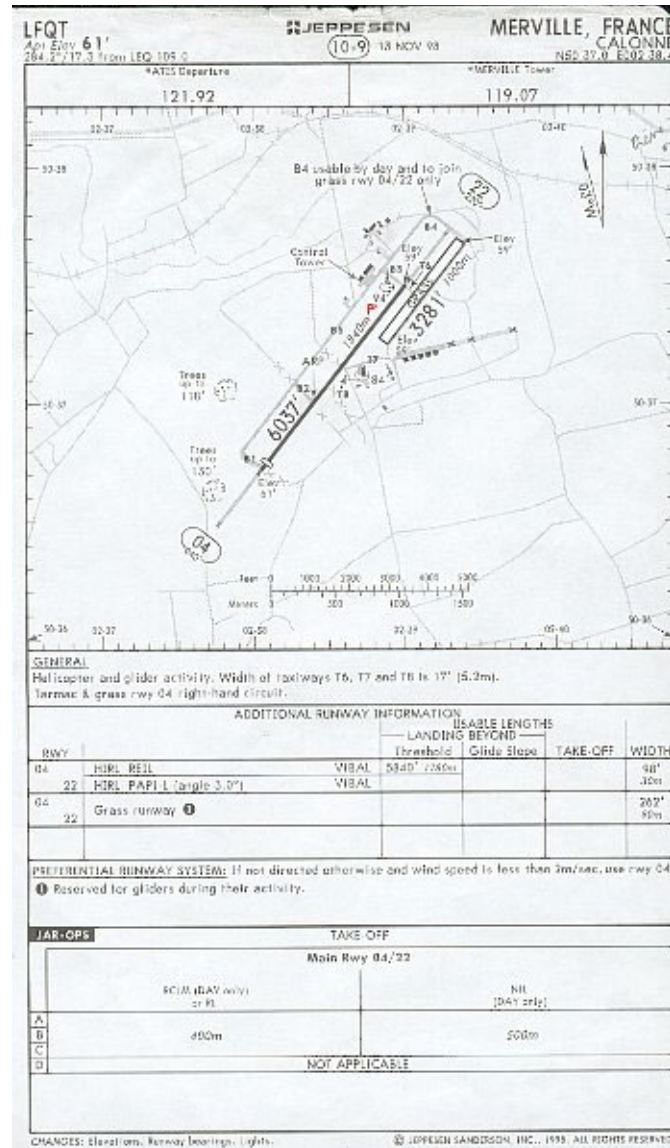
SID CMB LEQ



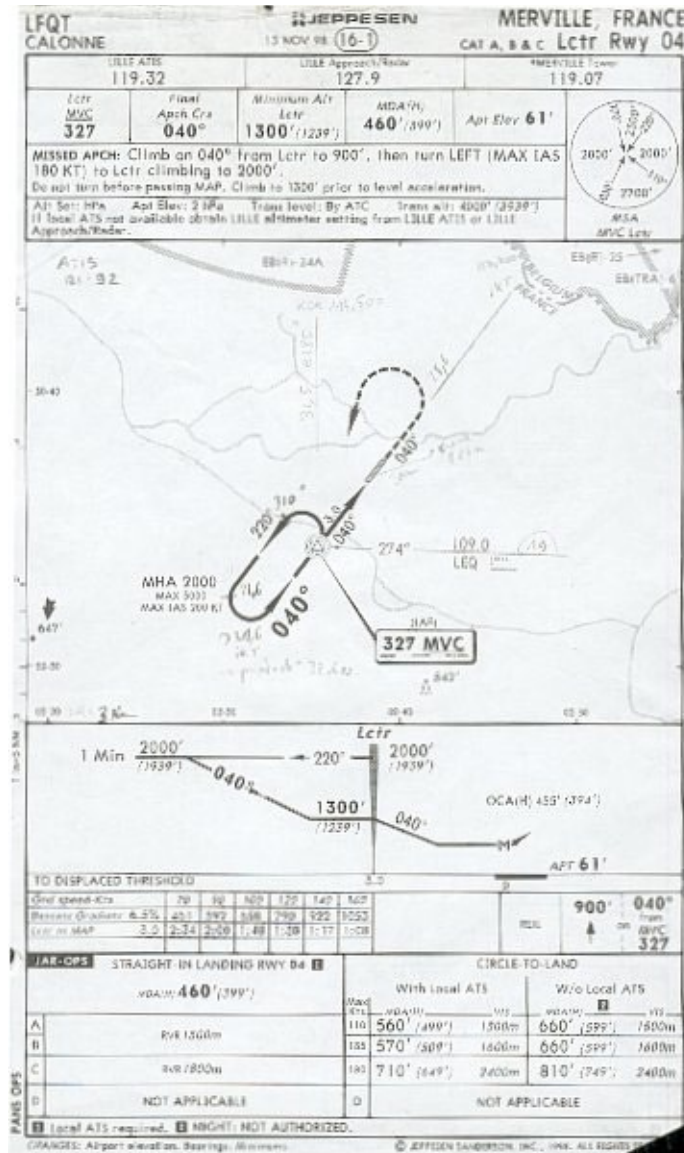
SID BNE TRACA



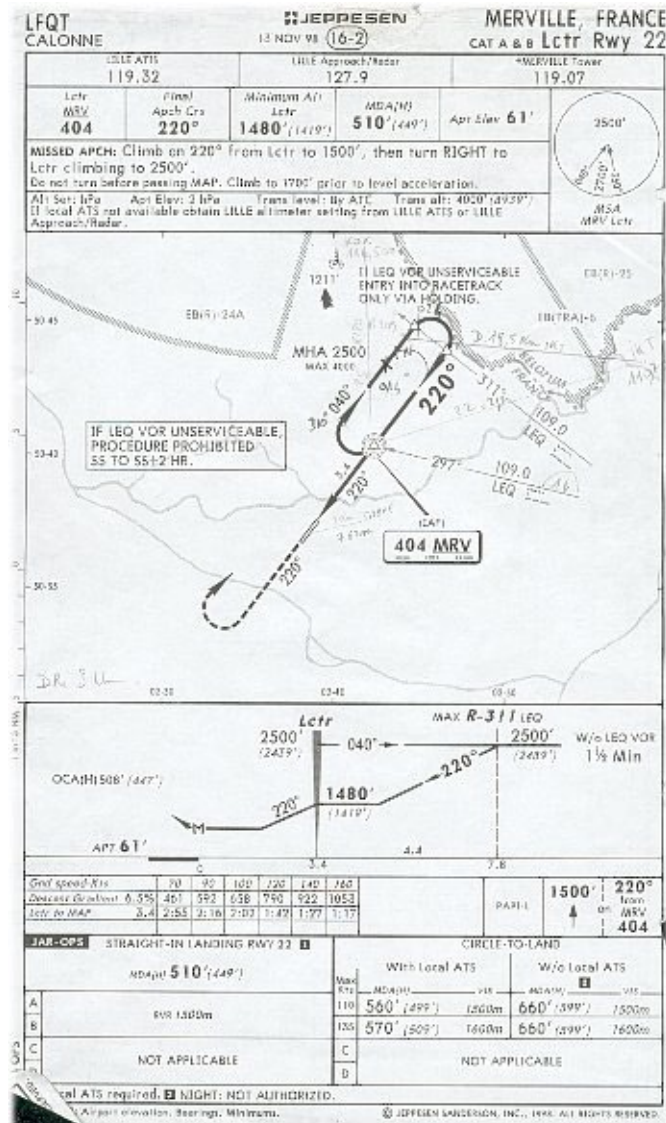
Aérodrome



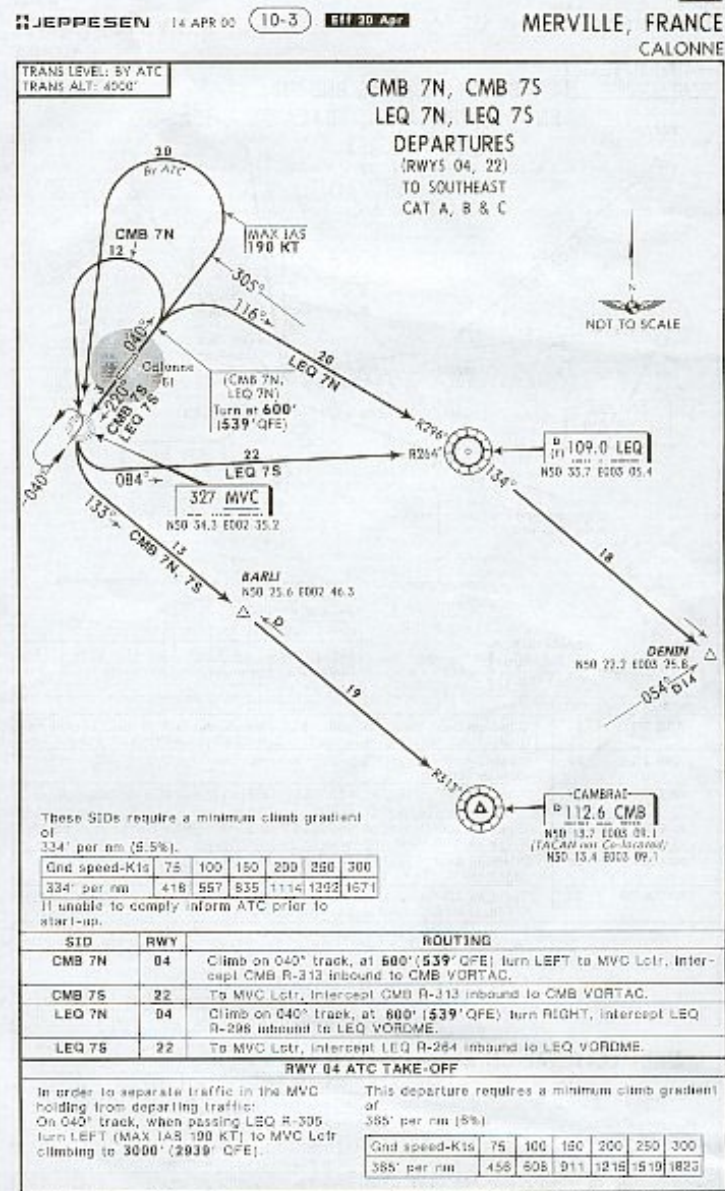
Procédure Lctr 04



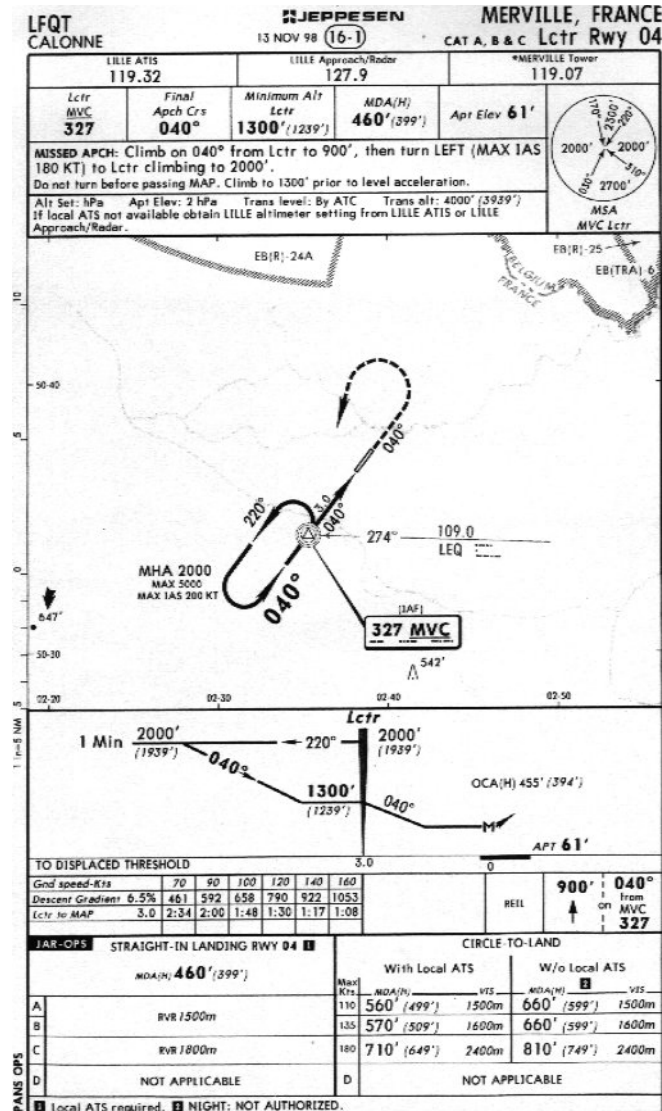
Procédure Lctr 22



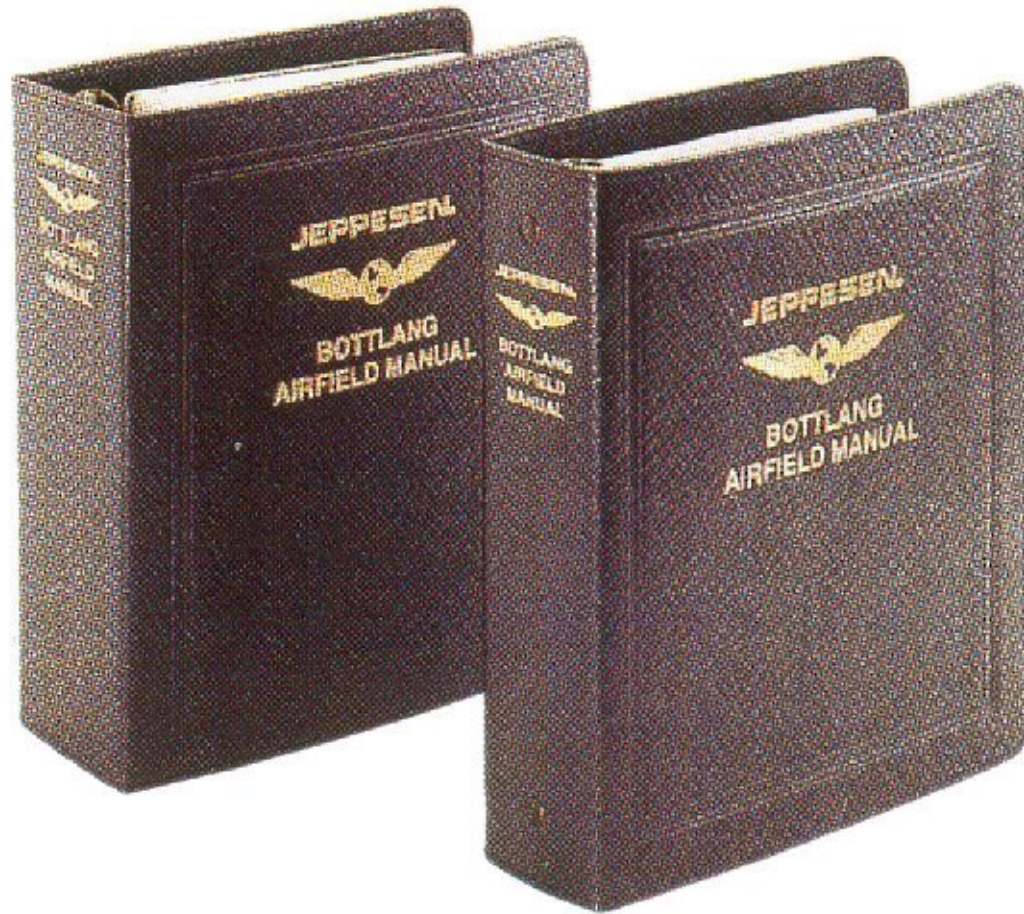
Merville SID CMB



Merville Lctr Rwy 04



Jeppesen manuals



Presentation

- 2 manuals

- First manual:

- you can find EN ROUTE charts, and different information about Meteorology, the airport, the radio aids, etc...

- Second manual

- You can find TERMINAL CHARTS (departure, arrival, different procedures...)

Tome 1

- Introduction
- Chart notams
 - En Route
 - Radio Aids
 - Meteorology
- Tables and Codes
- Air Traffic Control
- Entry Requirements
 - Emergency
 - Airport directory

Annual Update

JEPPESEN GMBH

FRANCE COV. FRA04

This a complete listing of all effective sheets appearing in your Jeppesen Airway Manual.

Record Sheets required on the Checklist Handling Form(s). Return only the Checklist Handling Form(s) with recorded pages. Do not return the checklist.

Please note tabs are not included as a checklist item.

Whenever you have a page with a more recent date than the date listed on the checklist, retain that chart and do not reorder. Manual sheets that are not shown on this checklist but that are in your manual should be removed and destroyed. If you need less than 40 sheets per binder, they may be obtained at no charge. *

* If this completed checklist indicates that you require more than 40 sheets per binder, new contents priced at EURO 76.50 (Postage extra) per binder will be required.

THIS CHECKLIST INCLUDES REVISION NO. F 26-02. PLEASE BE SURE TO COMPLETE THE CHECKLIST *B E F O R E* INSERTING THE ENCLOSED REVISION NO. F 27-02.

NOTE: THIS CHECKLIST CAN NOT BE FILLED IF RECEIVED AFTER 02 AUG 02.

	TITLE PAGE (FRA VOLUME 1)	22 FEB 85	()
	TITLE PAGE (FRA VOLUME 2)	22 FEB 85	()
	RECORD OF REVISIONS (F)	04 JAN 02	()
	BULLETIN FRA 98-A/98-A2	30 JAN 98	()
	BULLETIN FRA 98-A3/98-A4	30 JAN 98	()
	BULLETIN FRA 99-J/99-J2	12 NOV 99	()
	BULLETIN FRA 01-A/01-A2	27 APR 01	()
	BULLETIN FRA 01-A3/01-A4	27 APR 01	()
	INTRODUCTION		
	INTRODUCTION I/II	03 AUG 01	()
L	INTRODUCTION JAR-OPS 1/2 (JAA)	10 SEP 99	()
	INTRODUCTION 1/2	26 MAR 99	()
	INTRODUCTION 3/4	27 JUN 97	()
	INTRODUCTION 5/6	04 FEB 00	()
	INTRODUCTION 7/8	04 FEB 00	()
	INTRODUCTION 9/10	27 JUN 97	()
	INTRODUCTION 11/12	27 JUN 97	()
	INTRODUCTION 13/14	14 MAY 99	()
	INTRODUCTION 41/42	06 JUL 01	()
	INTRODUCTION 43/44	06 JUL 01	()
	INTRODUCTION 45/46	03 AUG 01	()

Weekly Update



Airway Manual Services Record of Revisions

FRANKFURT PRODUCED REVISIONS

CUSTOMER NUMBER: _____

ASSIGNED TO: _____ LOCATION: _____

REVISION NUMBER	REVISION DATE	REVISION NUMBER	REVISION DATE	REVISION NUMBER	REVISION DATE
1	04-01-02	24	16-06-02	47	
2	11-01-02	25	21-06-02	48	
3	18-01-02	26		49	
4	25-01-02	27		50	
5	01-02-02	28	12-07-02	51	
6	08-02-02	29	19-07-02	52	
7	15-02-02	30	26-07-02	53	
8	22-02-02	31	02-08-02	54	
9	01-03-02	32	09-08-02	55	
10	08-03-02	33	16-08-02	56	
11	15-03-02	34	23-08-02	57	
12	22-03-02	35	30-08-02	58	
13	29-03-02	36	06-09-02	59	

Control slip

Customer Comment Slip

Thank you for purchasing Jeppesen's Flight Information Services, they are the best and most comprehensive service available. We keep it that way by listening to our customers and continuously improving our service.

We solicit your ideas and suggestions on all aspects of the service you are receiving and will commit to implement as many suggestions as feasible.

This comment slip is provided for your convenience, and may also be used for communicating with your Customer Service Representative.

Your comments:

To enable us to identify your subscription, and to contact you where required, please copy the details from your mailing label and add your phone and/or fax numbers. Capital letters would be appreciated.

Name: _____ Customer #: _____

Street: _____ City: _____

Country: _____ Telephone: _____

Fax: _____ e-mail: _____

Briefing bulletin



JEPPESSEN[®]
BRIEFING
BULLETIN

JEP 02-C

VERTICAL NAVIGATION (VNAV) - Concept Extension

Background

Beginning in December 1999, Jeppesen started to include Vertical Navigation (VNAV) descent information on selected non-precision instrument approach charts. VNAV information had previously been provided in Jeppesen's electronic navigation database (NavData) only.

VNAV path information illustrates the geometric descent path with a descent angle from the Final Approach Fix (FAF) to the Threshold Crossing Height (TCH). The inclusion of VNAV angles on non-precision approach charts was done on a limited basis.

For those non-precision approach procedures for which the State authority did not specify a descent gradient or did not provide a recommended DME/Altitude table, a descent angle **derived** from the Jeppesen NavData database is to be shown on the corresponding approach chart.

This angle, if used by certified VNAV-capable avionics equipment, will ensure a stable, constant rate of descent clearing all intervening altitude restrictions (step-down fixes) established by the State authority.

Extending the Concept

As next step of the VNAV implementation plan, Jeppesen will start providing, on a routine basis, the VNAV elements for straight-in non-precision operations when a descent gradient and/or recommended DME/Altitude information table has been specified by the controlling State authority.

In these cases, a descent angle will be **converted** from the State specified

Introduction: Table of contents

TABLE OF CONTENTS

Below is a complete list of the standard contents of Airway Manual. *Limited or special coverages may not contain all items, but that material which is included should be arranged in the order outlined.*

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Introduction: Chart glossary

CHART GLOSSARY

CHART NOTAMS — Jeppesen Chart NOTAMs include significant information changes affecting Enroute, Area, and Terminal charts. Entries are published until the temporary condition no longer exists, or until the permanent change appears on revised charts. Enroute chart numbers / panel numbers / letters and area chart identifiers are included for each entry in the enroute portion of the chart NOTAMs. To avoid duplication of information in combined Enroute and Terminal Chart NOTAMs, navaid conditions, except for ILS components, are listed only in the Enroute portion of the Chart NOTAMs. All times are local unless otherwise indicated. Arrows indicate new or revised information. Chart NOTAMs are only an *abbreviated* service. Always ask for pertinent NOTAMs prior to flight.

COMMON TRAFFIC ADVISORY FREQUENCY (CTAF) (USA) — A frequency designed for the purpose of carrying out airport advisory practices while operating to or from an uncontrolled airport. The CTAF may be a UNICOM, Multicom, FSS, or tower frequency.

COMMUNITY AERODROME RADIO STATION (CARS) — An aerodrome radio that provides weather, field conditions, accepts flight plans and position reports.

COMPULSORY REPORTING POINTS — Reporting points which must be reported to ATC. They are designated on aeronautical charts by solid triangles or filed in a flight plan as fixes selected to define direct routes. These points are geographical locations which are defined by navigation aids/fixes. Pilots should discontinue position reporting over compulsory reporting points when informed by ATC that their aircraft is in "radar contact."

CONDITIONAL ROUTES (CDR) — (Europe) — Category 1,2,3.

Category 1: Permanently plannable CDR during designated times.

Category 2: Plannable only during times designated in the Conditional Route Availability Message (CRAM) published at 1500Z for the 24 hour period starting at 0600Z the next day.

CONTROLLED AIRSPACE — An airspace of defined dimensions within which air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification.

NOTE: Controlled airspace is a generic term which covers ATS airspace Classes A, B, C, D, and E.

CONTROL ZONE (ICAO) — A controlled airspace extending upwards from the surface of the earth to a specified upper limit.

COURSE —

1. The intended direction of flight in the horizontal plane measured in degrees from north.
2. The ILS localizer signal pattern usually specified as front course or back course.
3. The intended track along a straight, curved, or segmented MLS path.

CRITICAL HEIGHT — Lowest height in relation to an aerodrome specified level below which an approach procedure cannot be continued in a safe manner solely by the aid of instruments.

DECISION ALTITUDE/HEIGHT (DA/H) (ICAO) — A specified altitude or height (A/H) in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

NOTE 1: Decision altitude (DA) is referenced to mean sea level (MSL) and decision height (DH) is referenced to the threshold elevation.

NOTE 2: The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path.

DECISION HEIGHT (DH) (USA) — With respect to

Introduction: Abbreviations

ABBREVIATIONS

A/A	Air to Air	AOE	Airport/Aerodrome of Entry
AAF	Army Air Field	AOR	Area of Responsibility
AAL	Above Aerodrome Level	APAPI	Abbreviated Precision Approach Path Indicator
AAS	Airport Advisory Service	APC	Area Positive Control
AB	Air Base	APCH	Approach
ABM	Abeam	APP	Approach Control
ABN	Aerodrome Beacon	APT	Airport
AC	Air Carrier	ARB	Air Reserve Base
ACA	Arctic Control Area	ARINC	Aeronautical Radio, Inc.
ACA	Approach Control Area	AHO	Aerodrome Reporting Officer
ACAS	Airborne Collision Avoidance System	ARR	Arrival
ACARS	Airborne Communications Addressing and Reporting System	ARTCC	Air Route Traffic Control Center
		ASDA	Accelerate Stop Distance Available
ACC	Area Control Center	ASOS	Automated Surface Observation System
ACFT	Aircraft	ASR	Airport Surveillance Radar
ACN	Aircraft Classification Number	ATA	Actual Time of Arrival
AD	Aerodrome	ATCAA	Air Traffic Control Assigned Airspace
ADA	Advisory Area	ATCC	Air Traffic Control Center
ADF	Automatic Direction Finding	ATCT	Air Traffic Control Tower
ADIZ	Air Defense Identification Zone	ATD	Actual Time of Departure
ADR	Advisory Route	ATF	Aerodrome Traffic Frequency
ADV	Advisory Area	ATFM	Air Traffic Flow Management
AFIS	Aeronautical Enroute Information Service	ATIS	Automatic Terminal Information Service
AEH	Approach End Runway	ATS	Air Traffic Service
AERADIO	Air Radio	ATZ	Aerodrome Traffic Zone
AERO	Aerodrome	AUTH	Authorized
AF Aux	Air Force Auxiliary Field	AUW	All-up Weight
AFB	Air Force Base	AUX	Auxiliary
AFIS	Aerodrome Flight Information Service	AVBL	Available
AFN	American Forces Network	AWIB	Aerodrome Weather Information Broadcast
AFRS	Armed Forces Radio Stations	AWOS	Automated Weather Observing
AFS	Air Force Station		

Introduction: En route chart legend

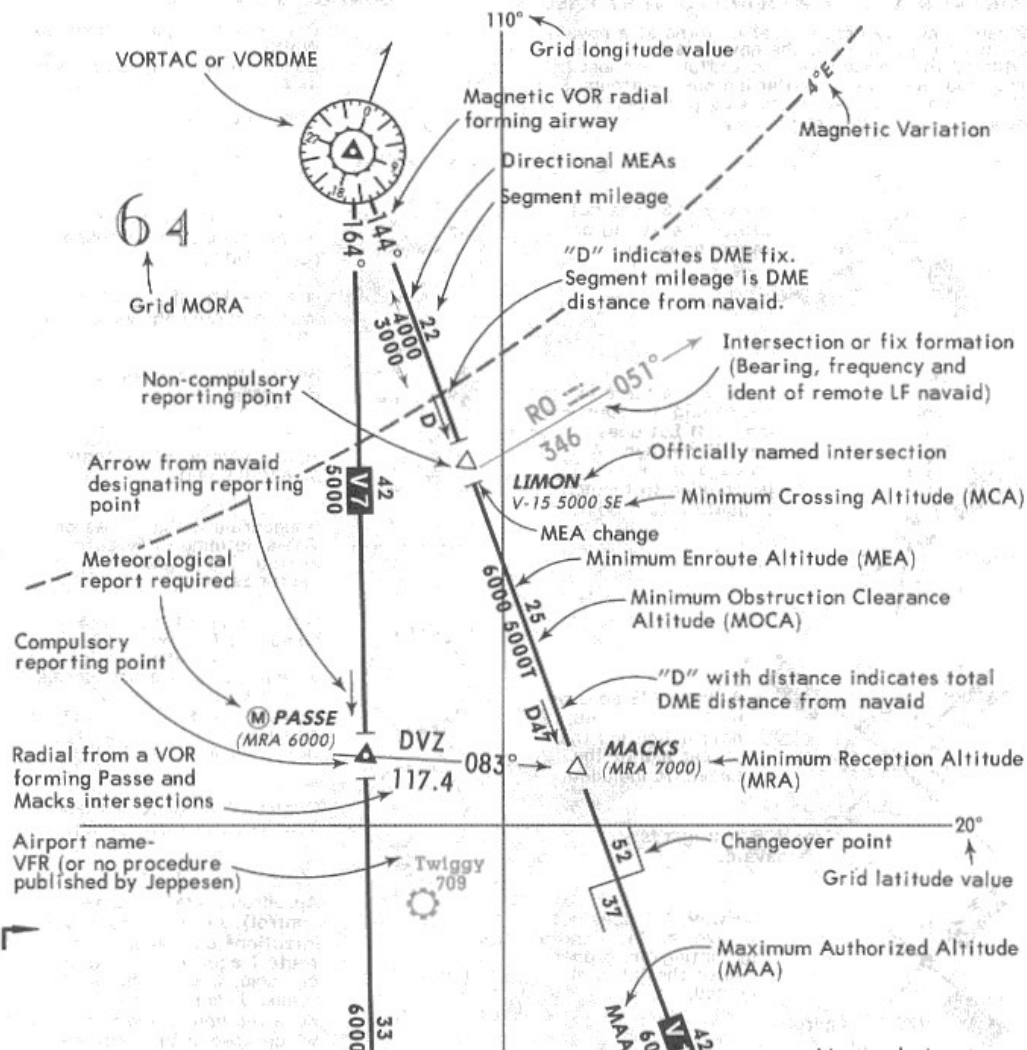
JEPPESEN

INTRODUCTION

3 APR 98

57

ENROUTE CHART LEGEND



Introduction: En route legend details

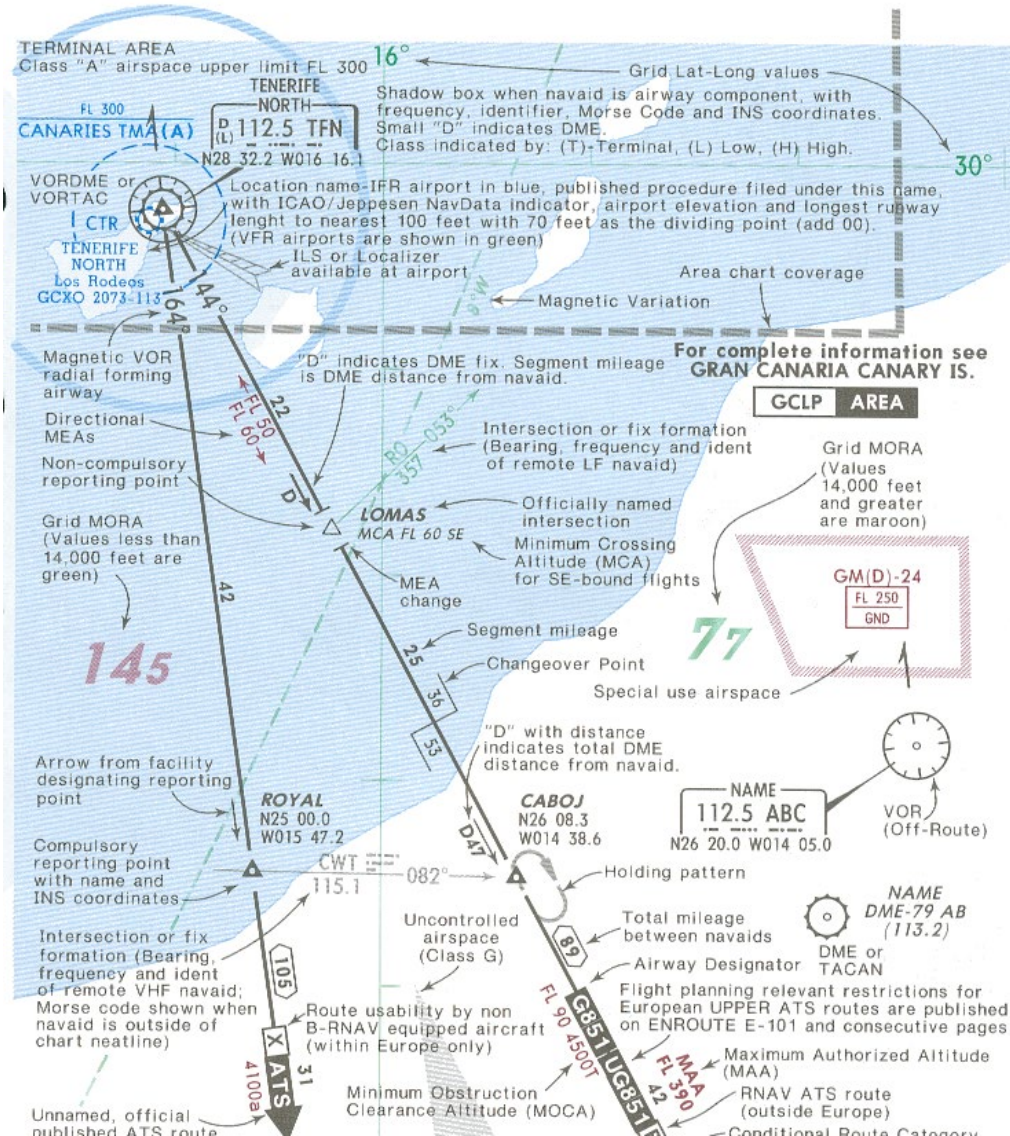
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INTRODUCTION

25 FEB 00

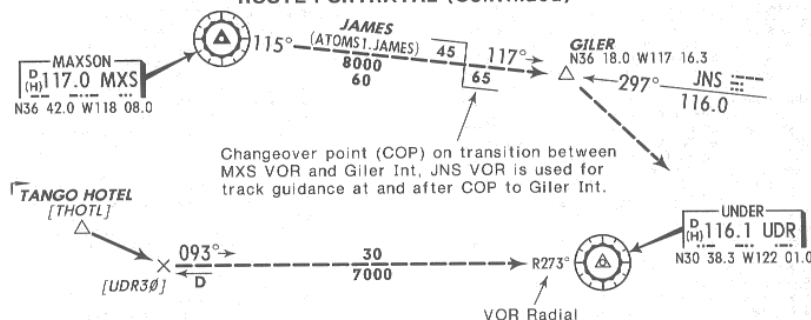
58A

LOW & HIGH/LOW ALTITUDE ENROUTE CHART LEGEND

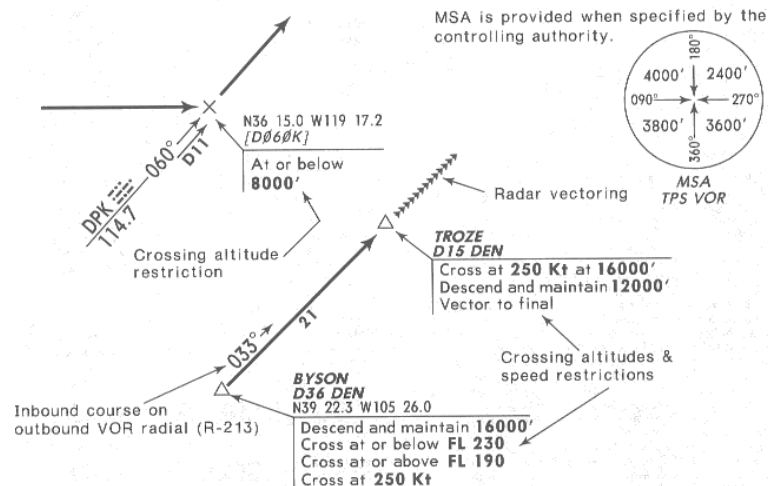


Introduction: SID/STAR legend

SID, DP, AND STAR LEGEND GRAPHIC (Continued) ROUTE PORTRAYAL (Continued)



Database identifiers are enclosed in brackets [UDR30]. Database identifiers are officially designated by the controlling state authority or are derived by Jeppesen. In either case, these identifiers have no ATC function and are not to be used in filing flight plans nor are they to be used when communicating with ATC. Database identifiers are shown only to enable the pilot to maintain orientation when using charts in concert with database navigation systems.



SID/DP CLIMB GRADIENT/CLIMB RATE TABLE

This SID/DP requires a minimum climb gradient of 330' per NM to 9000'.

Climb gradient converted to climb rate in feet per minute at specified ground speeds.

Required climb gradient

Arrival/departure airport, highlighted with circular screen.

Introduction: Approach chart legend

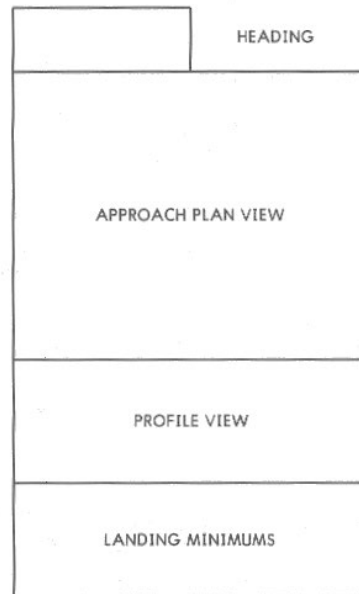
APPROACH CHART LEGEND

Approach charts are graphic illustrations of instrument approach procedures prescribed by the governing authority. All charts meet FAA requirements for aeronautical charts. The following legend pages briefly explain symbology used on approach charts throughout the world. *Not all items apply to all locations.* The approach chart is divided into specific areas of information as illustrated below.

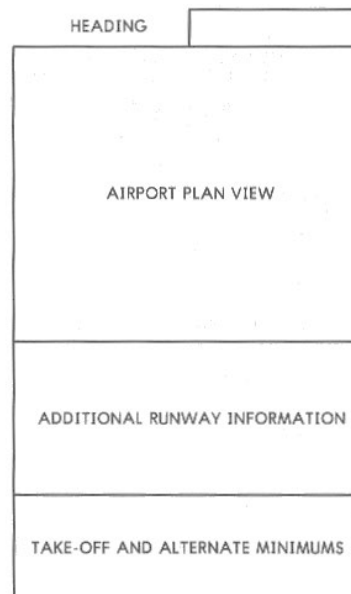
FORMATS

The first approach procedure published for an airport has the procedure chart published on the front side with the airport chart on the back side. On major airports, the airport chart may precede the first approach procedure. These locations will have expanded airport information that may occupy more than one side. When an airport has more than one published approach procedure, they are shown front and back on additional sheets. Blank pages will indicate "INTENTIONALLY LEFT BLANK".

APPROACH PROCEDURE CHART FORMAT



AIRPORT CHART FORMAT



On charts dated on and after 10 MAR 95, key information is displayed in bold type. Key information includes Communication frequencies, Primary NAVAID frequency and identifier, Procedure bearings and Altitudes, Airport and runway end elevation, Decision Altitude and Minimum Descent Altitude, and Missed Approach turn limit and direction, course and altitude.

Introduction: VASI:PAPI

JEPPESEN

INTRODUCTION

MAR 1-91

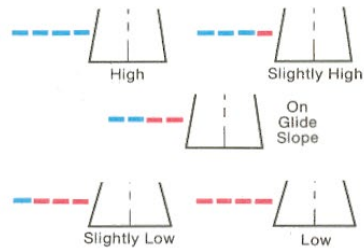
123

APPROACH CHART LEGEND LIGHTING SYSTEMS (continued)

PRECISION APPROACH PATH INDICATOR (PAPI)

PAPI is normally installed on the LEFT side of the runway.

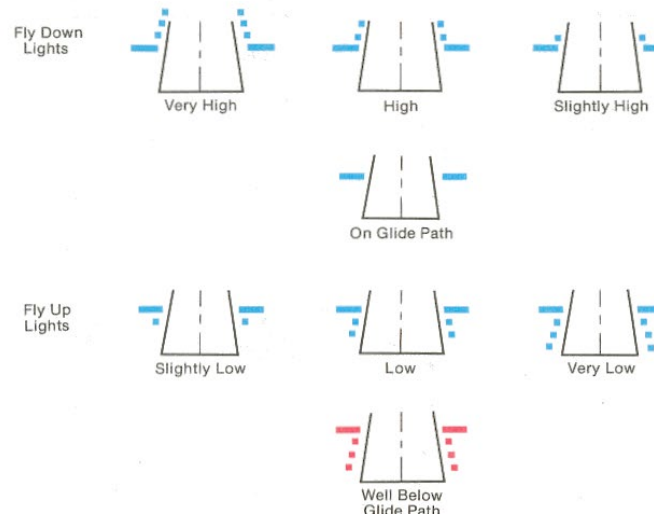
— Red PAPI Lights — White PAPI Lights



VISUAL APPROACH SLOPE INDICATOR (T-VASI)

T-VASI may be installed on the LEFT, RIGHT or BOTH sides of the runway.

— Red T-VASI Lights — White T-VASI Lights



Introduction: ICAO airport signs

162

JUL 17-92

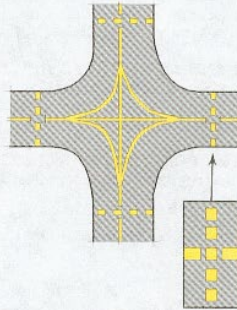
INTRODUCTION

JEPPESEN

ICAO RECOMMENDED AIRPORT SIGNS AND RUNWAY MARKINGS

TAXIWAY INTERSECTION MARKING

A taxiway intersection marking is displayed at the intersection of two paved taxiways.



RUNWAY INTERSECTION SIGNS

15-33

Taxiway intersects runway 15-33. Runway 15 threshold is to the left, runway 33 threshold is to the right. Arrangement of numbers indicates direction to the corresponding runway threshold.

33

Taxiway intersects the beginning of the take-off runway 33. (The sign at each take-off end contains only the runway number for the take-off runway while all others contain both runway numbers).

CAT II

Indicates a category II taxi-holding position.

CAT III

Indicates a category III taxi-holding position.

CAT II/III

Indicates a joint category II/III taxi-holding position.

NO ENTRY

Indicates that entrance to an area is prohibited.

TYPICAL SIGNS AT TAXI-HOLDING POSITIONS
ASSOCIATED TAXIWAY MARKING ALSO SHOWN

Chart Notams

JEPPESEN

CHART NOTAMS

15 NOV 02

E-1

EUROPE

Jeppesen Chart NOTAMS highlight only **significant** changes affecting Jeppesen Charts, also regularly updated at www.jeppesen.com

IMPORTANT: ASK FOR OTHER PERTINENT NOTAMS PRIOR TO FLIGHT

ENROUTE CHARTS

GENERAL

BELGIUM

Operations of state a/cft without FM immunized VHF receivers are only allowed when a/cft are equipped and certified according RNP/5 requirements without the use of VOR; non-FM immune state a/cft without RNP/5 requirements still get the opportunity to fly as OAT along TACAN Routes or under radar control.

Italy replaces the UK as provider of ATS, AIS and MET Service on behalf of the United Nations mission in Kosovo (UNMIK). Civil Operators should note that the MII procedures may not be compatible with ICAO standards. NOTAM are issued from LIAANYX as LYPR series K to disseminate supplementary information or changes to UNMIK AIC 2/2000. All a/cft opr into Kosovo are to note that only NOTAM for Kosovo issued by NOF Italy (LIAANYX) are valid. All other NOTAM from other agencies are to be disregarded. UNMIK AIC 2/2000 will remain avbl on the UK AIS Website. ...

U.K.

STANDARD ROUTES - UNITED KINGDOM previously published on Enroute pages E-24 to

AUSTRIA

ATS ROUTES changed:

B5, Klagenfurt VORDME to Salzburg VORDME. SFOR flights exempted from restrictions ufn.

E LO-12/6ACD.

L604, Graz VORDME to Linz VORDME. SFOR flights exempted from restrictions ufn.

E LO-12/6ABD.

L156 GOLVA (N4842.5 E01539.1) via Gleichenberg NDB, Sollenau VORDME and Wagram VORDME to LANUX, also available for NON-RNAV equipped aircraft. **LOWW.**

I 725 extended: Sollenau VORDME to ABLOM, 068°/218°, 34NM, MEA 6000'. **LOWW.**

R23, FIR [FIR18] (N4816.3 E01654.5) replaced by NCRP TOVKA at N4816.2 E01655.6. **LOWW.**

T161, Linz VORDME to SUBEN, SFOR flights exempted from restrictions ufn. **E LO-12/6A.**

UB5, Klagenfurt VORDME to Salzburg VORDME. SFOR flights exempted from restrictions ufn. **E HI-9/2&3, E HI-11/2&3.**

UI 604, Graz VORDME to Linz VORDME, SFOR flights exempted from restrictions ufn.

E HI-9/2&3.

UT161, Linz VORDME to SUBEN, SFOR flights exempted from restrictions ufn. **E HI-8/1.**

E HI-9/2.

BELARUS, REP OF

ATS ROUTES changed:

→ A68, SOMIS (N5227.9 E02937.0) withdrawn;

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CHART NOTAMS

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UW640, OKBIS (N4947.1 E02624.3) withdrawn; Zolochiv NDB - SORON, 114NM. **E HI-15/7.**
UW689, MALBE (N4849.4 E02222.5) to BUKOV, redesignated UN190. **E HI-9/5, E HI-15/7&8.**

YUGOSLAVIA

ATS ROUTES changed:

B5 changed/extended: NCRP LOKSI (N4434.3 E01910.8) - Valjevo VORDME, 113°/293°, 34NM, MEA 7000'; Valjevo VORDME to

VEMAK, changed to two-way. **E LO-13/4AB.**

G16 extended: Niksic NDB - NCRP PERAN, 321°/141°, 19NM, MEA 10000'. **E LO-13/4C.**

G23 extended: Valjevo VORDME - NCRP BATRU, 246°/066°, 24NM, MEA 8000'. **E LO-13/4AB.**

UG16, GAC rplcd by CRP Gacko NDB at same position. **E HI-11/5.**

TERMINAL CHARTS

GENERAL

FRANCE

For all Circle-to-land with or without prescribed flight tracks proc, CEIL values no longer required for standard operational minima.

ITALY

Coded designators introduced for all Italian STARs and SIDs which terminate at a radio navigation facility. The coded designator consists of the three-letter radio navigation facility identification followed by the validity and the route indicator. Example: BOLZANO 5R (BZO 5R). Level acceleration portion note in missed a/cft to be deleted from all Italian IAP, PANS OPS 4 applies.

SWEDEN

All IAPs: Acceleration portion in missed a/cft no longer applicable, PANS OPS 4 applies.
Omnidirectional dep procedures avbl at certain airports.
SIDs, minimum climb gradient of 400' per NM

BELGIUM

State a/cft not equipped with FM immune VHF receivers operating in civ TMAs are subject to PPR, those a/cft may expect an approval provided they operate under VMC or provided that the respective ATC unit has the possibility to assist in the approach by radar surveillance without VOR/ILS or is able to offer a NDB approach procedure; non-FM immune state a/cft operating within mil TMAs are not allowed to use their VOR/ILS equipment.

A Coruna, Spain, (10-9A) Low Visibility Procedures not avbl.

Aalborg, Denmark, GND svc 'AALBORG Handling' avbl on freq 131.55.

Albenga, Italy, (10-6/General Rules) Ufn Idg on rwy 09 is allowed by day, following a left-hand circuit, north of rwy at an altitude not below 2000'.
(11-1) VMC minimums chgd: Ceiling not below 2500' by day and 3000' by night.

Alderney, U.K., (16-1/16-2) IAPs no longer estbl outside controlled airspace.

(16-2) Until 30 NOV 02 IAP Cat A NDB rwy 26 suspended, except when Radar Vectors provided.

Alghero, Italy, (Fertilia) ARRs 10-2, MINKA - x-break Alghero R-067/24 DME - D17 ALG, MEA FL100/5000'.

Ufn ILS OM rwy 21 u/s.

Alicante, Spain, Ufn twy C4 clsd.

→ **Alta, Norway, Ufn (chart 13-3) IAP VORDME 185°** suspended.

Amsterdam, Netherlands, (Schiphol) After obtaining and read-back enroute clearance pilots are strongly requested to switch to 'SCHIPHOL Start-up' on freq 119.90 without ATC instructions. Caution: Due to construction of new rwy 18/36 2100m northwest of rwy 01L/19R, do not mistake Highway for rwy 18/36 or 01L/19R ufn.
Daily 2300-0600LT rwy 09/27 clsd for T/O.

Daily 2300-0600LT straight-in approaches rwy 22 and rwy 24 not permitted.
'SCHIPHOL Delivery' 121.97(5) solely ops as clearance delivery and provides no longer start-up svc.

Radio Aids: General information

JEPPESEN

RADIO AIDS

JUL 8-88

GENERAL INFORMATION

FREQUENCY ALLOCATION

Frequency allocation is established to provide a clear channeling between the various functions performed by aeronautical aids and communications facilities. Although a general allocation plan is recognized on a world-wide basis, variations may occur within certain ranges. The listing below is intended to provide that allocation most generally used by civil operators.

NAVIGATION AIDS

200 - 415 kHz	Non-directional Beacon (low power) and Radio Range (low power).
200 - 1750 kHz	Non-directional Beacon (standard).
75 MHz	Marker Beacon.
108.0 - 118.0 MHz	VOR test facility (VOT).
108.10 - 111.95 MHz	ILS localizer (on odd-tenths decimal frequencies, 108.1, 108.3 etc.)
108.0 - 111.8 MHz	VOR (even tenths of MHz).
112.0 - 117.95 MHz	VOR (even and odd tenths of MHz).
329.15 - 335.00 MHz	ILS glide slope.
960.0 - 1215.0 MHz	DME and TACAN.

AIRBORNE STATIONS

410 kHz	International DF (outside continental USA).
475 kHz	Working frequency exclusively for aircraft on sea flights desiring an intermediate frequency
500 kHz	International frequency for aircraft and ships over the seas. Transmission on this frequency (except for urgent and safety messages and signals) must cease twice each hour, for three minute periods beginning at 15 and 45 minutes past each hour.
3281 kHz	Lighter-than-air craft.
121.5 MHz	Universal simplex clear channel frequency for use by aircraft in distress or condition of emergency. Emergency Locator Beacons transmit on 121.5 MHz.
122.9 & 123.1 MHz	Air-to-air or air-to-ground communications with aeronautical search and rescue stations when engaged in search and rescue operations

118.0 - 135.975 MHz AIR TRAFFIC CONTROL OPERATIONS

NOTE: The radio spectrum between 118.0 and 136.0 MHz on 50 kHz channeling (i.e., 50 kHz spacing) could contain 360 channels. 100 kHz protection is provided to 121.5 MHz (emergency), so the number is 358. 25 kHz channeling is implemented in some areas and could afford 720 channels - - with 121.5 MHz, 100 kHz protection it will afford 714 channels. Jeppesen charts only "Two" decimal places; i.e., xxx.025 is shown as xxx.02, xxx.050 as xxx.05, xxx.075 as xxx.07, etc.

121.6 - 121.925 MHz	Airport utility frequencies (ground control). In addition, may be used for control of airport lights by transmissions of brief keyed RF signals from aircraft.
121.975 - 123.075 MHz	Also available to private aircraft for air traffic control operations.
123.175 - 123.425 MHz & 3281 kHz	Available for ground and aircraft flight test

Radio data: identifier

JEPPesen

RADIO AIDS

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NAVIGATION AIDS LISTED BY IDENTIFIER - EUROPE

Ident Name

A

A ALFA (JYVASKYLA), FINLAND
A TIRANA, ALBANIA
AA BARAJAS (MADRID), SPAIN
AAL AALBORG, DENMARK
AB ABSAM, AUSTRIA
AB AKRABERG, FAROE IS
AB ALBI, FRANCE
AB HAGFORS, SWEDEN
ABB ABBEVILLE, FRANCE
ABD ABYAD (DAMASCUS), SYRIA
ABN ALBENGA, ITALY
ABU ABU ARGUB, LIBYA, SPA JAMAHIRIYA
ABU ALTENBURG, GERMANY
AC GLASGOW, UNITED KINGDOM
AC HULTSFRED, SWEDEN
ACD ALCOBENDAS, SPAIN
AD ANDA (SANDANE), NORWAY
ADA ADANA, TURKEY
ADM AL-MASSIRA (AGADIR), MOROCCO
ADN ABERDEEN, UNITED KINGDOM
ADR ADRAR, ALGERIA
ADV ADVENT (SVALBARD), NORWAY
ADX ANDRAITX, SPAIN
AFI AFFLIGEM, BELGIUM
AG AGEN, FRANCE
AGB AUGSBURG, GERMANY
AGD ALTENBURG, GERMANY
AGH NEA ANCHIALOS, GREECE
AGN AGEN, FRANCE
AGO ANGOULEME, FRANCE
AH ALGHERO, ITALY
AH ANGELHOLM, SWEDEN
AHO ALGHERO, ITALY
AI ALICANTE, SPAIN
AJO AJACCIO, FRANCE
AK OUKACHA, MOROCCO
AKC TRABZON, TURKEY
AKI AKHISAR, TURKEY
AKR AKROTIRI, CYPRUS
AKU AKUJARVI (IVALO), FINLAND
AL ALENCON, FRANCE
AL LEPLOY (ALESUND), NORWAY
ALB ALBORAN, SPAIN
ALB ALLERSBERG, GERMANY
ALD ALDERNEY, UNITED KINGDOM
ALE ALEPPO, SYRIA

Ident Name

ANE ARLANDA (STOCKHOLM), SWEDEN
ANG ANGERS, FRANCE
ANK ANKARA, TURKEY
ANL NEA ANGHIALOS, GREECE
ANS ANSBACH, GERMANY
ANT ANTON, FINLAND
ANT ANTWERP (DEURNE), BELGIUM
ANW ARLANDA (STOCKHOLM), SWEDEN
ANY ANDOYA, NORWAY
AOG ROTA, SPAIN
AOR ASTOR, SWEDEN
AP ABERPORTH, UNITED KINGDOM
AQ ABERDEEN, UNITED KINGDOM
AQ AQABA, JORDAN
AQA AQABA, JORDAN
AQB AQABA, JORDAN
AQC AQABA, JORDAN
AR AURILLAC, FRANCE
ARA ARAXOS, GREECE
ARB TORTOLI, ITALY
ARF ARIFIYE (TOPEL), TURKEY
ARH EL ARISH, EGYPT
ARI AGRI, TURKEY
ARL ARLANDA (STOCKHOLM), SWEDEN
ARN ARBANCON, SPAIN
ARP ARPE, GERMANY
ARS AROS, SWEDEN
ARW NADOR, MOROCCO
ARX ARAXOS, GREECE
AS ALFA-SIERRA (KEMI), FINLAND
AS ANGERS, FRANCE
AS ARVIDSJAUER, SWEDEN
AS ASMARA, ERITREA
ASK ASKOY (BERGEN), NORWAY
ASM ASMARA, ERITREA
ASN ASWAN, EGYPT
AST ASYUT, EGYPT
ASW ARLANDA (STOCKHOLM), SWEDEN
AT ANNECY, FRANCE
ATA ALTA, NORWAY
ATF ABERDEEN, UNITED KINGDOM
ATH ATHENS, GREECE
ATL ASTIPALAI, GREECE
ATN AUTUN, FRANCE
ATR ALCANTARILLA (MURCIA), SPAIN
AU STAUNING, DENMARK
AUB AUBENAS, FRANCE

Radio data: list by state

JEPPESSEN		1 NOV 02		RADIO AIDS		FRANCE-1	
NAVIGATION AIDS							
Name	Ident	Freq.	Class	INS Coordinates		Var.	Elev.
France							
Abbeville	ABB	108.45	V D H W	N50 08.1	E001 51.3	W003	220
Agen	AG	400.0	H L W	N44 09.0	E000 40.4	W003	204
Agen	AGN	114.8	V D H W	N43 53.3	E000 52.4	W002	896
Aix-Les-Milles	ALM	430.0	H L W	N43 30.5	E005 21.6	W001	367
Ajaccio	AJO	114.8	V D H W	N41 46.2	E008 46.5	E000	2142
Ajaccio	CT	387.0	H L W	N41 47.7	E008 43.4	E000	18
Ajaccio	IS	341.0	H W	N41 53.9	E008 36.7	E000	
Ajaccio	RB	365.0	H L W	N41 54.9	E008 48.1	E000	18
Albi	AB	323.0	H L W	N43 54.8	E002 03.9	W001	564
Alencon	AL	380.0	H L W	N48 26.1	E000 06.7	W003	479
Ales	DA	402.0	H L W	N44 03.1	E004 08.5	W002	712
Amberieu	AMU	116.3	T H	N45 59.3	E005 19.9	W001	820
Amboise	AMB	113.7	V D H W	N47 25.7	E001 03.9	W002	387
Amboise	AMB	341.0	H W	N47 25.1	E001 02.5	W002	
Amiens	GI	339.0	H L W	N49 50.5	E002 29.1	W003	197
Angers	ANG	113.0	V H W	N47 32.2	W000 51.1	W003	
Angers	AS	392.0	H L W	N47 34.6	W000 09.1	W003	194
Angouleme	AGO	404.0	H L W	N45 42.6	E000 25.6	W003	436
Annecy	AT	384.0	H L W	N45 51.7	E006 01.2	W000	1521
Aubenas	AUB	427.0	H L W	N44 26.5	E004 21.7	W001	919
Auch	LMT	420.0	H L W	N43 43.1	E000 36.2	W003	410
Aurillac	AR	343.0	H L W	N44 57.3	E002 22.1	W002	2096
Autun	ATN	114.9	V D H W	N46 48.4	E004 15.5	W001	2244
Auxerre	AX	417.0	H L W	N47 55.2	E003 30.1	W001	523
Avignon	AVN	112.3	V U W	N43 59.7	E004 44.8	W001	148
Avord	AV	306.5	H L W	N46 52.9	E002 55.7	W001	577
Avord	AVD	110.6	T H	N47 03.5	E002 37.8	W001	597
Avord	AVD	288.5	H W	N47 07.2	E002 47.1	W002	577
Basle-Mulhouse	BLM	117.45	V D U W	N47 37.1	E007 29.1	W000	886
Basle-Mulhouse	BN	353.0	H M W	N47 39.7	E007 28.7	W000	864
Basle-Mulhouse	BS	376.0	H L W	N47 34.8	E007 32.2	W000	885
Bastia	BP	369.0	H L W	N42 25.5	E009 32.2	E000	26
Bastia	BTB	114.15	V D H W	N42 34.4	E009 28.5	E000	36
Beauvais	BV	391.0	H L W	N49 29.5	E002 01.8	W002	359
Beauvais	BVS	115.9	V H W	N49 26.2	E002 09.2	W002	
Bergerac	BGC	374.0	H L W	N44 49.2	E000 37.2	W002	171
Besancon	BSV	370.0	H M W	N47 16.0	E006 12.2	W000	1271
Beziers	ZR	397.0	H L W	N43 19.6	E003 16.1	W002	55
Biarritz	BTZ	114.15	V D H W	N43 27.1	W001 30.6	W002	259
Biarritz	BZ	341.0	H M W	N43 28.2	W001 24.2	W002	245
Biscarrosse	BRS	358.0	H L W	N44 21.1	W001 07.7	W003	98
Blois	BLB	397.0	H L W	N47 40.4	E001 12.1	W003	398

Meteorology: General Informations

METEOROLOGICAL SERVICE FOR INTERNATIONAL AIR NAVIGATION – ANNEX 3

NOTE: In this part of the METEOROLOGY section, selected Chapters and paragraphs have been extracted from ICAO Annex 3 – Meteorological Service for International Air Navigation. Chapter and paragraph numbers reflect those contained in the Annex.

CHAPTER 1 – DEFINITIONS

1.1 DEFINITIONS

When the following terms are used in the Standards and Recommended Practices for Meteorological Service for International Air Navigation, they have the following meanings:

AERODROME CLIMATOLOGICAL SUMMARY – Concise summary of specified meteorological elements at an aerodrome, based on statistical data.

AERODROME CLIMATOLOGICAL TABLE – Table providing statistical data on the observed occurrence of one or more meteorological elements at an aerodrome.

AERODROME METEOROLOGICAL OFFICE – An office, located at an aerodrome, designated to provide meteorological service for international air navigation.

AERONAUTICAL FIXED TELECOMMUNICATION NETWORK (AFTN) – A world-wide system of aeronautical fixed circuits provided, as part of the aeronautical fixed service, for the exchange of messages and/or digital data between aeronautical fixed stations having the same or compatible communications characteristics.

AERONAUTICAL METEOROLOGICAL STATION – A station designated to make observations and meteorological reports for use in international air navigation.

AIRCRAFT OBSERVATION – The evaluation of one or more meteorological elements made from an aircraft in flight.

AIRMET INFORMATION – Information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather phenomena which may affect the safety of low-level aircraft operations and which was not

AREA OF RESPONSIBILITY (World Area Forecast System) – A geographical area for which a regional area forecast centre prepares significant weather forecasts.

AUTOMATIC DEPENDENT SURVEILLANCE (ADS) – A surveillance technique in which aircraft automatically provide, via a data link, data derived from on-board navigation and position-fixing systems, including aircraft identification, four-dimensional position and additional data as appropriate.

BRIEFING – Oral commentary on existing and/or expected meteorological conditions.

CONSULTATION – Discussion with a meteorologist or another qualified person of existing and/or expected meteorological conditions relating to flight operations; a discussion includes answers to questions.

EXTENDED RANGE OPERATION – Any flight by an aeroplane with two turbine power-units where the flight time at the one power-unit inoperative cruise speed (in ISA and still air conditions), from a point on the route to an adequate alternate aerodrome, is greater than the threshold time approved by the State of the Operator.

FLIGHT DOCUMENTATION – Written or printed documents, including charts or forms, containing meteorological information for a flight.

FORECAST – A statement of expected meteorological conditions for a specified time or period, and for a specified area or portion of airspace.

GAMET AREA FORECAST – An area forecast in abbreviated plain language for low-level flights for a flight information region or sub-area thereof, prepared by the meteorological office designated by the meteorological authority concerned and exchanged with meteorological offices in adjacent flight informa-

TAF decode

APPENDIX 4. TECHNICAL SPECIFICATIONS FOR AERODROME FORECASTS IN THE TAF CODE FORM

Table A4-1. Template for Aerodrome Forecasts in the TAF Code Form

Key:

M = inclusion mandatory, part of every message

C = inclusion conditional, dependent on meteorological conditions or method of observation

O = inclusion optional

Element as specified in Chapter 6	Detailed content	Template(s)	Examples
Identification of the type of report (M)	Type of report (M)	TAF or TAF AMD	TAF TAF AMD
Location indicator (M)	ICAO location indicator (M)	nnnn	YUDO ¹
Date and time of origin of forecast (M)	Date and time of the origin of the forecast in UTC (M)	nnnnnnZ	160000Z
Date and period of validity of forecast (M)	Date and period of the validity of the forecast in UTC (M)	nnnnnn	160624 080918
Surface wind (M)	Wind direction (M)	nnn or VRB ³	24015KMH; VRB06KMH (24008KT); (VRB03KT) 19022KMH (19011KT)
	Wind speed (M)	[P]nn[n]	00000KMH (00000KT) 140P199KMH (140P99KT)
	Significant speed variations (C) ²	G[P]nn[n]	12012G35KMH (1206G18KT)
	Units of measurement (M)	KMH or KT	24032G54KMH (24016G27KT)
Visibility (M)	Minimum visibility (M)	nnnn C A V O K	0350 7000 9000 9999 CAVOK

Met Broadcast

JEPPESSEN

METEOROLOGY

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E-3

AVAILABILITY OF MET BROADCASTS - EUROPE Radiotelephony

Identify location for which weather is desired and find station(s) disseminating broadcast. Contents of broadcast of each station starts on page E-6.

WEATHER FOR:	AVAILABLE FROM STATIONS:	WEATHER FOR:	AVAILABLE FROM STATIONS:
Aalborg	Copenhagen	Bolzano	Zugspitze
Aberdeen	Scottish	Bournemouth	London (South)
Adana	Royal Air Force	Bremen	Hannover
Agadir	Casablanca, Las Palmas	Brest	Santiago
Aigen/Ennstal	Gerlitz	Brindisi	Brindisi, Royal Air Force
Ajaccio	Marseille	Bristol	London (South)
Akrotiri	Royal Air Force	Brize Norton	Royal Air Force
Alexandria	Cairo	Bruggen	Royal Air Force
Algiers	Algiers, Alicante	Brussels	Amsterdam, Brussels, Frankfurt, London (Main), Paris, Shannon
Alicante	Alicante, Madrid		Belgrade, Istanbul
Allentsteig	Rauchenwarth	Bucharest	
Alpe Rauz	Zugspitze	(Otopeni)	
Alta	Bodo	Cairo	Athens, Beirut, Cairo
Altenrhein	Zugspitze	Cardiff	London (South)
Amman	Beirut, Tel Aviv	Casablanca (Anfa)	Las Palmas, Seville (San Pablo)
Amsterdam	Amsterdam, Brussels, Frankfurt, Hannover, London (Main), Paris, Shannon	Casablanca	Casablanca
Ancona	Royal Air Force	(Mohammed V)	
Andoya	Bodo	Catania	Malta, Rome
Andraveda	Athens, Brindisi	(Fontanarossa)	
Angelholm	Jonkoping	Cologne-Bonn	Brussels, Frankfurt, Hannover, Shannon
Ankara	Ankara, Beirut, Istanbul	Coltishall	Royal Air Force
(Esenboga)		Constantine	Algiers
Annaba	Algiers	Copenhagen	Amsterdam, Berlin (Schonefeld), Copenhagen, Hannover, Oslo, Shannon, Stockholm, Warsaw
Antalya	Istanbul		Dublin
Asturias	Santiago	Cork	Royal Air Force
Aswan	Cairo	Cranwell	Beirut, Cairo, Nicosia
Athens	Athens, Nicosia	Damascus	Tunis
(Eleftherios		Djerba	Berlin (Schonefeld)
Venizelos Intl)		Dresden	Dublin, London (Main), Shannon
Athens	Athens, Brindisi, Cairo, Istanbul	Dublin	Zagreb
(Hellinikon)		Dubrovnik	Amsterdam, Brussels, Frankfurt, Shannon
Aviano	Royal Air Force	Dusseldorf	London (North)
Barcelona	Algiers, Barcelona, Bordeaux, Madrid, Marseille, Santiago, Shannon	East Midlands	Scottish
Bardufoss	Bodo, Royal Air Force	Edinburgh	see Harstadt-Narvik
Basle-Mulhouse	Frankfurt, Geneva, Paris, Pisa, Zurich	Evenes	Lisbon, Santiago, Seville (San
		Faro	Pablo)

Met broadcast

JEPPESEN

METEOROLOGY

16 NOV 01

E-7

MET BROADCASTS IN PLAIN LANGUAGE - EUROPE Radiotelephony

NOTE: c as available

STATION	IDENT	FREQS.	BROADCAST TIMES		FORM	CONTENTS & SEQUENCE
			PERIOD	H+		
Brindisi	Volmet	127.60	H24	cont.	MET Report TREND (T)	Brindisi (T), Pisa (T), Rome (Fiumicino) (T), Rome (Ciampino), Naples (Capodichino), Athens (Hellinikon) (T), Thessaloniki (Makedonia)(T), Kerkira (Ioannis Kapodistrias) (T), Andravida (T)
Brussels	Volmet	127.80	H24	cont.	MET Report QNH TREND	Brussels, Ostend, London (Heathrow), Luxembourg, Amsterdam, Paris (Orly), Frankfurt, Cologne-Bonn, Dusseldorf
		132.47				Brussels
	BUB BUN	114.6 110.6				
Cairo	Volmet	126.2	H24	cont.	MET Report TREND	Cairo, Alexandria, Aswan, Luxor, Hurghada (Intl), Beirut, Damascus, Lamaca, Athens (Hellinikon), Benghazi, Khartoum, Jeddah
Casablanca	Casa- blanca Met	127.60	H24	cont.	MET Report QNH (Q) TREND (T)	Casablanca (Mohamed V) (Q) (T), Rabat(T), Marrakech (T), Tanger (T), Agadir(T), Oujda, Gran Canaria, Malaga, Seville (San Pablo)
Copenhagen (Kastrup)	Volmet	127.00	H24	cont.	MET Report QNH (Q) TREND (T) c	Copenhagen (Kastrup) (Q) (T), Billund, Aalborg (T), Hamburg (T), Malmo (T), Goteborg (T), Stockholm (Arlanda) (T), Oslo (Garder- moen) (T), Stavanger (T)
Dublin	Volmet	127.00	H24	cont.	MET Report TREND	Dublin, Shannon, Cork, Belfast, Glasgow, Prestwick, Manchester, London (Heathrow), London (Gatwick)
Fischamend	FMD	110.4	H24	cont.	SIGMET	Vienna FIR
Frankfurt	Volmet	127.60	H24	cont.	MET Report QNH TREND	Frankfurt, Brussels, Amsterdam, Zurich, Geneva, Basle-Mulhouse, Vienna, Prague, Paris (Charles de Gaulle)

ATIS

JEPPESSEN

METEOROLOGY

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E-13

AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS) - EUROPE

The listing below includes airports served by an ATIS within Europe Charts coverage. Airports are listed alphabetically, under the name shown on the Jeppesen Instrument Approach Charts, with civil airports by city name, followed by airport name when different, or with military airport/facility name. ATIS information for frequencies lower than 137.00 MHz is also provided on Jeppesen Instrument Approach Charts and Jeppesen Enroute Charts communications tabulations.

AIRPORT LOCATION	VOICE FACILITY	ATIS FREQ (MHz)	INFORMATION BROADCAST	HOURS UTC
Aalborg	VHF	120.47	Arr & Dep	H24
Aberdeen (Dyce)	ADN VOR	114.30	Arr & Dep	Mon-Fri: 0645-2150 ☉
	VHF	121.85	Arr & Dep	Sat/Sun: 0700-2150 ☉
Agen (La Garenne)	VHF	129.60	Arr & Dep	H24
Ajaccio (Campo dell'Oro)	VHF	126.92	Arr & Dep	H24
Alesund (Vigra)	VHF	128.65	Arr & Dep	HR of ATS OPS
Alta	VHF	118.17	Arr & Dep	Mon-Fri: 0600-2200 ☉ Sat: 0600-1300 ☉ Sun: 1300-2200 ☉
Amman (Queen Alia Intl)	VHF	127.60	Arr & Dep	H24
Amsterdam	SPL VOR	108.40	Arr	H24
	VHF	132.97	Arr	H24
	VHF	122.20	Dep	H24
	VHF	130.15	Arr & Dep	Peak hrs.
Ankara (Esenboga)	VHF	123.60	Arr & Dep	H24
Antalya	VHF	118.27	Arr & Dep	H24
Athens (Eleftherios Venizelos)	VHF	136.12	Arr & Dep	H24
Athens (Hellinikon)	VHF	123.40	Arr & Dep	H24
Augsburg	VHF	124.57	Arr & Dep	Mon-Sat: 0530-2000 ☉ Sun: 0800-2000 ☉
Avignon	VHF	120.82	Arr & Dep	H24
Barcelona	VHF	118.65	Arr & Dep	H24
Bardufoss	VHF	129.72	Arr & Dep	Mon-Fri: 0550-1850 ☉
Bastle-Mulhouse	VHF	127.87	Arr & Dep	H24
Bastia (Poretta)	VHF	125.92	Arr & Dep	H24
	Tel: +33 (0)4 95591940			
Beauvais(Tille)	VHF	118.37	Arr & Dep	0600 1800
Beirut	KAD VOR	112.60	Arr & Dep	H24
Belfast (Aldergrove)	VHF	128.20	Arr & Dep	H24
Belfast (City)	VHF	136.62	Arr	HR of TWR OPS
Bergen (Flesland)	VHF	125.25	Arr & Dep	H24
Berlin (Schonefeld)	VHF	126.12	Arr & Dep	0500-2300 ☉
Berlin (Tegel)	TGL VOR	112.30	Arr & Dep	0500-2300 ☉
	VHF	125.90	Arr & Dep	0500-2300 ☉
Bern (Belp)	VHF	125.12	Arr & Dep	0450-1820 ☉
	Tel: +42 (0)31 9615015/16			
Beziers (Vias)	VHF	127.52	Arr & Dep	PTO

Operation of Met stations

E-52

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METEOROLOGY

JEPPESSEN

TELEPHONE/FAX NUMBERS AND HOURS OF OPERATIONS OF MET STATIONS - EUROPE

Station	Telephone/ Fax Number	Hours	Station	Telephone/ Fax Number	Hours
CYPRUS			FINLAND		
Larnaca (Intl)	(04) 630394	H24	Enontekio	016-5548420	H24
Paphos (Intl)	(06) 236816	H24	Haili	03-18166435	H24
DENMARK			Helsinki (Vantaa)	09-82773804, 82773331	H24
Aalborg	98171117	Mon-Thu: 0600-1400 ☉, Fri: 0600-1300 ☉ exc Hol; O/T	Helsinki (Malmi)	09-82774031, 0600-9-3808 (forecaster)	H24
Aarhus	39157272	O/T	Ivalo	016-6758620	H24
Billund	39157272	H24	Joensuu	Fax: 6758692	H24
Bornholm	39157272	H24	Jyvaskyla	013-2727024	H24
(Ronne)				014-4455820,	H24
Copenhagen	39157272,	H24	Kajaani	08-6897520	H24
(Kastrup)	32516611		Kauhava	06-1813309	H24
Copenhagen	39157272	H24	Kemi (Tornio)	016-2118720	H24
(Roskilde)			Kittila	016-3668920	H24
Esbjerg	39157272	H24	Kokkola-	06-8696820	H24
Herning	39157272	H24	Pietarsaari		
Karup	97101550	Mon-Thu: 0500-1430 ☉ Fri (Exc Hol): 0500-1300 ☉ O/T	(Kruunupy)		
			Kuopio	017-477211, 0600-9-3805 (forecaster)	HO
	97101795	O/T	Kuusamo	08-8518820	H24
Kolding	39157272	H24	Lappeenranta	05-4156428	H24
(Vamdrup)			Mariehamn	018-634420	H24
Lolland Falster	39157272	H24	Mikkeli	015-366427	HO
(Maribo)			Oulu	08-5207802	H24
Odense	39157272	H24		Fax: 5207798	
Sindal	39157272	H24	Pori	02-61006055	H24
Skive	39157272	H24	Rovaniemi	016-3636608, 0600-9-3807 (forecaster/H24)	H24
Sonderborg	39157272	H24			
Stauning	39157272	H24	Savonlinna	015-523376	H24
Thisted	39157272	H24	Seinajoki	06-4219620	HO
Vojens/	74590900	Mon-Thu: 0500-1430 ☉, Fri: 0500-1300 ☉ exc Hol; O/T		Fax: 4219699	
Skrydstrup	Ext. 3451		Sodankyla	016-610109	H24
	97101795		Tampere-	03-2835521,	HO
EGYPT			Pirkkala	0600-9-3803 (forecaster/H24)	
Abu Simbel	(097) 440528	SR/SS & O/R	Turku	02-2714803	H24
Alexandria (Borg	(03) 4278774	H24	Utti	05-18125354	H24
El Arab Intl)			Vaasa	06-2126130	H24
Alexandria (Intl)	(03) 4278774	H24	Varkaus	017-571278	H24
Aswan (Intl)	(097) 480320	H24	Ylivieska	08-421866	H24
	Ext. 421		FRANCE (also see Meteorology E-41 for locations not listed)		
Asyut (Intl)	(088) 325648	H24	International airports		
Cairo (Intl)	(02) 4157348	H24	Agen	(0)553771206	0445-1645 ☉
Dakhla	(092) 820635	H24	(La Garenne)		
El Arish (Intl)	(068) 320856	H24	Ajaccio	(0)495237675	0500-1730 ☉
El Kharga	(092) 920485	H24	(Campo dell'Oro)		
El Tor	(062) 770252	H24			
Giza (Embakla)	(02) 7100441	0600-SS			

Phonetic alphabet and morse code

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TABLES AND CODES

JEPPesen

PHONETIC, ALPHABET AND MORSE CODE

LETTER	CODE	WORD	LATIN ALPHABET REPRESENTATION
A	• —	Alfa	AL FAH
B	— •••	Bravo	BRAH VOH
C	— ••• •	Charlie	CHAR LEE or SHAR LEE
D	— ••	Delta	DELL TAH
E	•	Echo	ECK OH
F	•• — •	Foxtrot	FOKS TROT
G	— — •	Golf	GOLF
H	••••	Hotel	HOH TELL
I	••	India	IN DEE AH
J	•• — — —	Juliett	JEW LEE ETT
K	— • —	Kilo	KEY LOH
L	• — ••	Lima	LEE MAH
M	— —	Mike	MIKE
N	— •	November	NO VEM BER
O	— — —	Oscar	OSS CAH
P	• — — •	Papa	PAH PAH
Q	— — • —	Quebec	KEH BECK
R	•• •	Romeo	ROW ME OH
S	•••	Sierra	SEE AIR RAH
T	—	Tango	TANG GO
U	•• —	Uniform	YOU NEE FORM or OO NEE FORM
V	••• —	Victor	VIK TAH
W	• — —	Whiskey	WISS KEY
X	— •• —	X-ray	ECKS RAY
Y	— • — —	Yankee	YANG KEY
Z	— — ••	Zulu	ZOO LOO
NUMERAL OR NUMERAL ELEMENT	CODE	PRONUNCIATION	
1	• — — — —	WUN	
2	•• — — —	TOO	
3	••• — —	TREE	
4	•••• —	FOW-er	
5	•••••	FIFE	
6	— ••••	SIX	
7	— — •••	SEV-en	
8	— — — ••	AIT	
9	— — — — •	NIN-er	
0	— — — — —	ZE-RO	

Conversions

JEPPesen

TABLES AND CODES

MAY 4-90

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CONVERSIONS

METERS PER SECOND to FEET PER MINUTE (mps = 196.85 fpm)

MPS	FPM	MPS	FPM	MPS	FPM	MPS	FPM
1	197	6	1181	11	2165	16	3150
1.5	295	6.5	1279	11.5	2263	16.5	3248
2	394	7	1378	12	2362	17	3346
2.5	492	7.5	1476	12.5	2460	17.5	3444
3	591	8	1575	13	2559	18	3543
3.5	689	8.5	1673	13.5	2657	18.5	3641
4	787	9	1772	14	2756	19	3740
4.5	885	9.5	1870	14.5	2854	19.5	3838
5	984	10	1969	15	2953	20	3937
5.5	1082	10.5	2067	15.5	3051		

METERS PER SECOND to KNOTS (1 mps = 1.9438 knots)

Meters p/sec.	0	1	2	3	4	5	6	7	8	9
	Knots									
0	-	1.9	3.9	5.8	7.8	9.7	11.7	13.6	15.6	17.5
10	19.4	21.4	23.3	25.3	27.2	29.2	31.1	33.0	35.0	36.9
20	38.9	40.8	42.8	44.7	46.6	48.6	50.5	52.5	54.4	56.4
30	58.3	60.3	62.2	64.1	66.1	68.0	70.0	71.9	73.9	75.8
40	77.8	79.7	81.6	83.6	85.5	87.5	89.4	91.4	93.3	95.2
50	97.2	99.1	101.1	103.0	105.0	106.9	108.8	110.8	112.7	114.7
60	116.6	118.6	120.5	122.5	124.4	126.3	128.3	130.2	132.2	134.1
70	136.1	138.0	140.0	141.9	143.8	145.8	147.7	149.7	151.6	153.6

TEMPERATURES (CELSIUS/FAHRENHEIT)

°C	°F	°C	°F	°C	°F
-40	-40.0	-8	17.6	24	75.2
-39	-38.2	-7	19.4	25	77.0
-38	-36.4	-6	21.2	26	78.8
-37	-34.6	-5	23.0	27	80.6
-36	-32.8	-4	24.8	28	82.4
-35	-31.0	-3	26.6	29	84.2
-34	-29.2	-2	28.4	30	86.0
-33	-27.4	-1	30.2	31	87.8
-32	-25.6	0	32.0	32	89.6
-31	-23.8	1	33.8	33	91.4

WEIGHT

Lbs	Kgs	Lbs	Kgs
2.2046	1	.45359	1
4	2	1	1
7	3	1	1
9	4	2	2
11	5	2	2
13	6	3	3
15	7	3	3
18	8	4	4
20	9	4	4

Units of measurements

COUNTRY	TABLE	COUNTRY	TABLE
✈ Niger	ICAO (56) (60)	Sudan	Blue
Nigeria	Blue (63)	Suriname	... (36)
Niue Is	Blue	Swaziland	Blue (60)
Norfolk Is	Blue	Switzerland	SI/Non-SI
Okinawa	... (31)	Syria	Blue
Oman	Blue (60)	Taiwan	Blue
Pago Pago	Blue	Tanzania	Blue (60)
Panama	... (31) (62)	Thailand	SI/Non-SI
Papua New Guinea	Blue	Togo	ICAO (56) (60)
Paraguay	ICAO (30) (3/)	Tonga	Blue
Peru	ICAO/Blue (3) (37)	Transkei	Blue
Philippines	Blue (6) (36)	Tunisia	ICAO (60)
Poland	SI (64) (73)	Turkey	Blue
Portugal	SI (59) (63)	Tuvalu	Blue
Qatar	Blue (63)	Uganda	Blue
Reunion	ICAO (60)	United Arab Emirates	Blue
Romania	ICAO (44) (64)	United Kingdom	Blue (63)
Rwanda	ICAO (8) (60)	United States	Blue (33)
St. Helena I	.	Uruguay	ICAO (3/)
Sao Tome & Principe	...	USSR	ICAO (23) (34)
Senegal	ICAO (56) (60)	Vanuatu	Blue (60)
Seychelles	Blue (60)	Venezuela	Blue
Sierra Leone	ICAO (4) (60)	Vietnam	ICAO (60)
Singapore	Blue (27)	Western Samoa	Blue
Solomon Is	Blue (51) (60) (70)	Yemen	Blue (60)
Somalia	Blue	Yugoslavia	SI/Non-SI (60)
Spain	ICAO (63) (68)	Zaire	ICAO (57) (60)
✈ Sri Lanka	ICAO (51) (60)	Zambia	Blue
		Zimbabwe	Blue (60)

Notes:

- (1) ICAO table on request
- (3) ICAO for MET; Blue for AIC.

NOTAMS

JEPPESEN

TABLES AND CODES

JAN 10-92

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NOTAMS

NOTAMs promulgating significant information changes are disseminated from locations all over the world. NOTAMs are intended to supplement Aeronautical Information Publications and provide a fast medium for disseminating information at a short notice. The following format and codes are used in disseminating NOTAMs.

TYPICAL NOTAM — in the SYSTEM NOTAM Format

GG EDZZNAEG
020610 EGGNYNYX

(A0623/91 NOTAMN

Q) EGXX/QRDCA/IV/NBO/W/000/400/5510N00520W050

A) EGT/EGPX
B) 9104030730
C) 9104281500
D) APR 03 07 12 21 24 AND 28 0730 TO 1500
E) DANGER AREA DXX IS ACTIVE
F) GND
G) 12200M (40000 FT.) MSL.)

Priority, address, and date-time group for use by NOTAM office.

NOTAM series A number 0623 of 1991. New NOTAM.

Q) More than one FIR is affected; Danger Area activated; IFR/VFR; NOTAM is for the immediate attention of aircraft operators, for inclusion in pre-flight bulletins and operationally significant for IFR flights; navigational warning; ground; 40,000 ft; centered at N 5510 W 00520 for a radius of 50 NM.

A) ICAO four letter indicator for Scottish and London FIR/UIR.
B) Beginning 1991, 4th month, 3rd day, 0730 UTC.
C) Ending 1991, 4th month, 28th day, 1500 UTC.
D) Danger area is only active 4th month, day 03, 07, 12, 21, 24 and 28, between 0730 to 1500 hr.
E) Danger Area DXX is activated.
F) Lower limits beginning at ground level.
G) Upper limits to 12,200m (40,000 ft) MSL.

FORMAT EXPLANATION of SYSTEM NOTAM

NOTAMN — New NOTAM
NOTAMR — Replaces a previous NOTAM
NOTAMC — Cancels a NOTAM
NOTAMS — SNOWTAM (see page 41)

NOTAM format item Q is divided into eight separate qualifier fields.

- FIR — ICAO location indicator plus "XX" if applicable to more than one FIR.
- NOTAM CODE — For NOTAM code decode see page 33. If the subject of the NOTAM (second and

SNOWTAM

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TABLES AND CODES

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SNOWTAM

Extracted from ICAO Annex 15 — AERONAUTICAL INFORMATION SERVICES

ORIGINATION AND DISTRIBUTION

Notification of the presence or removal or significant changes in hazardous conditions due to snow, slush, ice or water on the movement area is to be made preferably by use of the SNOWTAM format, or the NOTAM Code and plain language.

Information concerning snow, ice and standing water on aerodrome pavements shall, when reported by means of a SNOWTAM, contain information in the order shown in the following SNOWTAM Format.

SNOWTAM FORMAT

{ COM heading }	{ PRIORITY INDICATOR }	{ ADDRESSES }		←=	
	{ DATE AND TIME OF FILING }		{ ORIGINATOR'S INDICATOR }		←=
{ Abbreviated heading }	{ SWAA* SERIAL NUMBER }	{ LOCATION INDICATOR }	{ DATE/TIME OF OBSERVATION }	{ OPTIONAL GROUP }	←=
S	W	*	*		

SNOWTAM	{ Serial number }	→→→
{ AERODROME LOCATION INDICATOR }	A)	→→→
{ DATE/TIME OF OBSERVATION (Time of completion of measurement in UTC) }	B)	→→→
{ RUNWAY DESIGNATORS }	C)	→→→
{ CLEARED RUNWAY LENGTH, IF LESS THAN PUBLISHED LENGTH (m) }	D)	→→→
{ CLEARED RUNWAY WIDTH, IF LESS THAN PUBLISHED WIDTH (m; if offset left or right of centre line add "L" or "R") }	E)	→→→
{ DEPOSITS OVER TOTAL RUNWAY LENGTH <i>(Observed on each third of the runway, starting from threshold having the lower runway designation number)</i> NIL — CLEAR AND DRY 1 — DAMP 2 — WET or water patches 3 — RIME OR FROST COVERED (depth normally less than 1 mm) 4 — DRY SNOW 5 — WET SNOW 6 — SLUSH 7 — ICE 8 — COMPACTED OR ROLLED SNOW 9 — FROZEN RUTS OR RIDGES	F)	→→→
{ MEAN DEPTH (mm) FOR EACH THIRD OF TOTAL RUNWAY LENGTH }	G)	→→→
{ FRICTION MEASUREMENTS ON EACH THIRD OF RUNWAY AND FRICTION MEASURING DEVICE MEASURED OR CALCULATED COEFFICIENT or ESTIMATED SURFACE FRICTION 0.40 and above GOOD — 5 0.35 to 0.36 MEDIUMGOOD — 4 0.35 to 0.30 MEDIUM — 3 0.25 to 0.26 MEDIUMPOOR — 2 0.25 and below POOR — 1 9 - unreliable UNRELIABLE — 9	H)	→→→
<i>(When quoting a measured coefficient use the observed two figures, followed by the abbreviation of the friction measuring device used. When quoting an estimate use single digit)</i>		→→→
{ CRITICAL SNOWBANKS (if present, insert height (cm)/distance from the edge of runway (m) followed by "L", "R" or "LR" if applicable) }	J)	→→→

Sunset/Sunrise tables

SUNRISE										LAT	SUNSET									
January											January									
2	5	8	11	14	17	20	23	26	29		2	5	8	11	14	17	20	23	26	29
h m	h m	h m	h m	h m	h m	h m	h m	h m	h m		h m	h m	h m	h m	h m	h m	h m	h m	h m	
■	■	■	■	■	■	■	■	12 04	11 03	N 72	■	■	■	■	■	■	■	■	12 25	13 24
■	■	■	■	■	12 05	11 11	10 46	10 25	10 06	70	■	■	■	■	■	12 21	13 11	13 39	14 01	14 21
■	11 34	11 13	10 56	10 41	10 27	10 13	09 59	09 46	09 33	68	■	12 36	13 00	13 20	13 37	13 54	14 09	14 25	14 39	14 54
10 27	10 21	10 14	10 06	09 58	09 49	09 39	29	09 19	09 09	66	13 41	13 49	13 59	14 09	14 20	14 32	14 43	14 55	15 06	15 18
09 49	09 45	09 41	09 35	29	22	09 15	09 07	08 59	08 50	64	14 19	14 25	14 32	14 40	14 49	14 58	15 07	15 17	27	36
22	20	09 16	09 12	09 07	09 02	08 56	08 49	43	35	62	14 45	14 50	14 57	15 03	15 10	15 18	26	35	43	52
09 02	09 00	08 58	08 54	08 50	08 46	08 40	08 35	08 29	08 22	N 60	15 06	15 10	15 16	15 22	15 28	15 35	15 42	15 49	15 57	16 05
08 45	08 44	42	39	36	32	27	22	17	11	58	22	27	32	37	43	49	55	16 02	16 09	15
31	30	28	26	23	20	16	12	08 07	08 02	56	37	41	45	15 50	15 55	16 01	16 06	12	19	25
19	18	17	15	12	09	08 06	08 02	07 58	07 53	54	15 49	15 53	15 57	16 01	16 06	11	16	22	28	34
08 08	08 07	08 06	08 05	08 02	08 00	07 57	07 54	50	46	52	16 00	16 03	16 07	11	16	20	25	30	36	41
07 58	07 58	07 57	07 56	07 54	07 52	07 49	07 46	07 42	07 39	N 50	16 09	16 13	16 16	16 20	16 24	16 29	16 33	16 38	16 43	16 48
38	38	38	37	36	34	32	30	27	24	45	29	32	35	39	42	46	50	54	58	17 03
22	22	22	22	21	20	18	16	14	12	40	16 46	16 48	16 51	16 54	16 57	17 01	17 04	17 08	17 11	15
07 08	07 08	07 08	07 08	07 08	07 07	07 06	07 05	07 03	07 01	35	17 00	17 02	17 04	17 07	17 10	13	16	19	22	25
06 56	06 57	06 57	06 57	06 57	06 56	06 56	06 55	06 54	06 52	30	12	14	16	19	21	24	26	29	32	34
06 35	06 36	06 37	06 37	06 38	06 38	06 38	06 38	06 37	06 37	N 20	17 32	17 34	17 36	17 38	17 40	17 42	17 44	17 46	17 48	17 50
17	18	19	20	21	22	22	22	23	23	N 10	17 50	17 52	17 54	17 55	17 57	17 58	18 00	18 01	18 02	18 04
06 00	06 02	06 03	06 04	06 05	06 06	06 07	06 08	06 09	06 10	0	18 07	18 09	18 10	18 11	18 12	18 14	14	15	16	17
05 43	05 44	05 46	05 48	05 49	05 51	05 52	05 54	05 55	05 56	S 10	25	26	27	28	29	29	29	30	30	30
24	26	28	30	32	34	36	38	40	42	20	18 43	18 44	18 45	18 45	18 46	18 46	18 46	18 45	18 45	18 44
05 03	05 05	05 07	05 10	05 12	05 15	05 17	05 20	05 23	05 25	S 30	19 05	19 05	19 06	19 06	19 05	19 05	19 04	19 03	19 02	19 00

ATC définitions

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RVSM

Reduced Vertical Separation Minimum (RVSM) EUR

1. AREA OF APPLICABILITY

RVSM shall be applicable in that volume of airspace between FL 290 and FL 410 inclusive in the flight information regions (FIR/UIR) as shown on ATC page E-7.

RVSM CRUISING FLIGHT LEVELS

FL430*	
FL400	FL410
FL380	FL390
FL360	FL370
FL340	FL350
FL320	FL330
FL300	FL310
FL280*	FL290

*non RVSM level

AIRCRAFT EQUIPMENT

The minimum equipment list (MEL) fulfilling the MASPS consists of: (see JAA TGL6)

Two independent altitude measurement systems each equipped with:
cross-coupled static/source system with

compliance with the criteria of the RVSM minimum aircraft systems performance specifications (MASPS).

2. They have instituted procedures in respect of continued airworthiness (maintenance and repair) and programs, and
3. They have instituted flight crew procedures for operations in the EUR RVSM airspace (dimensions specified on ATC page E-7).

Note1:

An RVSM approval is not restricted to a specific region instead, it is valid globally on the understanding that any operating procedures specific to a given region in this case the EUR Region, should be stated in the operations manual or appropriate crew guidance.

Note 2:

Aircraft that have received State approval for RVSM operations will be referred to as 'RVSM approved aircraft'.

Note 3:

Aircraft that have not received State approval for RVSM operations will

Flight procedures

3 ARRIVAL AND APPROACH SEGMENTS

3.2 STANDARD INSTRUMENT ARRIVALS

3.2.1 When necessary or where an operational advantage is obtained, arrival routes from the en-route phase to a fix or facility used in the procedure are published. When arrival routes are published, the width of the associated area decreases from the "en-route" value until the "initial approach" value with a convergence angle of 30° each side of the axis. This convergence begins at 46 km (25 NM) before the IAF if the length of the arrival route is greater than or equal to 46 km (25 NM). It begins at the starting point of the arrival route if the length of the arrival route is less than 46 km (25 NM). The arrival route normally ends at the initial approach fix. Omnidirectional or sector arrivals can be provided taking into account minimum sector altitudes (MSA) (see 1.2.5).

3.2.2 Terminal radar is a suitable complement to published arrival routes. When terminal radar is employed the aircraft is vectored to a fix, or onto the

intermediate or final approach track, at a point where the approach may be continued by the pilot through reference to the instrument approach chart.

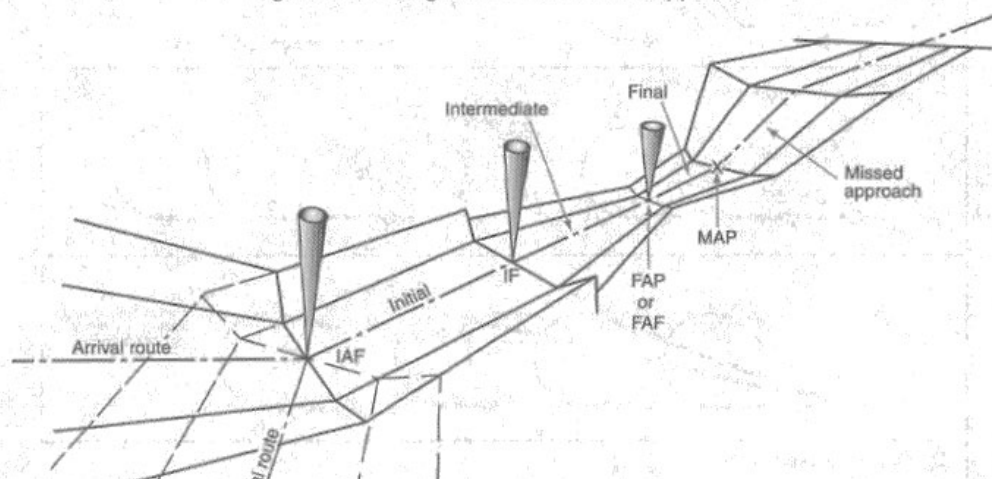
3.3 INITIAL APPROACH SEGMENT

3.3.1 General

3.3.1.1 The initial approach segment commences at the initial approach fix (IAF) and ends at the intermediate fix (IF). In the initial approach, the aircraft has departed the en-route structure and is manoeuvring to enter the intermediate approach segment. Aircraft speed and configuration will depend on the distance from the aerodrome, and descent required. The initial approach segment provides at least 300m (984 ft) of obstacle clearance in the primary area.

3.3.1.3 Where no suitable initial approach fix or intermediate fix is available to construct the instrument procedure in the form shown in Figure III-3-1, a reversal procedure, racetrack or holding pattern is required.

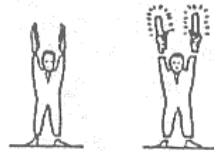
Figure III-3-1. Segments of Instrument Approach



ICAO rules of the air

ICAO RULES OF THE AIR ANNEX 2

5.1.1.2 This Bay — Arms above head in vertical position with palms facing inward.



5.1.1.3 Proceed To Next Signalman — Right or left arm down, other arm moved across body and extended to indicate direction of next signalman.



5.1.1.4 Move Ahead — Arms a little aside, palms facing backward and repeatedly moved upward-backward from shoulder height.



5.1.1.5 Turn —

- a. *Turn to your left:* Right arm downward, left arm repeatedly moved upward-backward. Speed of arm movement indicating rate of turn.



5.1.1.6 Stop — Arms repeatedly crossed above head (the rapidity of the arm movement should be related to the urgency of the stop; i.e., the faster the movement the quicker the stop).



5.1.1.7 Brakes —

- a. *Engage brakes:* Raise arm and hand, with fingers extended, horizontally in front of body then clench fist.

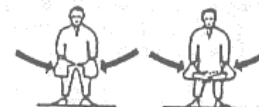


- b. *Release brakes:* Raise arm, with fist clenched, horizontally in front of body, then extend fingers.



5.1.1.8 Chocks —

- a. *Chocks inserted:* Arms down, palms facing inwards, move arms from extended position inwards.



- b. *Chocks removed:* Arms down, palms facing

FRANCE-1

Cat A - 2.2 NM

Entry requirements

1. PASSPORT:

Required.

2. VISA:

Required, except of nationals of the following countries:

Andorra, Argentina, Austria, Belgium, Brunei, Canada, Chile, Cyprus, Czech, Denmark, Finland, Germany, Great Britain, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Republic of Korea, Liechtenstein, Luxembourg, Malta, Monaco, Netherlands, New Zealand, Norway, Poland, Portugal, San Marin, Singapore, Slovakia, Slovenia, Spain, Sweden, Switzerland, U.S.A., and Vatican.

Transit visas are required from nationals of the following countries:

Afghanistan, Albania, Angola, Bangladesh, Ethiopia, Ghana, Haiti, Iran, Iraq, Liberia, Libya, Nigeria, Pakistan, Sierra Leone, Somalia, Sri Lanka and the Democratic Republic of Congo, except if they are holders of a diplomatic or service passport or are in possession of a normal entry visa for France.

3. HEALTH:

Neither a certificate of vaccination against smallpox nor against cholera is required of passengers arriving on international flights.

4. AIRCRAFT ENTRY REQUIREMENTS:

a) Scheduled Air Traffic

Scheduled operations are governed by interstate bilateral air agreements or specific clearances granted by the French Government.

Overflights of other airlines require a clearance from the Ministry of Foreign Affairs requested through diplomatic channels.

b) Non-Scheduled Air Traffic

For overflights and technical landings of

- 8 days in advance in all other cases.

Long series of flights (i.e. within the Europe-Mediterranean region at least 4 flights per month over a time period of two to twelve months with the same charterer; for other destinations 2 flights per month over a time period of two to twelve months with the same charterer)

- 2 months in advance.

Short series of flights (i.e. flights do not conform either with single flights or with long series mentioned above)

- 8 days in advance (Europe-Mediterranean region);
- 15 days (other destinations).

ii) Commercial landings of aircraft registered in a country which has ratified the Multilateral Agreement on Commercial Rights of Non-Scheduled Air Services in Europe are subject to the following conditions:

1. The flight is intra-European (all landings are effected in a country member of the European Civil Aviation Organization) and conforms with the following categories of flights:

- humanitarian flights;
- taxi flights with a seating capacity of not more than six passengers;
- flights where the whole capacity is hired by a single person (except tourist agencies for transport of personnel and merchandise);
- non-regular flights for

Emergency procedures

INTERNATIONAL CIVIL AVIATION ORGANIZATION (ICAO)

3.3.3 When an aircraft subjected to an act of unlawful interference must depart from its assigned track or its assigned cruising level without being able to make radiotelephony contact with ATS, the pilot-in-command should, whenever possible:

- a. attempt to broadcast warnings on the VHF emergency frequency and other appropriate frequencies, unless considerations aboard the aircraft dictate otherwise. Other equipment such as on-board transponders, data links, etc., should also be used when it is advantageous to do so and circumstances permit; and
- b. proceed in accordance with applicable special procedures for in-flight contingencies, where such procedures have been established and promulgated in ICAO Document 7030 — *Regional Supplementary Procedures*; or
- c. if no applicable regional procedures have been established, proceed at a level which differs from the cruising levels normally used for IFR flight in the area by 300m (1,000') if above FL 290 or by 150m (500') if below FL 290.

(Annex 2, Attachment B, 2.2)

4 EMERGENCY DESCENT

4.1 INITIAL ACTION BY THE AIR TRAFFIC CONTROL UNIT

4.1.1 Upon receipt of advice that an aircraft is making an emergency descent through other traffic, all possible action shall be taken immediately to safeguard all aircraft concerned. When deemed necessary, air traffic control units shall immediately broadcast by means of the appropriate radio aids, or if not possible, request the appropriate communications stations immediately to broadcast an emergency message. (Doc 4444, Part III, 16.4.1)

4.2 ACTION BY THE PILOT-IN-COMMAND

5 DISTRESS AND URGENCY RADIOTELEPHONY COMMUNICATION PROCEDURES

5.1 GENERAL

5.1.1 The radiotelephony distress signal **MAYDAY** and the radiotelephony urgency signal **PAN PAN** shall be used at the commencement of the first distress and urgency communication respectively. (Annex 10, Vol II, 5.3.1.2)

5.1.1.1 At the commencement of any subsequent communication in distress and urgency traffic, it shall be permissible to use the radiotelephony distress and urgency signals. (Annex 10, Vol II, 5.3.1.2.1)

5.1.2 The originator of messages addressed to an aircraft in distress or urgency condition shall restrict to the minimum the number and volume and content of such messages as required by the condition. (Annex 10, Vol II, 5.3.1.3)

5.1.3 If no acknowledgement of the distress or urgency message is made by the station addressed by the aircraft, other stations shall render assistance, as prescribed in 5.2.2 and 5.3.2 respectively. (Annex 10, Vol II, 5.3.1.4)

NOTE: "Other stations" is intended to refer to any other station which has received the distress or urgency message and has become aware that it has not been acknowledged by the station addressed.

5.1.4 Distress and urgency traffic shall normally be maintained on the frequency on which such traffic was initiated until it is considered that better assistance can be provided by transferring that traffic to another frequency. (Annex 10, Vol II, 5.3.1.5)

State special procedures

JEPPESSEN

3 MAY 02

EMERGENCY

FRANCE-1

ICAO DIFFERENCES OR STATE SPECIAL PROCEDURES

GENERAL

In general, the Emergency, Unlawful Interference, Communications Failure, Interception and Search and Rescue procedures are in conformity with the Standards, Recommended Practices and Procedures in ICAO Annexes and Documents.

COMMUNICATIONS FAILURE

In VMC: ICAO Procedure

When the flight is executed in Class A airspace

the conditions of Class B will apply.

In IMC: ICAO Procedure, supplemented as follows:

DIFFERENCES TO THE RADIO COMMUNICATION FAILURE PROCEDURES EUROPE (EMERGENCY E-21)

para e) The point to refer is the Initial Approach Fix (IAF) instead of the designated navigational aid as this fix is the published approach procedure reference point to be notified in the flight plan.

If communication failure occurred during the arrival phase (STAR), the approach procedure phase to an airport, or the departure phase from an airport (SID), the pilot will comply with the specific procedures published for that airport, if any.

GÉNÉRAL LEAVING PROCEDURE

If unable to land at the airport of destination for any reason, the pilot shall leave TMA within 30 minutes and try to reach VMC applying appropriate procedure described under "LEAVING PROCEDURE".

Procedures for airports are listed under pertinent TMA name. (For cross-reference see below:)

Albi (Le Sequestre)	TOULOUSE TMA
Annecy (Meythet)	CHAMBERY TMA
Avignon (Caumont)	MARSEILLE/PROVENCE TMA

Beauvais (Tille)	PARIS TMA
Beziers (Vias)	MONTPELLIER TMA

Bourg (Ceyzeriat)	LYON TMA
Bretigney (St. Orpèr)	PARIS TMA

Melun (Villaroche)	PARIS TMA
Merville (Calonne)	LILLE TMA
Muret (L'Herm)	TOULOUSE TMA
Pau/Pyrenees	PYRENEES TMA
Pontoise	
(Cormeilles-en-Vexin)	PARIS TMA
Rochefort (St. Agnant)	LA ROCHELLE TMA

St. Etienne (Bouttheon)	LYON TMA
St. Nazaire (Montoir)	NANTES TMA
Tarbes	PYRENEES TMA

(Lourdes-Pyrenees)	
Toussus-Le Noble	PARIS TMA
Valence (Chabeuil)	LYON TMA
Villefranche (Tarare)	LYON TMA
Vichy	CLERMONT-FERRAND TMA

Villacoublay (Velizy)	PARIS TMA
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COMMUNICATION FAILURE PROCEDURES

AGEN TMA

a) Arrival Procedure

In case of a missed approach climb along extended runway centerline to 1000 ft, continue climb to intercept and follow 27 NM DME-arc of VORDME "AGN" until reaching 3500 ft, then turn right to NDB "AG".

b) Leaving Procedure

If the second attempt for landing fails, leave the TMA by following SID SECHE 1W at the minimum safe level and try to reach VMC.

c) After Departure

In VMC: Return for landing on the aerodrome.
In IMC: Continue flight to the TMA boundary at the last assigned flight level, thereafter continue climb to the cruising level.

AJACCIO TMA

1. AJACCIO (Campo Dell'Oro)

a) Leaving Procedure

Rejoin and follow track 236° from locator "RB", then intercept and follow radial 244 of VORDME "AJO" and try to reach VMC.

b) After Departure

Continue flight to the boundary of

Legend and explanation

LEGEND AND EXPLANATION

LOCATION (AIRPORT), Apt of Entry (if applicable)

Airports are listed alphabetically by location name, with airport name in parentheses when different than location name. A cross index by airport name is provided.

ELEVATION, JEPPESEN NavData (ICAO) IDENTIFIER, IATA IDENTIFIER (if applicable), TIME ZONE (1), COORDINATES.

- (1) Time Zone, in numeric format, observed by the airport as stated in the source and indicating the standard time differential of each zone from Universal Time Coordinated (UTC).
- * indicates that the airport observes Daylight Savings or Summer Time.

TELEPHONE/TELEFAX NUMBERS

Telephone/Telefax numbers are provided for contact with the airport, where available.

RUNWAY DATA and runway/approach lights

All usable runways are listed indicating the following items.

1. Runway designators.
2. Total runway length, excluding stopways, overruns or clearways.
3. TORA and LDA if not identical with total runway length. TODA and ASDA when longer than take-off run (TORA) and provided by controlling authority. For explanation see reverse.
1. Type of runway surface.
2. Runway bearing strength.
 - Load classification number (LCN) supplemented (if known) by
 - r (rigid pavement) - radius of relative strength in inches
 - f (flexible pavement) - thickness in inches
 - Load Classification Group (LCG)
 - Wheel and/or aircraft loads in thousands of pounds
 - SIWL - single isolated wheel load times number of main wheels = allowable aircraft weight.
 - ESWL - Equivalent Single Wheel Load, a calculated value for multi-wheel legs. The resultant value is considered to be the same as SIWL for determining LCN as indicated on page 4.
 - S or SW - (allowable aircraft weight) for single wheel per leg configuration.
 - T or DW - (allowable aircraft weight) for tandem or dual wheel per leg configuration.
 - TT or DDW - (allowable aircraft weight) for twin tandem or double dual wheel per leg configuration.
 - TDT - Runway weight bearing capacity for aircraft with twin delta tandem landing gear.
 - DDT - Runway weight bearing capacity for aircraft with double dual tandem type landing gear.

ICAO location indicators decode

JEPPESEN NavData (ICAO) LOCATION IDENTIFIERS DECODE - EUROPE

DA FIR/UIR	EBTY TOURNAI (MAUBRAY)	EDDL DUSSELDORF
DAAA ALGIERS FIR	EBUL URSEL (URSEL AB)	EDDM MUNICH
	EBWE WEELDE (WEELDE AB)	EDDN NURNBERG
DA ALGERIA	EBZH HASSELT	EDDP LEIPZIG-HALLE
DAAD BOU SAADA	EBZR ZOERSEL (OOSTMALLE AB)	EDDR SAARBRUCKEN
DAAE BEJAIA	EBZW GENK (ZWARTBERG)	EDDS STUTTGART
(SOUMMAM-ABANE RAMDANE)		EDDT BERLIN (TEGEL)
DAAG ALGIERS	ED FIR/UIR	EDDV HANNOVER
(HOUARI BOUMEDIENE)	EDBB BERLIN FIR	EDDW BREMEN
DAAJ DJANET (TISKA)	EDFF FRANKFURT FIR	EDEB BAD LANGENSALZA
DAAP ILLIZI (TAKHAMALT)	EDLL DUSSELDORF FIR	EDEH HERRENTEICH
DAAT TAMANRASSET (AGUENAR)	EDMM MUNICH FIR	EDEL LANGENLONSHEIM
DAAV JIJEL (FERHAT ABBAS)	EDUU RHEIN UIR	EDEM MOSENBERG
DABB ANNABA (RABAH BITAT)	EDVV HANNOVER UIR	EDEQ MUHLHAUSEN
DABC CONSTANTINE	EDWWW	EDEW WALLDURN
(MOHAMED BOUDIAF)	BREMEN FIR	EDFA ANSPACH/TAUNUS
DABS TEBESSA		EDFB REICHELNSHEIM
(CHEIKH LARBI TEBESSI)	ED GERMANY	EDFC ASCHAFFENBURG
DAOB TIARET	EDAB BAUTZEN	EDFD BAD NEUSTADT/SAALE
(A. BOUSSOUF AIN	EDAC ALTENBURG (NOBITZ)	(GRASBERG)
BOUCHEKIF)	EDAD DESSAU	EDFE EGELSBAACH
DAOI ECH-CHELIFF	EDAE EISENHUTTENSTADT	EDFG GELNHAUSEN
DAON TLEMCEN	EDAG GROSSRUCKERSWALDE	EDFH HAHN
(ZENATA-MESSALI EL HADJ)	EDAH HERINGSWALDE	EDFI HIRZENHAIN
DAOO ORAN (ES SENIA)	EDAI SEGELETZ	EDFJ LAGER HAMMELBURG
DAOV GHRIS	EDAJ GERA (LEUMNITZ)	EDFK BAD KISSINGEN
DATG IN GUEZZAM	EDAK GROSSENHAIN	EDFL GIESSEN (LUTZELLINDEN)
DATM BORDJ MOKHTAR	EDAL FURSTENWALDE	EDFM MANNHEIM (MANNHEIM CITY)
DAUA ADRAR	EDAM MERSEBURG	EDFN MARBURG-SCHONSTADT
(TOUAT-CH SIDI MOHAMED	EDAN NEUSTADT-GLEWE	EDFO MICHELSTADT
BELKEBIR)	EDAO NORDHAUSEN	EDFP OBER-MORLEN
DAUE EL GOLEA	EDAP NEUHAUSEN	EDFQ ALLENDORF/EDER
DAUG GHARDAIA	EDAQ HALLE (OPPIN)	EDFR ROTHENBURG O.D. TAUBER
(NOUMERAT-MOUFDI ZAKARIA)	EDAR PIRNA (PRATZSCHWITZ)	EDFS SCHWEINFURT
DAUH HASSI MESSAOUD	EDAT NARDT	(SCHWEINFURT SUD)
(OUED IRARA-KRIM BELKACEM)	EDAU RIESA (GOHLIS)	EDFT LAUTERBACH
DAUI IN SALAH	EDAV FINOW	EDFU MAINBULLAU
DAUK TOUGGOURT (SIDI MAHDI)	EDAW ROITZSCHJORA	EDFV WORMS
DAUO EL OUED (GUEMAR)	EDAX RECHLIN-LARZ	EDFW WURZBURG (SCHENKENTURM)
DAUT TIMIMOUN	EDAY STRAUSBERG	EDFX HOCKENHEIM
DAUU OUARGLA	EDAZ SCHONHAGEN	EDFY ELZ
DAUZ ZARZAITINE (IN AMENAS)	EDBA ALKERSLEBEN	EDFZ MAINZ (FINTHEN)
	(WULFERSHAUSEN)	EDGA AILERTCHEN
DT FIR/UIR	EDBC COCHSTEDT/SCHNEIDLINGEN	EDGB BREITSCHEID
DTTC TUNIS FIR	EDBD DEDELOW	EDGE EISENACH-KINDEL
	EDBF FEHRBELLIN	EDGF FULDA-JOSSA
DT TUNISIA	EDBG BURG	EDGH HETTSTADT
DTKA TABARKA (7 NOVEMBER)	EDBH BARTH	EDGI INGELFINGEN (BUHLHOF)
DTMB MONASTIR (HABIB BOURGUIBA)	EDBI ZWICKAU	EDGJ OCHSENFURT
DTTA TUNIS (CARTHAGE)	EDBJ JENA (SCHONGLEINA)	EDGK KORBACH
DTTF GAFSA (KSAR)	EDBK KYRITZ	EDGL LUDWIGSHAFEN
DTTI BORJ EL AMRI	EDBL LAUCHA	(LUDWIGSHAFEN
DTTJ DJERBA (ZARZIS)	EDBM MAGDEBURG	UNFALLKLINIK)
DTTQ EL BORMIA	EDBN OFUNA	EDBN OFUNA

Airport data

JEPPESSEN

1 NOV 02

AIRPORT DIRECTORY

FRANCE-1

Airport Availability Hours in France are generally H24. The hours published in this Directory are the operating hours of the Air Traffic Service Unit at the airport. Pilots intending to use the apt outside these hours are required to follow the "auto-information-service" procedures, valid within France (see France, ATC Section). Remarks are included covering the restricted use of any apt.

France

Abbeville Apt of Entry

220' LFOI +01:00° N50 08.6 E001 49.9
Apt Administration 0322242471, 0322240521; Fax 0322312438; Aeroclub 0322240848.
02/20 4101' CONCRETE. S/L 11, T/L 18, TT/L 42.
Rwy 20 (CONCRETE) Right-Hand Circuit.
02/20 2953' GRASS.
Rwy 20 (GRASS) Right-Hand Circuit.
13/31 2083' GRASS. LDA 13 2001'.
Rwy 13 Right-Hand Circuit.
Winter: Mon, Thu-Fri 1000-1315LT, 1445-1930LT. Sat Sun Hol 1000-1330LT, 1430-1930LT. Summer (15 APR-30 SEP): Mon-Fri 1100-1415LT, 1600-2015LT. Sat Sun Hol 1100-1415LT, 1545-2030LT. Customs O/R: Days, 24hr PN.
F-3. Fire 1.

Agen (La Garenne) Apt of Entry

204' LFBA AGF +01:00° N44 10.5 E000 35.4
TWR 0553770057. Apt Manager 0553770051, Fax 0553770052. Apt Operator 0553770088, Fax 0553960349.
Aeroclub 0553963667.
11/29 7103' BITUMEN. PCN 23/F/C/W/T. LDA 11 6562'. LDA 29 5791'. HIRL. HIALS 29.
Rwy 11 Right-Hand Circuit.
Mon-Fri 0600-2230LT, Sat 0630-1200LT, 1400-1930LT (Winter until 1830LT), Sun/Hol 0900-1200LT, 1400-1930LT (Winter until 1830LT), O/T PPR 24hr. Customs O/R: 24hr PPR to Apt Manager.
F-3. Jet. Fire 3.

Aire-Sur-L'Adour

259' LFDA +01:00° N43 42.6 W000 14.7
Airport Operator 0558716466 (except Mon and Tue).
12/30 3281' MACADAM. S/L 11. LDA 12 2740'. LDA 30 2182'.
Rwy 30 (MACADAM) Right-Hand Circuit.
12/30 3248' GRASS. LDA 12 2838'. LDA 30 2247'.
Rwy 30 (GRASS) Right-Hand Circuit.
O/R except Mon and Tue.

Albi (Le Sequestre) Apt of Entry

564' LFCI LBI +01:00° N43 54.8 E002 07.0
Airport Operator 0563494847, Fax 0563494840, AFIS 0563543849, Fax 0563542400. Aeroclub 0563542768.
09/27 5118' MACADAM. S/L 22, T/L 33, LDA 09 4872'. LDA 27 4560'. HIRL.
Mon-Fri 0730-1900LT, Sat, Sun & Hol not provided; O/T PPR 24 hr for Mon-Fri and PPR 48hr for Sat, Sun & Hol.
Customs O/R: O/R to AFIS; for Mon-Fri PPR 24hr, for Sat, Sun & Hol PPR on last preceding workday.
F-3. Jet. Fire U.

Alençon (Valframbert)

479' LFOF +01:00° N48 26.8 E000 06.6
Apt Operator 0233292586, 0233276183, Fax 0233319317.
07/25 2543' PAVED. AUW-13. LDA 07 1854'. LDA 25 1952'.
Rwy 25 (PAVED) Right-Hand Circuit.
07/25 2395' GRASS. LDA 07 1804'. LDA 25 1804'.
Rwy 25 (GRASS) Right-Hand Circuit.
Days O/R.
F-3. Fire 1.

Ales (Deaux)

669' LFMS +01:00° N44 04.4 E004 08.6
Apt Administration 0466784949. ATS 0466835222; Fax 0466836491. Aeroclub 0466835618.
01/19 4593' MACADAM. S/L 20, T/L 29. LDA 01 3937'.
TODA 01 4806'. RL.
Rwy 19 Right-Hand Circuit.
Mon-Thu 0830-1200LT & 1400-1800LT, Fri 0830-1200LT & 1400-1700LT. O/T for commercial, rescue and official flights only. Mon-Fri 24hr PPR, Sat Sun 48hr PPR. Pr larity for fire fighting acct.
F-3. Fire 1.

Amberieu (Amberieu AB)

823' LFXA +01:00° N45 58.8 E005 20.3
Airport 0474382152 ext 81229. Aeroclub ext 81388.
01/19 6562' ASPHALT. S/L 44. LDA 19 5899'. TODA 01 6890'. TODA 19 6890'. ASDA 01 6890'. ASDA 19 6890'.
Rwy 19 Right-Hand Circuit. Rwy 01/19 closed every Thu 0800-1200LT.
02/20 2625' GRASS. S/L 11. LDA 20 2231'.
Rwy 02 Right-Hand Circuit.
By NOTAM. Open to public air traffic.
F-3 O/R. Jet A-1. Oxygen. Fire 6.

Ambert (Le Poyet)

1847' LFHT +01:00° N45 31.0 E003 44.8
Apt Operator 0470000155. Aeroclub 0470000155.

How to serve Jeppesen

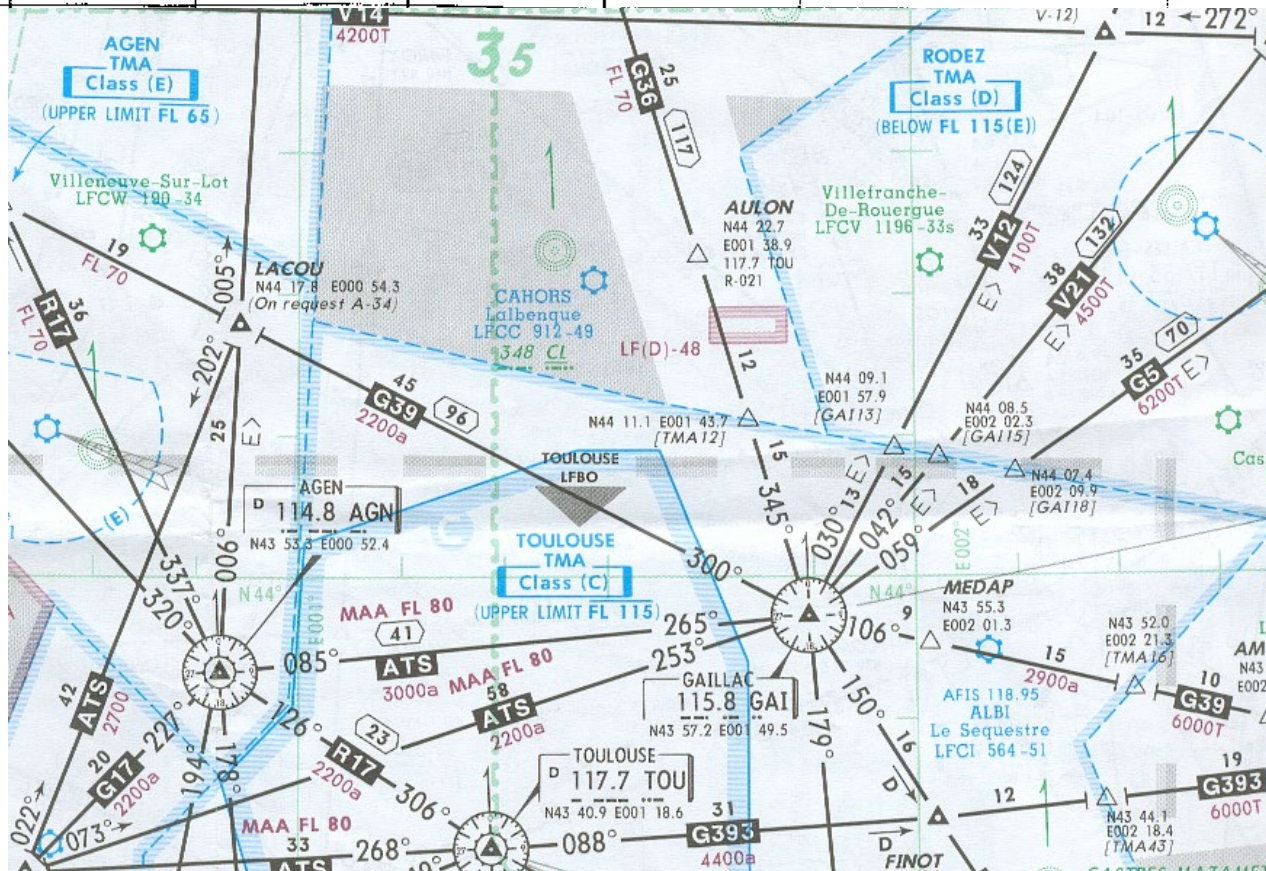
JAR-OPS 1.1045 Appendix I	Jeppesen Airway Manual
a) Minimum flight level Minimum flight altitude	1. Enroute-, Area-, SID/STAR charts. 2. INTRODUCTION section, page 57, Enroute Chart Legend.
b) Operating minima for departure-, destination- and alternate aerodromes	1. Approach-, Airport charts. 2. 10-9X pages(JAA-Minima). 3. ATC section, JAA AOM pages 601-611.
c) Communication facilities and navigation aids	1. RADIO AIDS section. 2. Approach-, Airport charts. 3. SID/STAR charts. 4. Enroute charts.
d) Runway data and aerodrome facilities	1. AIRPORT DIRECTORY. 2. Airport charts.
e) Approach, missed approach and departure procedures including noise abatement procedures	1. Approach charts. 2. SIDs & STARs. 3. Noise pages. 4. ATC section, chapter 8. a) Noise Abatement Procedures. b) State pages.

How to serve Jeppesen

f) COM-failure procedures	1. EMERGENCY section. 2. ATC section, Emergency and Communications Failure Procedures.
g) Search and rescue facilities in the area over which the aeroplane is to be flown	1. EMERGENCY section, Search and Rescue. 2. ATC section. 3. AIRPORT DIRECTORY.
h) A description of the aeronautical charts that must be carried on board in relation to the type of flight and the route to be flown, including the method to check their validity.	1. INTRODUCTION section, Chart Legend.
i) Availability of aeronautical info and MET services	1. METEOROLOGY section. 2. ENROUTE charts. 3. AIRPORT DIRECTORY
j) Enroute COM/NAV procedures	1. ATC section, State pages 2. ENROUTE section. 3. Enroute charts

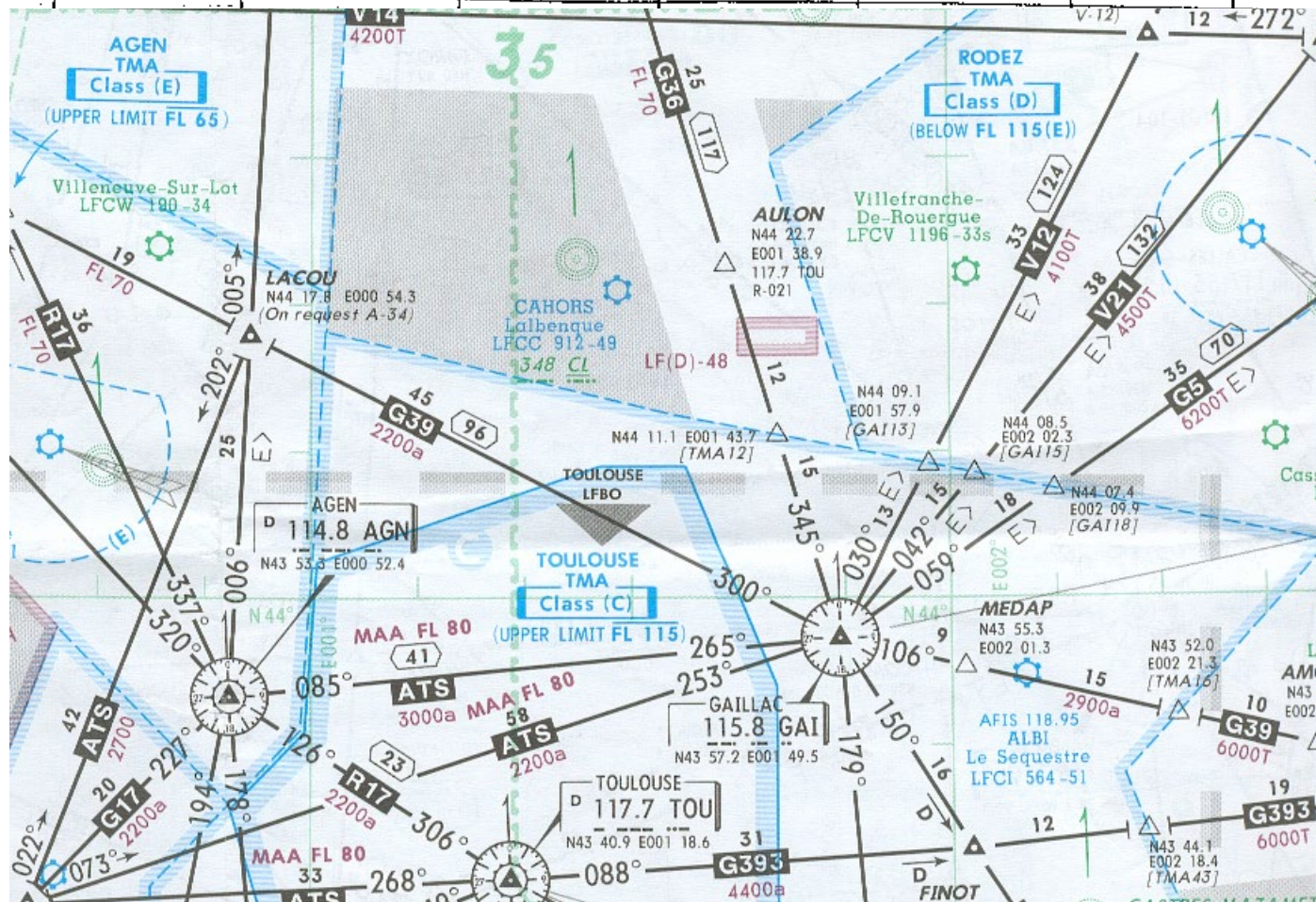
MEA

MEA	MINIMUM EN ROUTE IFR ALTITUDE		AIRWAY (E.A.C.) (2)	Demi largeur de l'airway + 5 NM de chaque côté du Centerligne	EN FRANCE	• +1000ft • +2000ft	<i>- Valeur OFFICIELLE</i> <i>Altitude ou Niveau le plus bas</i> <i>utilisable garantissant :</i> - Les réceptions radio COM et radio NAV - Passage des zones P.R.D - La sécurité par rapport au relief
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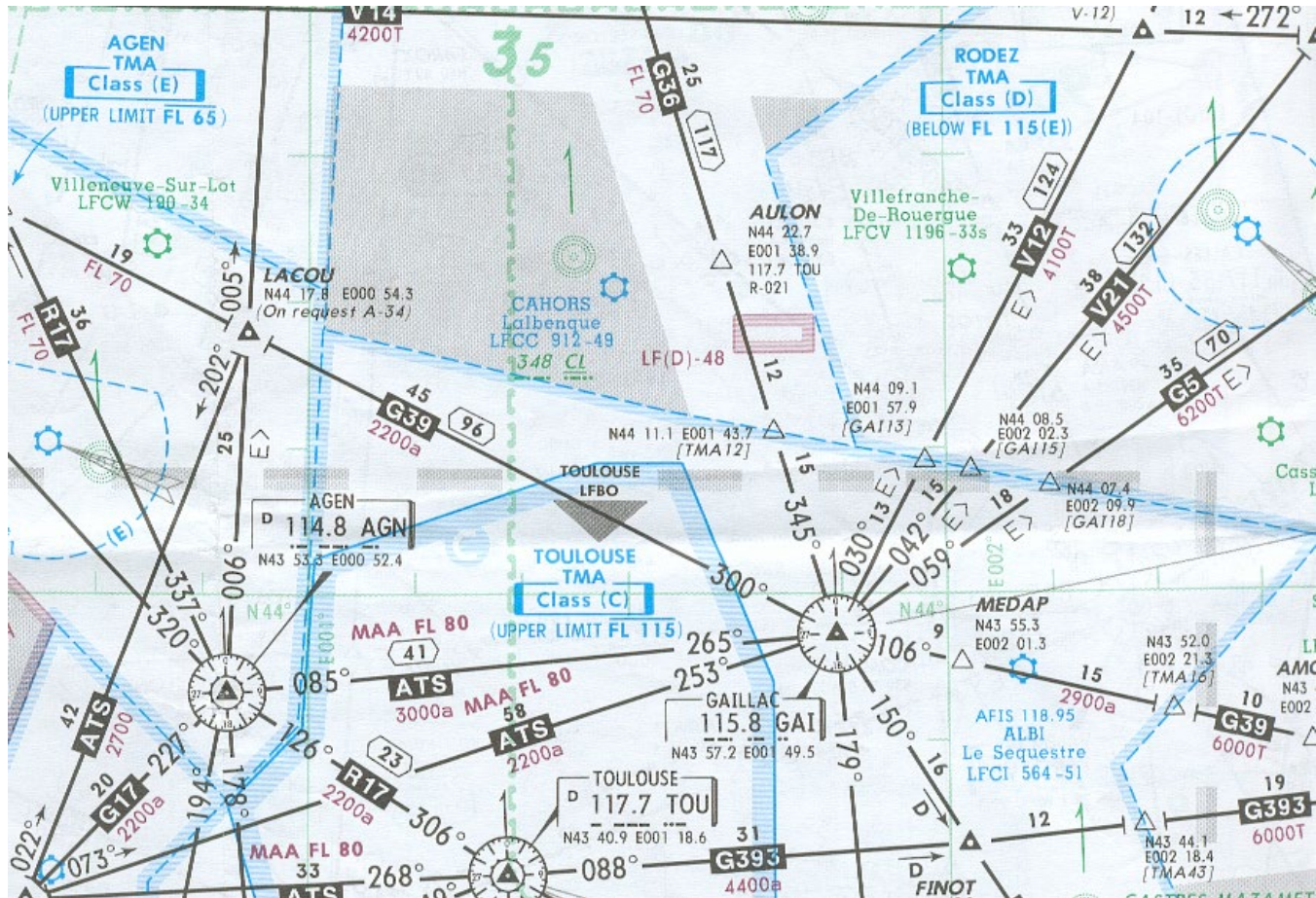
MOCA

MOCA	MINIMUM Obstruction Clearance Altitude	3300T	Sous AIRWAY (EANC) (2)	Demi largeur de l'airway + 5 NM de chaque côté du Centerligne	EN FRANCE	+1500 ft	-Valeur OFFICIELLE -Assure la sécurité par rapport au relief sur un couloir élargi.
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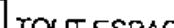


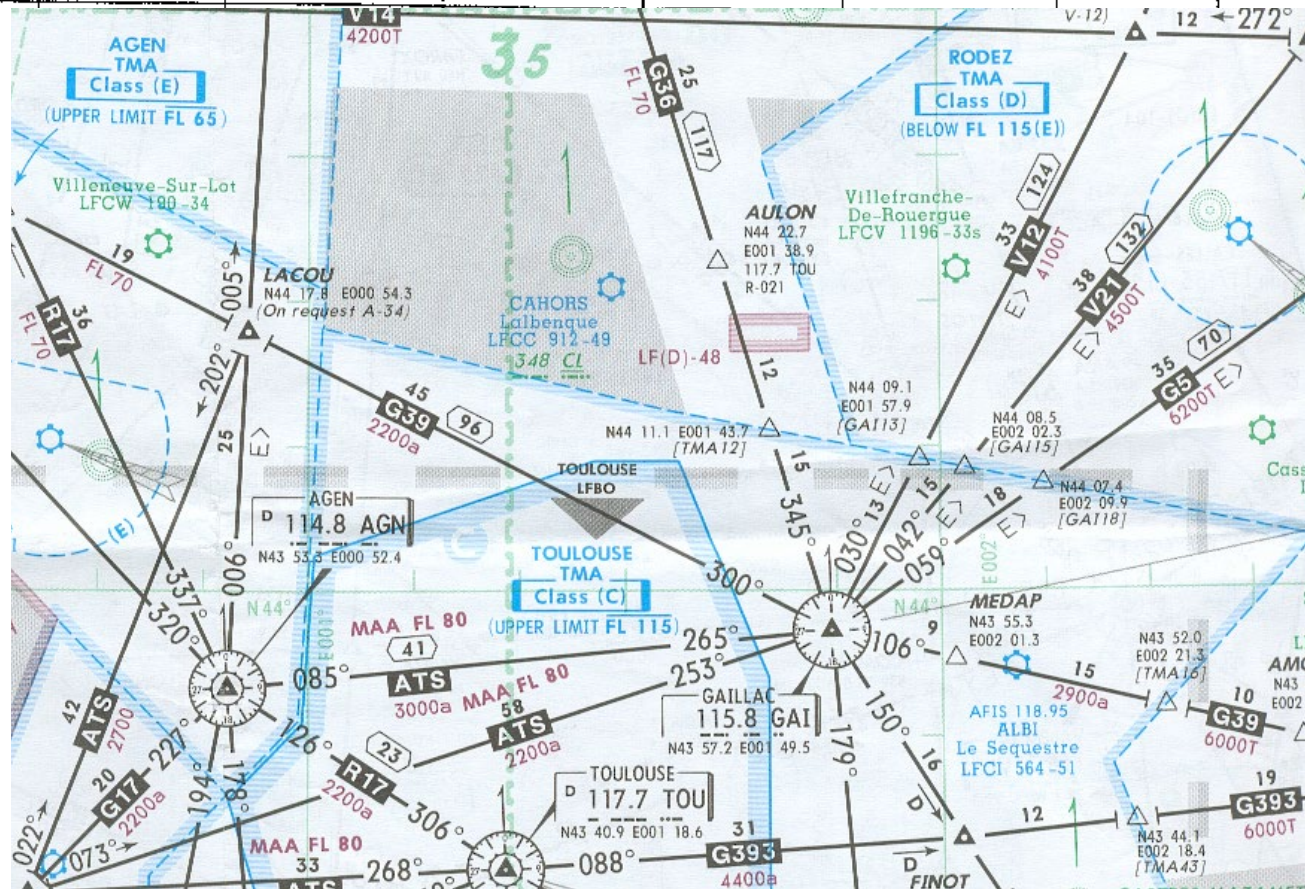
MORA

MORA (1) (Jeppesen)	MINIMUM OFF ROUTE ALTITUDE	3300a	SOUS AIRWAY (EANC) (2)	+/- 10 NM DU CENTERLINE DE L'AWY ASSOCIEE	De 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	→ Calcul JEPPESEN (3) Assure uniquement la protection sol.
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GRID MORA

GRID MORA (1) (Jeppesen)	MINIMUM OFF ROUTE ALTITUDE Dans Grille de 1° de longitude sur 1° de latitude		TOUT ESPACE (EAC-EANC)	1° de Longitude 1° de Latitude	de 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	- Calcul JEPPESEN (3) Permet le calcul des protections donc des niveaux de vol hors airways
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AMA

AMA (1) (Jeppesen)	AREA MINIMUM ALTITUDE	TMA +CTR (EAC)	Courbes de niveaux => variables	de 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	-Calcul JEPPESEN (3) Sur cartes AREA, altitudes minimales exprimées sous forme de courbes de niveau (couleur verte)
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